

5G RAN

Downlink Scheduling Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Added descriptions of cell requirements for the PDCCH rate matching switch. For details, see [8.9.3.4 Cells](#).
Added descriptions of the mutually exclusive relationship between rate-matching-pattern-configuration-based PDCCH rate matching and the number of symbols occupied by the PDCCH. For details, see [8.9.3.2.2 Mutually Exclusive Functions](#).
Added descriptions of the mutually exclusive relationship between "TRS rate matching optimization" and "decoupling of joint transmission from TRS rate matching". For details, see [8.9.3.2.2 Mutually Exclusive Functions](#).

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added DCCH type 1 scheduling. For details, see 8.11 DCCH Type 1 Scheduling .	Modified parameter: Added the DCCH_TYPE1_SCH_SW option to the NRDUCellDISchRes.DISchResAlgoSw parameter. Added the NRDUCellDISchRes.DcchType1SchUserNumThld parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added optimized scheduling of tail RBGs. For details, see 8.12 Optimized Scheduling of Tail RBGs .	Modified parameter: Added the TAIL_RBG_SCH_OPT_SW option to the NRDUCellDISchRes.DISchResAlgoSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between rate-matching-pattern-configuration-based PDCCH rate matching and time-domain power saving BWP. For details, see 8.9.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of downlink PDSCH coverage enhancement with rate-matching-pattern-configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate matching. For details, see 8.9.3.2.3 Function Impacts .	None	High - frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between rate-matching-pattern-configuration-based PDCCH rate matching and UL and DL Decoupling. For details, see 8.9.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between RB allocation for downlink experience optimization and RF module fast deep dormancy. For details, see 8.4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Added the definition of UEs using simplified SRS-based weights and description of the scenarios where downlink MCS index selection optimization for UEs using simplified SRS-based weights takes effect. For details, see [7.10.1 Principles](#) and [7.10.3.2.1 Prerequisite Functions](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010011	Scheduling	4 Scheduling Content and Type Determining
FBFD-010011	Scheduling	5 UE Scheduling
FBFD-010011	Scheduling	6 Rank Selection
FBFD-010011	Scheduling	7 MCS Selection
FBFD-010011	Scheduling	8 PDSCH Time-Frequency Resource Allocation
FBFD-010011	Scheduling	9 DMRS Resource Allocation

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Scheduling content and type determining	None	4 Scheduling Content and Type Determining
UE scheduling	None	5 UE Scheduling
Rank selection	None	6 Rank Selection
MCS selection	None	7 MCS Selection
PDSCH time-frequency resource allocation	None	8 PDSCH Time-Frequency Resource Allocation
DMRS resource allocation	None	9 DMRS Resource Allocation

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Scheduling content and type determining	None	4 Scheduling Content and Type Determining
UE scheduling	None	5 UE Scheduling
Rank selection	None	6 Rank Selection
MCS selection	None	7 MCS Selection
PDSCH time-frequency resource allocation	None	8 PDSCH Time-Frequency Resource Allocation
DMRS resource allocation	None	9 DMRS Resource Allocation

3 Overview

3.1 Basic Concepts and Definitions

The New Radio (NR) system uses shared channels for transmission, with time-frequency resources dynamically allocated among UEs. The Scheduling feature allows the gNodeB to achieve time-frequency resource allocation in the uplink and downlink and maximizes system capacity while guaranteeing user fairness.

Scheduler

Schedulers are located at the Media Access Control (MAC) layer of the NR system. The basic function of a scheduler is to allocate appropriate system resources in each slot to UEs for transmission and reception. Schedulers are classified by direction into downlink schedulers and uplink schedulers. This document describes downlink schedulers. For details about uplink schedulers, see *Uplink Scheduling*.

A downlink scheduler allocates physical downlink shared channel (PDSCH) resources to UEs and selects an appropriate MCS index for transmitting system information or UE data. It performs the following functions:

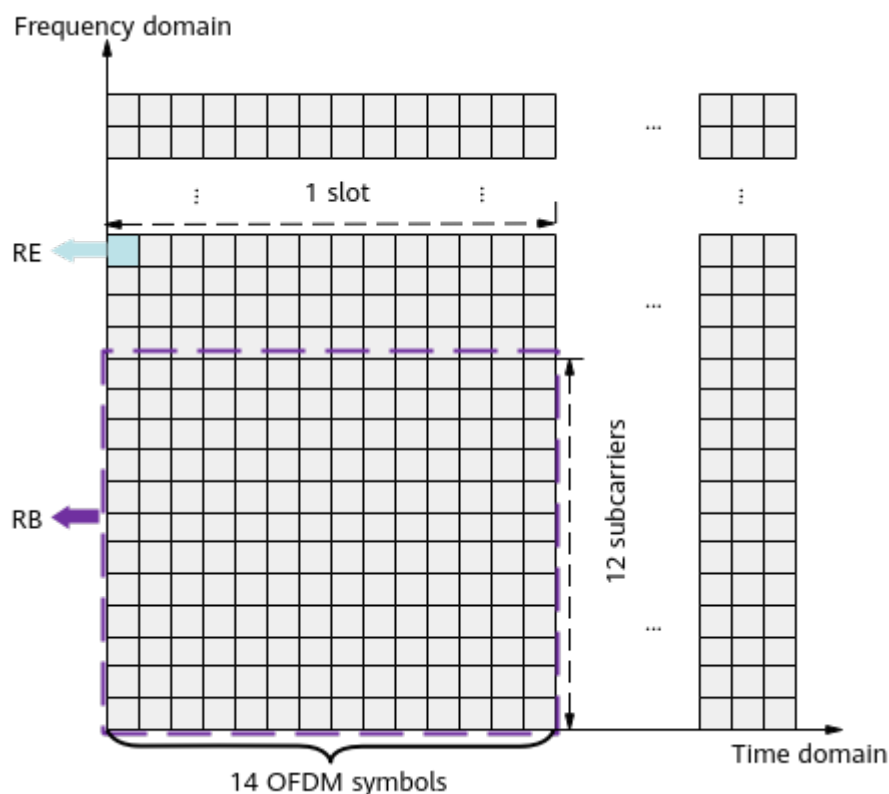
- Allocates PDSCH time-frequency resources to UEs.
- Allocates demodulation reference signal (DMRS) resources to UEs for PDSCH demodulation.
- Selects an appropriate MCS index for PDSCH data transmission.

RE and RB

A resource element (RE) is the smallest time-frequency resource unit in the NR system, comprising one subcarrier in the frequency domain and one OFDM symbol in the time domain.

An RB is the smallest unit of resources that can be allocated for data transmission at the physical layer in the NR system, composed of 12 subcarriers in the frequency domain and any number of symbols in the time domain.

Figure 3-1 illustrates an RE and an RB (in one slot in the time domain as an example).

Figure 3-1 RE and RB

A UE can be allocated multiple RBs in a slot. The maximum number of RBs that can be allocated correlates with the system bandwidth and subcarrier spacing. For example: Up to 273 RBs can be allocated in NR TDD when the bandwidth is 100 MHz and the subcarrier spacing is 30 kHz.

MCS

A modulation and coding scheme (MCS) is a combination of a modulation scheme and code rate. It defines the number of valid bits that can be carried by a resource element (RE). Different modulation schemes use different channel code rates. If the channel quality is good, a gNodeB selects an MCS with a more efficient modulation scheme and a higher code rate (with fewer redundant bits inserted). If the channel quality is poor, the gNodeB selects one with a less efficient modulation scheme and a lower code rate (with more redundant bits inserted). For details, see section 5.1 "Modulation mapper" in 3GPP TS 38.211 V15.5.0.

Rank

A rank determines the maximum number of layers or data streams in a multi-antenna system. The system adaptively selects a rank for downlink data transmission based on the UE-reported rank indicator (RI).

TBS

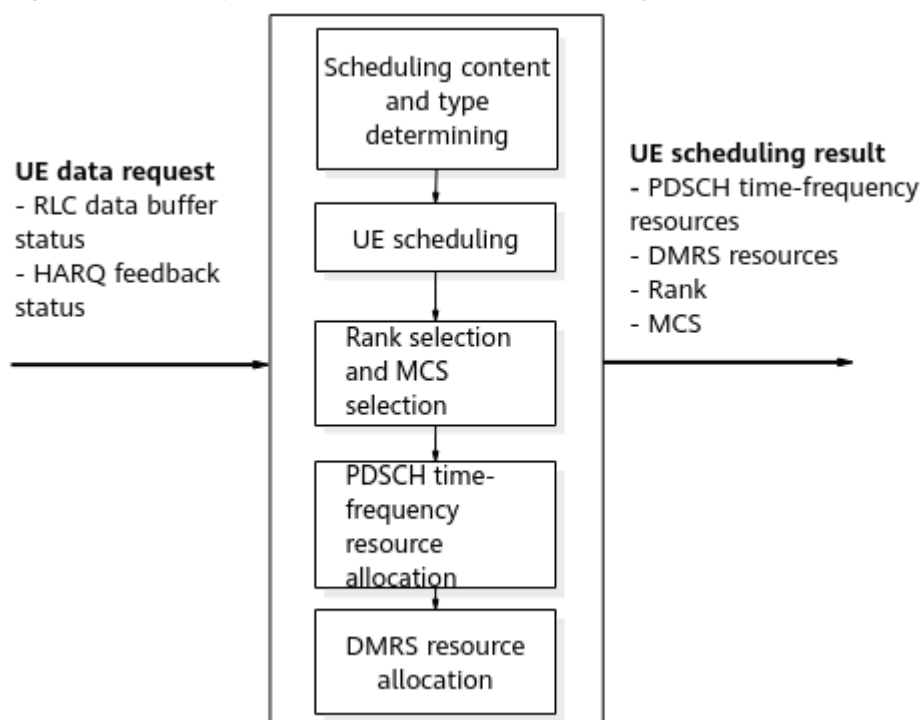
A transport block size (TBS) is the size of a transport block transmitted from the MAC layer to the PHY layer. Peak traffic volume calculation requires that the TBS

of all slots be obtained within a second. The TBS depends on the MCS and the number of resources allocated for data transmission.

3.2 Basic Procedure of Downlink Scheduling

Figure 3-2 shows the basic procedure of downlink scheduling. A scheduler uses UE data requests as the inputs and UE scheduling results as the outputs. For details, see chapter 5 "Physical downlink shared channel related procedure" in 3GPP TS 38.214 V15.4.0.

Figure 3-2 Basic procedure of downlink scheduling



The following describes the inputs and outputs of a downlink scheduler.

- UE data request
 - Radio Link Control (RLC) data buffer status: Refers to the amount of data to be scheduled in the RLC buffer.
 - Hybrid automatic repeat request (HARQ) feedback status: Indicates whether data has been correctly transmitted or retransmitted. The feedback can be an acknowledgment (ACK), negative acknowledgment (NACK), or discontinuous transmission (DTX).
- UE scheduling result
 - PDSCH time-frequency resources: Refer to the PDSCH time-domain resources and frequency-domain resources allocated by the scheduler to a UE.
 - DMRS resources: Refer to the DMRS resources allocated by the scheduler to a UE.
 - MCS: Refers to the modulation and coding scheme signaled by the scheduler to each successfully scheduled UE.

- Rank: Refers to the rank that the scheduler signals to each successfully scheduled UE.

 NOTE

Information such as the antenna quantity of the base station, UE capability, and downlink power also affects the scheduling result.

A downlink scheduler performs scheduling in each slot as follows:

1. Before scheduling data, the downlink scheduler needs to determine the scheduling content and type. For details, see [4 Scheduling Content and Type Determining](#).
2. Based on inputs, the scheduler determines the scheduling priorities of bearers and selects the UEs to be scheduled. This satisfies scheduling fairness requirements and maximizes system throughput. For details, see [5 UE Scheduling](#).
3. The gNodeB adaptively selects a rank for scheduling based on the UE-reported RI. For details, see [6 Rank Selection](#).
4. The scheduler adaptively selects an MCS for scheduling based on the UE-reported CQI. After receiving a CQI from a UE, the scheduler uses a certain mechanism to check the validity and reliability of the CQI. If the CQI is validated, the scheduler performs initial MCS selection. The initial MCS may not allow the downlink IBLER to close in on the target value. Therefore, the scheduler needs to adjust the MCS based on HARQ feedback to enable the downlink IBLER to close in on the target value, improving the downlink throughput of the system. For details, see [7 MCS Selection](#).
5. PDSCH time-frequency resources (time-domain and frequency-domain resources) are allocated. The number and positions of RBs to be allocated to a UE are determined based on the data volume and MCS. For details, see [8 PDSCH Time-Frequency Resource Allocation](#).
6. DMRS resources are allocated for UEs to demodulate the PDSCH. Section 7.4.1 "Reference signals" in 3GPP TS 38.211 V15.4.0 defines PDSCH DMRS, in terms of DMRS symbols in the time domain and DMRS configuration types in the frequency domain. For details, see [9 DMRS Resource Allocation](#).

4 Scheduling Content and Type Determining

4.1 Principles

Before scheduling data, a downlink scheduler needs to determine the scheduling content and type.

- Determining the scheduling content

In each slot, the content to be scheduled includes control-plane information and user-plane information, both of which are transmitted on the PDSCH.

- Control-plane information consists of both common and UE-level control information. Common control information includes broadcast messages, paging messages, random access response (RAR) messages, and RRC connection setup messages. UE-level control information includes such information as SRB1, SRB2, and SRB3.
- User-plane information is the service data transmitted between UEs and the radio network.

Generally, the gNodeB does not allow transmission of both UE-level control-plane information and user-plane information for a UE in the same slot. This is to improve transmission reliability of UE-level control-plane information. However, when CCEs are insufficient, concurrent scheduling of UE-level control-plane information and user-plane information is introduced to save CCEs. This function is controlled by the **NRDUCellDISch.DISrbDrbSchPolicy** parameter and is supported only in low frequency bands:

- If this parameter is set to **RELIABLE_CONCURRENT_SCH**, the gNodeB allows for scheduling both UE-level control-plane information and user-plane information of a UE in the same slot but with a low code rate. This setting is recommended when CCEs are insufficient.
- If this parameter is set to **HIGH_SE_CONCURRENT_SCH**, the gNodeB allows transmission of both UE-level control-plane information and user-plane information for a UE in the same slot but with a high code rate. This setting is recommended in extremely heavy-load scenarios (with the downlink PRB usage being greater than 95%).

- If this parameter is set to **SEPARATE_SCH**, concurrent scheduling of UE-level control-plane information and user-plane information does not take effect. That is, UE-level control-plane information and user-plane information are not scheduled in one slot.
- Determining the scheduling type
 - If there is data in the RLC buffer, the scheduler considers that there is new data to be transmitted and defines the scheduling type as initial transmission.
 - If the HARQ feedback status of the downlink data transmission to a UE is ACK, the data transmission is successfully scheduled and the data scheduling procedure ends. However, if unscheduled data remains in the RLC buffer for the UE, initial transmission scheduling continues.
 - If the HARQ feedback status of the downlink data transmission to a UE is NACK or DTX, the data transmission scheduling fails and the scheduler needs to re-schedule the data by performing retransmission scheduling.
 - If the gNodeB does not perform DTX detection on the PUSCH, it may mistakenly detect DTXs as ACKs. In this case, the retransmission delay increases for services in AM, causing the throughput to decrease. In addition, for voice services in UM, retransmissions are not performed at the MAC and RLC layers, causing downlink packet loss. Against this backdrop, PUSCH DTX detection can be enabled (by selecting the **PUSCH_ACK_DTX_DETECT_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter). This function enables the gNodeB to determine whether a UE is in the DTX state by checking ACKs (one or two bits) transmitted over the PUSCH. If a UE is in the DTX state, the gNodeB initiates initial PDSCH scheduling to improve downlink experience for the UE. This function applies only to low-frequency NR TDD cells and SUL cells.

4.2 Network Analysis

4.2.1 Benefits

A scheduler needs to schedule control-plane and user-plane information in each slot. However, the resources that can be allocated in each slot in the system are limited, and the information may not be completely scheduled in one slot. Therefore, before allocating resources, the scheduler needs to determine the scheduling content and type to ensure that important information is preferentially scheduled. This maximizes system capacity while ensuring fairness among users.

Other sub-functions have the following impacts on the network:

- **Concurrent scheduling of UE-level control-plane information and user-plane information**

After this function takes effect in low frequency bands in heavy-load or extremely heavy-load scenarios, there are fewer needs for CCE allocation triggered by user-plane information and a larger number of scheduled UEs in

the downlink. Therefore, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases and the average downlink UE throughput (**User Downlink Average Throughput (DU)**) may increase.

- **PUSCH DTX detection**

This function increases the **Cell Downlink Average Throughput (DU)** and VoNR mean opinion score (MOS).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Scheduling content and type determining is a basic function and has no impact on the network.

Other sub-functions have the following impacts on the network:

- **Concurrent scheduling of UE-level control-plane information and user-plane information**

None

- **PUSCH DTX detection**

This function causes the following impacts: (1) the probability of false detection of ACKs in the uplink decreases, (2) the downlink voice packet loss rate may increase, (3) the **RRC Setup Success Rate (CU)** may slightly increase, and (4) the **Cell Downlink Average Throughput (DU)**, downlink IBLER, number of downlink retransmissions, and downlink RBLER may fluctuate.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

Table 4-1 describes the parameters used for function optimization. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Downlink SRB DRB Schedule Policy	NRDUCellDLSch. DLSrbDrbSchPolicy	None	• When CCEs are insufficient, the value RELIABLE_CONCURRENT_SCH is recommended. • In extremely heavy-load scenarios (with the downlink PRB usage being greater than 95%), the value HIGH_SE_CONCURRENT_SCH is recommended. • In other scenarios, the value SEPARATE_SCH is recommended.
Uplink PUSCH Algorithm Switch	NRDUCellPusch. ULPuschAlgoSwitch	PUSCH_ACK_DTX_DETECT_SW	Select this option if PUSCH DTX detection is required according to the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Enabling concurrent scheduling of UE-level control-plane information and user-plane information (using a  
CCE insufficiency scenario as an example)  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISrbDrbSchPolicy=RELIABLE_CONCURRENT_SCH;  
//(Optional) Enabling PUSCH DTX detection  
MOD NRDUCELLPUSCH: NrDuCellId=7, UIPuschAlgoSwitch=PUSCH_ACK_DTX_DETECT_SW-1;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions. Example:

```
//Disabling PUSCH DTX detection  
MOD NRDUCELLPUSCH: NrDuCellId=7, UIPuschAlgoSwitch=PUSCH_ACK_DTX_DETECT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

4.4.3 Network Monitoring

This function is a basic function that ensures basic service performance. Separate monitoring is not required.

5 UE Scheduling

After determining the transmission scheduling type, the gNodeB performs scheduling based on the result. The basic procedure and optimization functions for UE scheduling are as follows:

- Basic procedure for UE scheduling: For details, see [5.1 Basic Procedure for UE Scheduling](#).
- Optimization functions for UE scheduling: In addition to the basic procedure, scheduling also needs to optimize policies (for example, adjust scheduling priorities for UEs or apply packet bundled scheduling) based on the actual data transmission process in different service scenarios to ensure better transmission performance. [Table 5-1](#) lists optimization functions for UE scheduling.

Table 5-1 Optimization functions for UE scheduling

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
CCE-based SU-MIMO PDSCH balanced scheduling	√	-	5.2 CCE-based SU-MIMO PDSCH Balanced Scheduling
Precise scheduling priority adjustment	√	-	5.3 Precise Scheduling Priority Adjustment
Delay-based scheduling optimization	√	-	5.4 Delay-based Scheduling Optimization

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Downlink scheduling optimization during RRC reconfiguration	√	-	5.5 Downlink Scheduling Optimization During RRC Reconfiguration
To-be-scheduled UE optimization	√	-	5.6 To-Be-Scheduled UE Optimization
Downlink CCE aggregation level-based scheduling optimization	√	-	5.7 Downlink CCE Aggregation Level-based Scheduling Optimization
Downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm	-	√	5.8 Downlink Scheduling Based on Beam Priorities Defined by the Hybrid EPF Algorithm
Downlink first-packet experience improvement	√	-	5.9 Downlink First-Packet Experience Improvement
Enhanced adaptive retransmission in the downlink	√	√	5.10 Enhanced Adaptive Retransmission in the Downlink

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Stopping downlink scheduling for downlink-abnormal UEs	√	√	5.11 Stopping Downlink Scheduling for Downlink-Abnormal UEs
Downlink basic scheduling enhancement	√	-	5.12 Downlink Basic Scheduling Enhancement
Downlink UE scheduling status pre-evaluation	√	-	5.13 Downlink UE Scheduling Status Pre-Evaluation

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

5.1 Basic Procedure for UE Scheduling

5.1.1 Principles

After determining the transmission scheduling type, the gNodeB performs scheduling based on the result.

- Initial transmission scheduling

Downlink initial transmission scheduling uses the enhanced proportional fair (EPF) algorithm to prioritize multiple services with downlink transmission requests, scheduling them in descending order of priority. The EPF formula is as follows:

$$\text{priority} = \frac{\text{eff}}{r} \times Y_{\text{conn}}, \text{ where:}$$

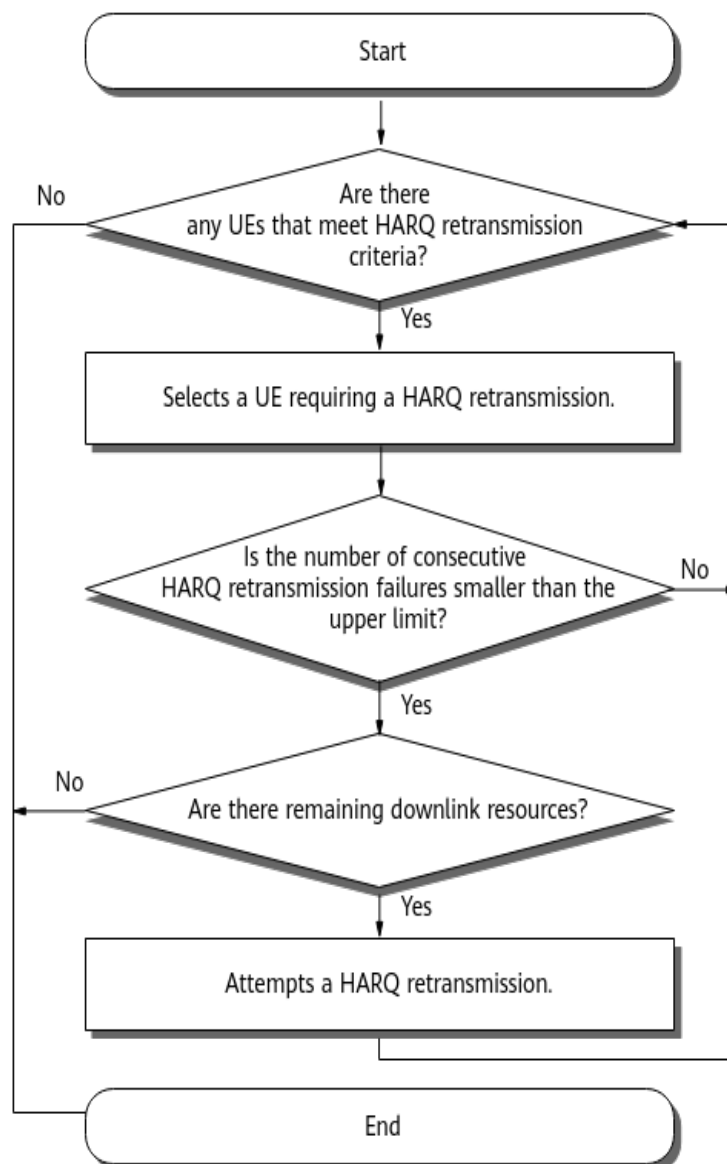
- eff**: Indicates the channel quality of a UE.
- r**: Indicates the historical transmission rate of a UE.
- Y_{conn}**: Indicates the priority weighting factor of different service types. For details about **Y_{conn}**, see *QoS Management*.

- Retransmission scheduling

Downlink retransmission scheduling uses asynchronous adaptive HARQ.

Figure 5-1 shows the scheduling procedure for HARQ retransmissions in each slot.

Figure 5-1 HARQ retransmission scheduling procedure



- a. The scheduler checks for UEs that meet the HARQ retransmission criteria. The scheduler checks information such as whether the current slot is within the inter-frequency measurement gap to determine whether to add UEs with retransmission requests to the retransmission scheduling queue.
 - For a UE that satisfies the retransmission criteria, the scheduler adds the UE to the queue and proceeds to step 2.
 - For a UE that does not satisfy the retransmission criteria, the scheduler ends the retransmission scheduling procedure.

 NOTE

A measurement gap is a time period during which the UE performs measurements on a neighboring frequency of the serving frequency.

- b. The scheduler selects a UE requiring a HARQ retransmission.
The scheduler selects the UE with the highest HARQ retransmission priority.
- c. The scheduler checks whether the number of consecutive retransmission failures is less than the upper limit (specified by the **NRDUCellPdsch.MaxDLHarqTransNum** parameter).
 - If it is, the scheduler attempts to allocate resources for the UE requiring the retransmission and proceeds to step 4.
 - If it is not, the scheduler abandons this retransmission and releases the UE HARQ data.
- d. The scheduler checks whether there are remaining downlink resources.
 - If there are, the scheduler attempts a HARQ retransmission for the UE with the highest priority in the queue and proceeds to step 1.
 - If there are not, the scheduler ends the retransmission scheduling procedure.

5.1.2 Network Analysis

5.1.2.1 Benefits

With the basic procedure for UE scheduling, the scheduler determines whether to perform initial transmission scheduling or retransmission scheduling based on the HARQ feedback status of downlink data transmission. In addition, the scheduler determines the scheduling priorities of UEs in different scheduling types to ensure UE fairness.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

Basic procedure for UE scheduling has no impact on the network.

5.1.3 Requirements

5.1.3.1 Licenses

There are no license requirements for basic functions.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

None

5.1.3.2.2 Mutually Exclusive Functions

None

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowersavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LTE and NR bundled scheduling delays scheduling when certain conditions are met, differing from the original scheduling mechanism.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	The downlink common control information (such as the PBCH and SS information) for each transmission reception point (TRP) within a hyper cell is scheduled in a unified manner. System information for each TRP is the same and occupies the same time-frequency resources. The downlink UE-dedicated signaling and data for each TRP within a hyper cell are independently scheduled and allocated by the serving TRP.
Low-frequency TDD	Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	The downlink common control channels and signals (such as the PBCH and SS) are centrally scheduled for all TRPs in a combined cell. The same information is transmitted for each TRP, occupying the same time-frequency resources.

5.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

Table 5-2 describes the parameters used for function optimization. This section does not describe parameters related to cell establishment.

Table 5-2 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Maximum Downlink HARQ Transmission Number	NRDUCellPdsch. MaxDLHarqTrans Num	None	Use the recommended value.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to **5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the maximum number of downlink HARQ transmissions
MOD NRDUCCELLPDSCH: NrDuCellId=0, MaxDLHarqTransNum=5;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

5.1.4.3 Network Monitoring

This function is a basic function that ensures basic service performance. Separate monitoring is not required.

5.2 CCE-based SU-MIMO PDSCH Balanced Scheduling

5.2.1 Principles

In heavy-load scenarios, the throughput of SU-MIMO cells decreases when PDCCH resources are limited. CCE-based SU-MIMO PDSCH balanced scheduling is introduced to increase cell throughput. The **CCE_BASED_SU_PDSCH_BAL_SCH_SW** option of the **NRDUCellPdsch.DlPdschAlgoSwitch** parameter specifies whether to enable this function. After this function is enabled, when the number of UEs in RRC_CONNECTED mode is greater than or equal to the threshold specified by the **NRDUCellPdsch.SuBalSchUeNumThld** parameter, the scheduler increases the scheduling priority of UEs performing large-packet services in SU-MIMO scenarios. This improves the spectral efficiency and cell throughput. CCE-based SU-MIMO PDSCH balanced scheduling is supported only in low frequency bands, not in high frequency bands.

5.2.2 Network Analysis

5.2.2.1 Benefits

After this function takes effect in low frequency bands, the cell spectral efficiency and average downlink cell throughput increase.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

This function has the following impacts in low-frequency scenarios:

- The number of scheduled RBs in the cell increases, downlink interference increases, and the service drop rate in the cell may slightly increase.

- The scheduling delay of downlink small-packet services increases. If the downlink transmission delay increases at a rate higher than that of the downlink traffic volume, the downlink cell traffic volume decreases.

5.2.3 Requirements

5.2.3.1 Licenses

There are no license requirements for basic functions.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

None

5.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

[Table 5-3](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DIPdschAlgoSwitch	CCE_BASED_SU_PDSCH_BAL_SCH_SW	When PDCCH resources and cell throughput are limited, select this option; otherwise, you are advised to deselect this option.
CCE-based SU PDSCH Balanced Sch UE Num Thld	NRDUCellPdsch.SuBalSchUeNumThld	None	In low-frequency TDD scenarios, you are advised to set this parameter to the value of the cell bandwidth (MHz) multiplied by 4.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for CCE-based SU-MIMO PDSCH balanced scheduling
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=CCE_BASED_SU_PDSCH_BAL_SCH_SW-1;
//Configuring the threshold of the number of UEs for triggering CCE-based SU-MIMO PDSCH balanced scheduling
MOD NRDUCELLPDSCH: NrDuCellId=0, SuBalSchUeNumThld=200;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for CCE-based SU-MIMO PDSCH balanced scheduling
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=CCE_BASED_SU_PDSCH_BAL_SCH_SW-0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.2.4.3 Network Monitoring

The indicators listed in [5.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.3 Precise Scheduling Priority Adjustment

5.3.1 Principles

In downlink heavy-load scenarios, the accuracy of downlink scheduling priority calculation for UEs is low. If there is a large number of UEs performing small-packet services in the cell, the number of times UEs performing large-packet services are scheduled decreases. As a result, the average downlink cell throughput decreases. As such, the precise scheduling priority adjustment function is introduced. This function is controlled by the **SCH_PRI_PRECISE_ADJ_SW** option of the **NRDUCellPdsch.DIPdschAlgoSwitch** parameter. This function improves the accuracy of scheduling priority calculation for UEs and increases the number of times UEs performing large-packet services are scheduled, thereby increasing the average downlink cell throughput.

5.3.2 Network Analysis

5.3.2.1 Benefits

After precise scheduling priority adjustment takes effect in downlink low-frequency heavy-load scenarios, the UE scheduling priority calculation is more accurate and the number of times UEs performing large-packet services are scheduled is increased, improving the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**).

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

After this function takes effect in low-frequency scenarios, the average downlink user-perceived rate may decrease when the proportion of UEs performing large-packet services is high.

5.3.3 Requirements

5.3.3.1 Licenses

There are no license requirements for basic functions.

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

None

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

None

5.3.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.3.3.4 Others

None

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

[Table 5-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DIPdschAlgoSwitch	SCH_PRI_PRECISE_ADJ_SW	You are advised to select this option when the number of RRC_CONNECTED UEs in the cell is greater than 200 and the downlink PRB usage in the cell is greater than 95%. In other scenarios, you are advised to deselect this option.

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for precise scheduling priority adjustment (recommended when the number of RRC_CONNECTED UEs in a cell is greater than 200 and the downlink PRB usage in the cell is greater than 95%)
MOD NRDUCCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=SCH_PRI_PRECISE_ADJ_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for precise scheduling priority adjustment
MOD NRDUCCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=SCH_PRI_PRECISE_ADJ_SW-0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.3.4.3 Network Monitoring

The indicators listed in [5.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.4 Delay-based Scheduling Optimization

5.4.1 Principles

For low-frequency TDD cells with at most 8T8R, delay-based scheduling optimization can be used to increase the average downlink UE throughput. This function is controlled by the **DL_DELAY_SCH_OPT_SW** option of the **NRDUCellPdsch.DIPdschAlgoSwitch** parameter. It takes effect for a UE only when the buffer size of the UE is less than the value of the **NRDUCellPdsch.DIDelaySchBufferThld** parameter.

5.4.2 Network Analysis

5.4.2.1 Benefits

After delay-based scheduling optimization takes effect, the average downlink user-perceived rate increases.

5.4.2.2 Impacts

The impacts between this function and other functions are described in [5.4.3.2.3 Function Impacts](#).

Low-frequency NR TDD: Delay-based scheduling optimization increases the **User Downlink Average Throughput (DU)**.

5.4.3 Requirements

5.4.3.1 Licenses

There are no license requirements for basic functions.

5.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAIgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Delay-based scheduling optimization requires that PDSCH MU-MIMO be enabled.

5.4.3.2.2 Mutually Exclusive Functions

None

5.4.3.2.3 Function Impacts

None

5.4.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.4.3.4 Others

None

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

[Table 5-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. <i>DlPdschAlgoSwit ch</i>	DL_DELAY_SCH_O PT_SW	You are advised to select this option.

5.4.4.1.2 Using MML Commands

Before using MML commands, refer to [5.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for delay-based scheduling optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=DL_DELAY_SCH_OPT_SW-1;
```

Deactivation Command Examples

```
//Turning off the switch for delay-based scheduling optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=DL_DELAY_SCH_OPT_SW-0;
```

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.4.4.3 Network Monitoring

The indicators listed in [5.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.5 Downlink Scheduling Optimization During RRC Reconfiguration

5.5.1 Principles

Downlink scheduling optimization during RRC reconfiguration is introduced to reduce the consumption of common CCEs. This function is controlled by the **RRC_RECFG_DL_SCH_OPT_SW** option of the **NRDUCellDlsch.DlschAlgoSwitch** parameter. After this function is enabled, the base station allocates RBs in DCI format 1-1 during RRC reconfiguration, reducing the number of common CCEs consumed in heavy-traffic scenarios and improving CCE usage. This function is supported only in low frequency bands, not in high frequency bands.

5.5.2 Network Analysis

5.5.2.1 Benefits

After this function is enabled, fewer common CCEs are consumed in low-frequency heavy-load scenarios (with a large number of UEs or common CCE congestion), improving CCE usage. In addition, this function relieves common CCE congestion in heavy-load scenarios and may improve indicators as follows:

- The number of RRC_CONNECTED UEs increases.
- The average downlink cell throughput increases.
Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
- The downlink delay decreases, and the uplink cell throughput may increase.
Packet DL Delay in MAC = **N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num**
Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**

5.5.2.2 Impacts

The impacts between this function and other functions are described in [5.5.3.2.3 Function Impacts](#).

After this function is enabled in low-frequency scenarios, during RRC reconfiguration, the downlink IBLER may increase and stopping downlink data transmission scheduling may cause the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) to decrease.

5.5.3 Requirements

5.5.3.1 Licenses

There are no license requirements for basic functions.

5.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

None

5.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RRC reconfiguration procedure optimization	RRC_RECFG_PROCESS_OPT_SW option of the gNodeBParam.ProcessSwitch parameter	None	RRC reconfiguration procedure optimization and downlink scheduling optimization during RRC reconfiguration are mutually exclusive, indicating that they cannot be both enabled.

5.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DC_I_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithm parameter	<i>Multi-Frequency Smart Aggregation</i>	When downlink scheduling optimization during RRC reconfiguration is in effect, the downlink bit error rate of UEs with single DCI in effect may slightly increase during RRC reconfiguration.

5.5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.5.3.4 Others

None

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

Table 5-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCellDISch. DISchAlgoSwitch	RRC_RECFCG_DL_S CH_OPT_SW	You are advised to select this option.

5.5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink scheduling optimization during RRC reconfiguration  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=RRC_RECFCG_DL_SCH_OPT_SW-1;
```

Deactivation Command Examples

```
//Disabling downlink scheduling optimization during RRC reconfiguration  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=RRC_RECFCG_DL_SCH_OPT_SW-0;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.5.4.3 Network Monitoring

The indicators listed in [5.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.6 To-Be-Scheduled UE Optimization

5.6.1 Principles

When there is a relatively large number of RRC_CONNECTED UEs and a relatively high proportion of UEs in the downlink DRX state, information about UEs in the downlink DRX state may not be updated promptly. This will delay scheduling of other UEs and affect the average downlink UE throughput. The to-be-scheduled UE optimization function can be enabled using the **TO_BE_SCHEDULED_UE_OPT_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter to more promptly update scheduling information of to-be scheduled UEs with buffered data, thereby improving the average downlink UE throughput. This function applies only to low-frequency NR TDD cells and SUL cells.

5.6.2 Network Analysis

5.6.2.1 Benefits

When there is a relatively large number of RRC_CONNECTED UEs, the number of UEs scheduled in the uplink may increase after this function takes effect. This function has the following impacts on the network:

- The uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**), uplink interference (**N.UL.NI.Avg**), and average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) may increase.
- The channel conditions of new UEs in uplink scheduling may differ greatly from those of existing UEs. Therefore, the average uplink UE throughput (**User Uplink Average Throughput (DU)**), average uplink MCS index, average uplink rank, average number of paired layers for uplink MU pairing, average proportion of PRBs for uplink MU pairing, and PDCCH CCE usage may fluctuate.

When the proportion of RRC_CONNECTED UEs in the downlink DRX state is high, the number of UEs scheduled in the downlink may also increase. There are also the following impacts on the network:

- The downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**), average downlink UE throughput (**User Downlink Average Throughput (DU)**), and average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) may increase.
- The average downlink MCS index, average downlink rank, average number of paired layers for downlink MU pairing, and average proportion of PRBs for downlink MU pairing may fluctuate.
- When TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video) are used, the number of uplink TCP response packets increases after the number of UEs scheduled in the downlink increases. The network impacts are the same as those of the increase in the number of UEs scheduled in the uplink.

5.6.2.2 Impacts

The impacts between this function and other functions are described in [5.6.3.2.3 Function Impacts](#).

The network impacts produced by to-be-scheduled UE optimization are described in benefit analysis. For details, see [5.6.2.1 Benefits](#).

5.6.3 Requirements

5.6.3.1 Licenses

There are no license requirements for basic functions.

5.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.6.3.2.1 Prerequisite Functions

None

5.6.3.2.2 Mutually Exclusive Functions

None

5.6.3.2.3 Function Impacts

None

5.6.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.6.3.4 Others

None

5.6.4 Operation and Maintenance

5.6.4.1 Data Configuration

5.6.4.1.1 Data Preparation

[Table 5-7](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Service Experience Algorithm Switch	NRDUCellServExp.ServiceExpAlgoSwitch	TO_BE_SCHEDULED_UE_OPT_SW	You are advised to select this option when there is a large number of RRC_CONNECTED UEs.

5.6.4.1.2 Using MML Commands

Before using MML commands, refer to [5.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the to-be-scheduled UE optimization switch (recommended only when there is a relatively large number of RRC_CONNECTED UEs)  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=TO_BE_SCHEDULED_UE_OPT_SW-1;
```

Deactivation Command Examples

```
//Turning off the to-be-scheduled UE optimization switch (recommended only when there is a relatively large number of RRC_CONNECTED UEs)  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=TO_BE_SCHEDULED_UE_OPT_SW-0;
```

5.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.6.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.6.4.3 Network Monitoring

The indicators listed in [5.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.7 Downlink CCE Aggregation Level-based Scheduling Optimization

5.7.1 Principles

In heavy-load or extremely heavy-load scenarios (with the downlink PRB usage being higher than 70%), UEs at the cell edge have relatively high CCE aggregation levels and occupy a relatively large number of CCEs. This increases the CCE allocation failure rate for UEs near the cell center and affects the average downlink cell throughput. To address this issue, downlink CCE aggregation level-based scheduling optimization is introduced. This function is controlled by the **NRDUCellDisch.DLCceBasedSchPriFactor** parameter. When this parameter is set to **OFF**, this function does not take effect. When this parameter is set to a value other than **OFF**, this function takes effect and the gNodeB decreases the scheduling priority of UEs with high CCE aggregation levels, which increases the CCE allocation success rate for UEs near the cell center and the average downlink cell throughput. A larger value of this parameter results in a larger decrease in the scheduling priority of UEs with high CCE aggregation levels. This function is supported only in low-frequency NR TDD scenarios.

5.7.2 Network Analysis

5.7.2.1 Benefits

After this function takes effect in low-frequency NR TDD scenarios with heavy load or extremely heavy load, the CCE allocation success rate of UEs near the cell center, the number of scheduled UEs in the downlink, and the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increase.

5.7.2.2 Impacts

The impacts between this function and other functions are described in [5.7.3.2.3 Function Impacts](#).

In low-frequency NR TDD scenarios with heavy load or extremely heavy load, this function has the following impacts on the network:

- The **User Downlink Average Throughput (DU)** may fluctuate, depending on the traffic model of UEs. For example, if UEs with high CCE aggregation levels contribute a large proportion of traffic, the average downlink UE throughput may decrease because the scheduling priority of such UEs is lowered.
- An increase is caused in the number of scheduled UEs in the downlink, scheduling times of uplink response packets, and consumed PUSCH resources. As a result, the **User Uplink Average Throughput (DU)** may decrease in the case of uplink congestion.

5.7.3 Requirements

5.7.3.1 Licenses

There are no license requirements for basic functions.

5.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.7.3.2.1 Prerequisite Functions

None

5.7.3.2.2 Mutually Exclusive Functions

None

5.7.3.2.3 Function Impacts

None

5.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.7.3.4 Others

None

5.7.4 Operation and Maintenance

5.7.4.1 Data Configuration

5.7.4.1.1 Data Preparation

Table 5-8 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink CCE-based Sch Priority Factor	NRDUCellDLSch. DLCceBasedSchPriFactor	None	You are advised to use this parameter in heavy-traffic scenarios. The recommended configurations are as follows: • In low-frequency NR TDD scenarios with extremely heavy load (with the downlink PRB usage being greater than 95%), the value PRIORITY_16 is recommended. • In low-frequency NR TDD scenarios with heavy load (with the downlink PRB usage being about 70% to 95%), the value PRIORITY_8 is recommended. In other scenarios, the value OFF is recommended.

5.7.4.1.2 Using MML Commands

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink CCE aggregation level-based scheduling optimization (using the setting of PRIORITY_8 as an example)  
MOD NRDUCELLDLSCH: NrDuCellId=0, DlCceBasedSchPriFactor=PRIORITY_8;
```

Deactivation Command Examples

```
//Disabling Downlink CCE aggregation level-based scheduling optimization  
MOD NRDUCELLDLSCH: NrDuCellId=0, DlCceBasedSchPriFactor=OFF;
```

5.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.7.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.7.4.3 Network Monitoring

The indicators listed in [5.7.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.8 Downlink Scheduling Based on Beam Priorities Defined by the Hybrid EPF Algorithm

5.8.1 Principles

In high frequency bands, NR TDD uses the analog-digital hybrid beamforming architecture. The scheduler preferentially allocates analog beam resources to UEs and then allocates digital beam resources. Due to the restriction that RB resources of one analog beam cannot be used by UEs of another beam, when a UE performing small-packet services exclusively occupies an analog beam, remaining RB resources of the beam cannot be used by UEs of other beams, resulting in a

waste of resources. To address this issue, downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm is introduced. This function is controlled by the **NRDUCellDlsch.DlBeamPriBasedSch** parameter. When this parameter is set to **HEPF**, this function is enabled to calculate the priority of each beam based on the EPF priority and required RB quantity of UEs in each beam. Then, scheduling among beams is performed by beam priority, but within a beam, the EPF priority of each UE applies. Therefore, the scheduling priorities of the beams serving UEs with medium- or large-packet services increase, increasing the downlink PRB usage, average downlink UE throughput, and average downlink cell throughput. This function is supported only in high-frequency NR TDD scenarios.

5.8.2 Network Analysis

5.8.2.1 Benefits

In high-frequency NR TDD with medium or heavy load (with the downlink PRB usage of the cell being about 50% to 90%) when the small-packet and large-packet hybrid burst traffic model is used, after downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm takes effect, downlink PRB usage increases, and therefore the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) increase. In other scenarios, this function may not deliver any benefits.

5.8.2.2 Impacts

The impacts between this function and other functions are described in [5.8.3.2.3 Function Impacts](#).

In high-frequency NR TDD, this function has the following impacts on the network:

- Processing delay at the MAC layer and RLC layer in the gNodeB may decrease.
- When both uplink and downlink loads are medium or heavy, enabling "downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm" may decrease the **User Uplink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, uplink data volume at the RLC layer, and average uplink MCS index. If "uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm" is enabled afterwards, the preceding indicators may improve.
- The values of the following indicators may fluctuate, depending on the proportions of UEs near the cell center, at a medium distance from the cell center, and at the cell edge as well as the inbound traffic models of UEs:
 - Average downlink CQI, average downlink MCS index, average downlink rank, downlink IBLER, downlink RBLER, and downlink retransmission proportion
 - Proportion of using the 64QAM MCS index table in the downlink, and proportion of using 64QAM, 16QAM, or QPSK in downlink scheduling
 - Downlink CCE aggregation level, PDCCH CCE usage, average number of paired layers for downlink MU pairing, and average proportion of PRBs for downlink MU pairing

- Number of active UEs in the downlink and number of scheduled UEs in the downlink
- Proportion of using each modulation scheme in the uplink

5.8.3 Requirements

5.8.3.1 Licenses

There are no license requirements for basic functions.

5.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.8.3.2.1 Prerequisite Functions

None

5.8.3.2.2 Mutually Exclusive Functions

None

5.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Multi-beam FDM	VOL_BASED_BEAM_MULTIPLEX_SW option of the NRDUCellAlgoSwitch.BeamMultiplexSwitch parameter	<i>MIMO (TDD)</i>	When both "multi-beam FDM" and "downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm" are enabled, the proportion of multi-beam FDM taking effect decreases, and the benefits produced by the latter function may decrease.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink Hybrid EPF	NRDUCellUlsch.UlBeamPriorityBasedSchedSch set to HEPF	<i>Uplink Scheduling</i>	Enabling "uplink scheduling based on beam priorities defined by the Hybrid EPF algorithm" increases the average uplink cell throughput. However, if "downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm" is enabled afterwards and most UEs with downlink large-packet services are located at the cell edge, the number of uplink response packets increases and the uplink data volume at the RLC layer, average uplink cell throughput, and average uplink UE throughput may decrease.

5.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

5.8.3.4 Others

None

5.8.4 Operation and Maintenance

5.8.4.1 Data Configuration

5.8.4.1.1 Data Preparation

Table 5-9 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Beam Priority-based Schedule	NRDUCellDLSch. DLBeamPriBasedSch	None	NR TDD in high frequency bands: In FWA scenarios where beam resources are insufficient prior to RB resources (when the TTI duty cycle is close to 100% and the downlink PRB usage is less than 90%), you are advised to set this parameter to HEPF . In other scenarios, you are advised to set this parameter to OFF .

5.8.4.1.2 Using MML Commands

Before using MML commands, refer to **5.8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm (only for high frequency bands in NR TDD)  
MOD NRDUCELLDLSCH: NrDuCellId=0, DLBeamPriBasedSch=HEPF;
```

Deactivation Command Examples

```
//Disabling downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm (only for high frequency bands in NR TDD)  
MOD NRDUCELLDLSCH: NrDuCellId=0, DlBeamPriBasedSch=OFF;
```

5.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.8.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.8.4.3 Network Monitoring

The indicators listed in [5.8.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.9 Downlink First-Packet Experience Improvement

5.9.1 Principles

In low-frequency NR TDD scenarios with medium or light load (with the downlink PRB usage being less than 50%), if there are few remaining PRBs, scheduling first-packet UEs with a large amount of buffered data may result in a relatively low average downlink UE throughput. To alleviate this issue, the downlink first-packet experience improvement function is introduced. This function is controlled by the **FIRST_PKT_EXP_IMPROVE_SW** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter and takes effect only when the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter is selected. After the first-packet experience improvement function is enabled, in the event of few remaining PRB resources, the gNodeB uses slots with more resources to schedule first-packet UEs with a large amount of buffered data so that these UEs obtain more PRB resources, increasing the average downlink UE throughput.

5.9.2 Network Analysis

5.9.2.1 Benefits

In low-frequency NR TDD scenarios with medium or light load (with the downlink PRB usage being less than 50%), after this function takes effect, the gNodeB uses slots with more resources to schedule first-packet UEs with a large amount of buffered data when there are few remaining PRB resources, so these UEs can obtain more PRB resources, increasing the average downlink UE throughput (**User Downlink Average Throughput (DU)**).

5.9.2.2 Impacts

The impacts between this function and other functions are described in [5.9.3.2.3 Function Impacts](#).

In low-frequency NR TDD, this function may increase the first-packet transmission delay, which is calculated as follows: $(\text{N.Traffic.DL.RlcFirstPktDelay.Time} / \text{N.Qos.DL.RlcFirstPktDelay.Num}) / 1000$.

5.9.3 Requirements

5.9.3.1 Licenses

There are no license requirements for basic functions.

5.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Downlink first-packet experience improvement can take effect only when distributed massive MIMO is enabled.

5.9.3.2.2 Mutually Exclusive Functions

None

5.9.3.2.3 Function Impacts

None

5.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.9.3.4 Others

None

5.9.4 Operation and Maintenance

5.9.4.1 Data Configuration

5.9.4.1.1 Data Preparation

Table 5-10 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	FIRST_PKT_EXP_IMPROVE_SW	In medium- or light-load scenarios (with the downlink PRB usage being less than 50%), you are advised to select this option. In other scenarios, you are advised to deselect this option.

5.9.4.1.2 Using MML Commands

Before using MML commands, refer to [5.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink first-packet experience improvement  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=FIRST_PKT_EXP_IMPROVE_SW-1;
```

Deactivation Command Examples

```
//Disabling downlink first-packet experience improvement  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=FIRST_PKT_EXP_IMPROVE_SW-0;
```

5.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.9.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.9.4.3 Network Monitoring

The indicators listed in [5.9.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.10 Enhanced Adaptive Retransmission in the Downlink

5.10.1 Principles

To more precisely utilize the retransmission resources, enhanced adaptive retransmission in the downlink is introduced. Enhanced adaptive retransmission in the downlink increases the retransmission bit rate to reduce the number of RBs used for retransmission scheduling. The saved RBs can be used for initial transmission scheduling to further improve the spectral efficiency. In high-frequency NR TDD scenarios and low-frequency NR TDD non-massive MIMO scenarios, enhanced adaptive retransmission in the downlink is enabled by

selecting the **DL_RETRANS_PRECISE_RB_CTRL_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter. In low-frequency NR TDD massive MIMO scenarios, this function is a part of the AHR feature. For details, see *Massive MIMO AHR (Low-Frequency TDD)*.

5.10.2 Network Analysis

5.10.2.1 Benefits

After this function takes effect, enhanced adaptive HARQ can be used in the downlink, which consumes fewer retransmission resources and improves the spectral efficiency when retransmissions occur.

5.10.2.2 Impacts

The impacts between this function and other functions are described in [5.10.3.2.3 Function Impacts](#).

5.10.3 Requirements

5.10.3.1 Licenses

There are no license requirements for basic functions.

5.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.10.3.2.1 Prerequisite Functions

None

5.10.3.2.2 Mutually Exclusive Functions

None

5.10.3.2.3 Function Impacts

None

5.10.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.10.3.4 Others

None

5.10.4 Operation and Maintenance

5.10.4.1 Data Configuration

5.10.4.1.1 Data Preparation

Table 5-11 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-11 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCellDLSch. <i>DLSchAlgoSwitch</i>	DL_RETRANS_PRE CISE_RB_CTRL_SW	High-frequency NR TDD: When the downlink PRB usage is greater than 30%, you are advised to select this option. In other scenarios, you are advised to deselect this option. Low-frequency NR TDD: • Non-massive MIMO scenarios: When the downlink PRB usage is greater than 30%, you are advised to select this option. In other scenarios, you are advised to deselect this option. • Massive MIMO scenarios: For details about how to enable this function, see <i>Massive MIMO AHR (Low-Frequency TDD)</i>.

5.10.4.1.2 Using MML Commands

Before using MML commands, refer to [5.10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling enhanced adaptive retransmission in the downlink  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_RETRANS_PRECISE_RB_CTRL_SW-1;
```

Deactivation Command Examples

```
//Disabling enhanced adaptive retransmission in the downlink  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=DL_RETRANS_PRECISE_RB_CTRL_SW-0;
```

5.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.10.4.2 Activation Verification

When retransmissions occur, if the downlink PRB usage (**Downlink Resource Block Utilizing Rate (DU)**) increases after this function is enabled, this function has taken effect.

5.10.4.3 Network Monitoring

The indicators listed in [5.10.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.11 Stopping Downlink Scheduling for Downlink-Abnormal UEs

5.11.1 Principles

The function of stopping downlink scheduling for downlink-abnormal UEs is introduced to effectively control the downlink RBLER in a cell. After this function is enabled, the gNodeB stops downlink scheduling for UEs with a consecutively high downlink BLER. This reduces the downlink RBLER, further improves spectral efficiency, and increases the average downlink cell throughput in resource insufficiency scenarios. This function is enabled by selecting the **ABN_UE_STOP_DLSCH_SW** option of the **NRDUCellPdsch.DIPdschAlgoSwitch** parameter.

5.11.2 Network Analysis

5.11.2.1 Benefits

After this function is enabled for a cell serving UEs with a consecutively high downlink BLER, the downlink RBLER of the cell decreases. In addition, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases in any of the following scenarios:

- Downlink CCE resources are insufficient.
- Downlink PRB resources are insufficient.
- Downlink beam resources in high frequency bands are insufficient.

Downlink RBLER of a cell (%) = $(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB) \times 100\%$

5.11.2.2 Impacts

The impacts between this function and other functions are described in [5.11.3.2.3 Function Impacts](#).

Low frequency: This function may decrease the downlink PRB usage of a cell because the gNodeB stops downlink scheduling for abnormal UEs.

High frequency: This function affects the beam priority and UE scheduling proportion because the gNodeB stops downlink scheduling for abnormal UEs. This may cause fluctuations in some indicators. Especially, the **User Downlink Average Throughput (DU)** decreases when the abnormal UEs have poor uplink channel quality. Such indicators are as follows:

- PRB usage
- Average uplink/downlink MCS index
- PDCCH CCE usage
- CQI-related indicators
- Number of active UEs
- BLER-related indicators
- Average uplink and downlink UE throughputs: **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**

In addition, when downlink scheduling is stopped for downlink-abnormal UEs, these UEs may not be released. This issue can be solved by setting **NRDUCellQciBearer.UelnactivityTimer** to a value other than 0.

5.11.3 Requirements

5.11.3.1 Licenses

There are no license requirements for basic functions.

5.11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.11.3.2.1 Prerequisite Functions

None

5.11.3.2.2 Mutually Exclusive Functions

None

5.11.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	UE link fault detection	NRDUCellAlgoSwitch.UeLinkAbnormalDetectSwitch	None	After the function of stopping downlink scheduling for downlink-abnormal UEs is enabled, link fault detection may fail for UEs for which scheduling has been stopped. As a result, these UEs cannot be released, but air interface resources can be saved.

5.11.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.11.3.4 Others

None

5.11.4 Operation and Maintenance

5.11.4.1 Data Configuration

5.11.4.1.1 Data Preparation

[Table 5-12](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DIPdschAlgoSwitch	ABN_UE_STOP_DL SCH_SW	In low-frequency scenarios, you are advised to select this option. In high-frequency scenarios, you are advised to select this option when there are UEs with a high BLER in a cell.

5.11.4.1.2 Using MML Commands

Before using MML commands, refer to [5.11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for stopping downlink scheduling for downlink-abnormal UEs
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=ABN_UE_STOP_DLSCH_SW-1;
```

Deactivation Command Examples

```
//Turning off the switch for stopping downlink scheduling for downlink-abnormal UEs
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=ABN_UE_STOP_DLSCH_SW-0;
```

5.11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.11.4.2 Activation Verification

When resources are insufficient, if the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

5.11.4.3 Network Monitoring

The indicators listed in [5.11.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.12 Downlink Basic Scheduling Enhancement

5.12.1 Principles

To improve the downlink scheduling capability of the gNodeB in heavy-load scenarios, the downlink basic scheduling enhancement function can be enabled to simplify the downlink scheduling algorithm. The **DL_SCHEDULE_CPU_DEC_SW** option of the **gNodeBParam.NrBoardPerformanceSw** parameter specifies whether to enable downlink basic scheduling enhancement.

5.12.2 Network Analysis

5.12.2.1 Benefits

After this function is enabled in low-frequency heavy-load scenarios, the number of UEs scheduled in the cell increases, but the downlink cell throughput, average downlink user-perceived rate, and average MCS index for downlink scheduling decrease.

5.12.2.2 Impacts

The impacts between this function and other functions are described in [5.12.3.2.3 Function Impacts](#).

The network impacts produced by this function are described in benefit analysis. For details, see [5.12.2.1 Benefits](#).

5.12.3 Requirements

5.12.3.1 Licenses

There are no license requirements for basic functions.

5.12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.12.3.2.1 Prerequisite Functions

None

5.12.3.2.2 Mutually Exclusive Functions

None

5.12.3.2.3 Function Impacts

None

5.12.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.12.3.4 Others

None

5.12.4 Operation and Maintenance

5.12.4.1 Data Configuration

5.12.4.1.1 Data Preparation

[Table 5-13](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
NR Board Performance Switch	gNodeBParam.NrBoardPerformanceSw	DL_SCHEDULE_CPU_DEC_SW	You are advised to select this option.

5.12.4.1.2 Using MML Commands

Before using MML commands, refer to [5.12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for downlink basic scheduling enhancement
MOD GNODEBPARAM: NrBoardPerformanceSw=DL_SCHEDULE_CPU_DEC_SW-1;
```

Deactivation Command Examples

```
//Turning off the switch for downlink basic scheduling enhancement  
MOD GNODEBPARAM: NrBoardPerformanceSw=DL_SCHEDULE_CPU_DEC_SW-0;
```

5.12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.12.4.2 Activation Verification

If the number of UEs scheduled in the downlink in a cell (**N.User.Schedule.DL.Sum**) increases after this function is enabled, this function has taken effect.

5.12.4.3 Network Monitoring

The indicators listed in [5.12.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

5.13 Downlink UE Scheduling Status Pre-Evaluation

5.13.1 Principles

UEs to be scheduled may not be scheduled in special periods (for example, during gap-assisted measurement). In medium- and heavy-load scenarios (for example, the downlink PRB usage is greater than 30%), the downlink scheduling capability of the gNodeB decreases when such UEs enter the scheduling process. In this regard, the downlink UE scheduling status pre-evaluation function is introduced. This function is controlled by the **DL_UE_SCH_STATE_PREEVAL_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter:

- If this option is selected, the gNodeB performs status pre-evaluation on UEs for which there is buffer data to be scheduled, and UEs that cannot be scheduled in the current slot will not enter the scheduling process of the cell. This ensures the downlink scheduling capability of the gNodeB.
- If this option is deselected, the gNodeB does not perform status pre-evaluation on UEs to be scheduled, and the gNodeB evaluates whether to stop the scheduling of such a UE after the UE enters the scheduling process of the cell.

This function is supported only in low frequency bands, not in high frequency bands.

5.13.2 Network Analysis

5.13.2.1 Benefits

After this function is enabled in low-frequency scenarios, the average number of scheduled UEs in the downlink in a cell may increase, and the average downlink UE throughput (**User Downlink Average Throughput (DU)**) may fluctuate depending on the traffic model.

5.13.2.2 Impacts

The impacts between this function and other functions are described in [5.13.3.2.3 Function Impacts](#).

This function is enabled by default. It may increase the average number of UEs scheduled in the downlink in a cell. Depending on the traffic model, the values of the following indicators may fluctuate:

- Indicators related to uplink and downlink scheduling, such as the average downlink rank, average downlink MCS index, **Cell Downlink Average Throughput (DU)**, **Downlink Traffic Volume (DU)**, downlink IBLER, downlink RBLER, **N.DL.Pwr.Avg**, average PDCCH aggregation level, CCE allocation failure rate, PDCCH CCE usage, average number of paired layers for downlink MU pairing, and **N.PR.B.DL.ExtendUsed.Avg**.
- Load-related indicators, such as the downlink PRB usage, **N.User.RRConn.Active.DL.Max**, **N.User.RRConn.Active.DL.Avg**, **N.User.RRConn.Active.Max**, and **N.User.RRConn.Active.Avg**
- Access-related indicators, such as the **QoS Flow Setup Success Rate (CU)**, **NGSIG Connection Setup Success Rate (CU)**, **RRC Setup Success Rate (CU)**, RRC connection reestablishment success rate, contention-based random access success rate, and non-contention-based random access success rate
- Delay-related indicators, such as the delay of the first packet at the RLC layer, RLC service transmission delay, **N.PDCP.DL.Cell5QI.Delay**, and **N.RLC.DL.Cell5QI.Delay**
- NSA networking-related indicators, such as the SgNB abnormal release rate in NSA networking
- Retainability-related indicators, such as the **Service Call Drop Rate (CU)** and voice service drop rate
- Other indicators, such as the proportion of DRX sleep time, **N.UECntx.AbnormRel**, **N.QosFlow.AbnormRel**, and **N.PDUSession.AbnormRel**

NOTE

RRC connection reestablishment success rate = $(N.RRC.ReEst.Succ + N.RRC.SetupSucc.ReEst) / N.RRC.ReEst.Att$

Contention-based random access success rate = $N.RA.Contention.Resolution.Succ / N.RA.Contention.Att$

Non-contention-based random access success rate = $N.RA.Dedicated.Msg3 / N.RA.Dedicated.Att$

SgNB abnormal release rate in NSA networking = $(N.NsaDc.SgNB.Rel - N.NsaDc.SgNB.Rel.SgNBTrigger - N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel + N.NsaDc.SgNB.AbnormRel) / N.NsaDc.SgNB.Rel \times 100\%$

Voice service drop rate = $N.QosFlow.AbnormRel.Cell5QI / (N.QosFlow.NormRel.Cell5QI + N.QosFlow.AbnormRel.Cell5QI)$

5.13.3 Requirements

5.13.3.1 Licenses

There are no license requirements for basic functions.

5.13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.13.3.2.1 Prerequisite Functions

None

5.13.3.2.2 Mutually Exclusive Functions

None

5.13.3.2.3 Function Impacts

None

5.13.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.13.3.4 Others

None

5.13.4 Operation and Maintenance

5.13.4.1 Data Configuration

5.13.4.1.1 Data Preparation

[Table 5-14](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-14 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCeLLDLSch. <i>DISchAlgoSwitch</i>	DL_UE_SCH_STAT E_PREEVAL_SW	Use the default value.

5.13.4.1.2 Using MML Commands

Before using MML commands, refer to [5.13.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink UE scheduling status pre-evaluation  
MOD NRDUCeLLDLSCH: NRDuCellId=0, DISchAlgoSwitch=DL_UE_SCH_STATE_PREEVAL_SW-1;
```

Deactivation Command Examples

```
//Disabling downlink UE scheduling status pre-evaluation  
MOD NRDUCeLLDLSCH: NRDuCellId=0, DISchAlgoSwitch=DL_UE_SCH_STATE_PREEVAL_SW-0;
```

5.13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.13.4.2 Activation Verification

In medium- and heavy-load scenarios (for example, the downlink PRB usage is greater than 30%), if the total number of UEs scheduled in the downlink in a cell (**N.User.Schedule.DL.Sum**) increases after this function is enabled, this function has taken effect.

5.13.4.3 Network Monitoring

The indicators listed in [5.13.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6 Rank Selection

6.1 Principles

The rank selection procedure by the gNodeB varies before and after the reception of CSI measurement results from UEs:

- Before the reception of CSI measurement results from UEs: Rank selection is required when a UE newly accesses a cell or is newly handed over to a cell.

In NR TDD:

- When a UE newly accesses a cell, the rank used for downlink transmission is specified by the **NRDUCellPdsch.DlInitRank** parameter.
- When a UE is newly handed over to a cell, the rank used for downlink transmission is specified by the **NRDUCellPdsch.DlInitRankAfterHo** parameter.

- After the reception of CSI measurement results from UEs:

In high-frequency NR TDD scenarios, after a UE reports the CSI measurement results of the local cell, the system selects a rank for UE data transmission based on the UE-reported rank indicator (RI).

In low-frequency NR TDD:

- In the event of using PMI-based weights in the downlink, after a UE reports the CSI measurement results of the local cell, the system selects a rank for UE data transmission based on the UE-reported RI.
- In the event of using SRS-based weights in the downlink, the RI in CSI-RS measurement results generally cannot reflect the optimal rank. In this case, the gNodeB needs to determine the downlink rank for a UE based on both CSI-RS and SRS measurements. Therefore, downlink rank adaptation with SRS-based weighting is introduced. This function is controlled by the **DL_RANK_ADAPT_SW** option of the **NRDUCellPdsch.DlLinkAdaptEnhancementSw** parameter.
 - When downlink rank adaptation with SRS-based weighting is disabled, the downlink rank for a UE is determined by the RI that the UE reports.

- When downlink rank adaptation with SRS-based weighting is enabled, the gNodeB adaptively determines the downlink rank for a UE based on both CSI-RS and SRS measurements.
 - If the SRS PreSINR of a UE is greater than the SRS PreSINR threshold for downlink rank adaptation (**NRDUCellDLRank.DIRankAdaptSrsPresinrThld**): See [Downlink Rank Adaptation with SRS-based Weighting: Scenario 1](#).
 - If the SRS PreSINR of a UE is less than or equal to the SRS PreSINR threshold for downlink rank adaptation (**NRDUCellDLRank.DIRankAdaptSrsPresinrThld**): See [Downlink Rank Adaptation with SRS-based Weighting: Scenario 2](#).

Downlink Rank Adaptation with SRS-based Weighting: Scenario 1

If the SRS PreSINR of a UE is greater than the SRS PreSINR threshold for downlink rank adaptation (**NRDUCellDLRank.DIRankAdaptSrsPresinrThld**), the spectral efficiency thresholds for increasing or decreasing the downlink rank are configurable after downlink rank adaptation with SRS-based weighting is enabled. [Table 6-1](#) describes the configuration of these thresholds.

Table 6-1 Configuration of spectral efficiency thresholds for increasing or decreasing the downlink rank

Type	Parameter	Description
DL Rank2 Increase Spectral Eff Thld	NRDUCellDLRank.DIRank2IncSpctEffThld	If the downlink rank for a UE is rank 2 and the spectral efficiency ratio of the target increased rank to that of rank 2 is greater than the threshold specified by this parameter, the base station adaptively increases the downlink rank for the UE.
DL Rank3 Increase Spectral Eff Thld	NRDUCellDLRank.DIRank3IncSpctEffThld	If the downlink rank for a UE is rank 3 and the spectral efficiency ratio of the target increased rank to that of rank 3 is greater than the threshold specified by this parameter, the base station adaptively increases the downlink rank for the UE.
DL Rank3 Decrease Spectral Eff Thld	NRDUCellDLRank.DIRank3DecSpctEffThld	If the downlink rank for a UE is rank 3 and the spectral efficiency ratio of the target decreased rank to that of rank 3 is greater than the threshold specified by this parameter, the base station adaptively decreases the downlink rank for the UE.

Type	Parameter	Description
DL Rank4 Decrease Spectral Eff Thld	NRDUCellIDIRank.DIRank4DecSpectEffThld	If the downlink rank for a UE is rank 4 and the spectral efficiency ratio of the target decreased rank to that of rank 4 is greater than the threshold specified by this parameter, the base station adaptively decreases the downlink rank for the UE.

After downlink rank adaptation with SRS-based weighting is enabled, if the switch for the downlink single-UE multi-layer power control function (**DL_SU_MULTI_LAYER_PWR_CTRL_SW** option of the **NRDUCellAlgoSwitch.SuMimoMultipleLayerSw** parameter) and the switch for the downlink single-UE multi-layer power control adaptation function (**DL_MULTI_LAYER_PC_ADAPT_SW** option of the **NRDUCellIDIRank.DIRankSelAlgoSw** parameter) are both turned on:

- For UEs with downlink single-UE multi-layer power control in effect, downlink rank adaptation with SRS-based weighting still uses the parameters in [Table 6-1](#) as the decision thresholds.
- For UEs on which downlink single-UE multi-layer power control has not taken effect, relevant thresholds are configured as follows in downlink rank adaptation with SRS-based weighting:
 - **NRDUCellIDIRank.DINonPcRankIncSpectEffThld** is used as the spectral efficiency threshold for rank increase when rank 2 or 3 is being used for scheduling.
 - **NRDUCellIDIRank.DINonPcRankDecSpectEffThld** is used as the spectral efficiency threshold for rank decrease when rank 3 or rank 4 is being used for scheduling.

NOTE

- If SRS-based weights are used in the downlink, the cell must be configured with four or more transmit antennas and receive antennas.
- To prevent consecutive block errors caused by using an excessively high downlink rank, downlink rank adaptation with SRS-based weighting limits the highest downlink rank for UEs based on the inter-layer signal strength difference in SRS measurements. The **NRDUCellIDIRank.DIMaxRankLimitLayerGapThld** parameter specifies the inter-layer signal strength difference threshold for limiting the highest downlink rank. When the inter-layer signal strength difference is greater than this threshold, the gNodeB limits the highest rank in downlink rank adaptation.
- When the switch for the downlink single-UE multi-layer power control function (**DL_SU_MULTI_LAYER_PWR_CTRL_SW** option of the **NRDUCellAlgoSwitch.SuMimoMultipleLayerSw** parameter) is turned on and the rank for a UE is greater than 2 in downlink scheduling, the MCS index must be greater than the MCS threshold for multi-layer power control (specified by the **NRDUCellIDIRank.MultiLayerPcHighRankMcsThd** parameter) to make the function take effect.

Downlink Rank Adaptation with SRS-based Weighting: Scenario 2

If the SRS PreSINR of a UE is less than or equal to the SRS PreSINR threshold for downlink rank adaptation (specified by the **NRDUCellDLRank.DLRankAdaptSrsPresinrThld** parameter), after downlink rank adaptation with SRS-based weighting is enabled, the gNodeB adjusts the downlink rank for UEs through downlink rank adaptation for cell edge UEs running small-packet services with SRS-based weighting (controlled by the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDLRank.DLRankSelAlgoSw** parameter).

6.2 Network Analysis

6.2.1 Benefits

This function enables appropriate rank selection for UEs to be scheduled, ensuring system transmission performance and further improving user experience.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

6.3 Requirements

6.3.1 Licenses

There are no license requirements for basic functions.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details, see [4.3.3 Hardware](#).

6.3.4 Others

None

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

NR TDD: [Table 6-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Initial Rank	NRDUCellPdsch. <i>DLInitRank</i>	None	It is recommended that Huawei engineers configure this parameter based on network requirements.
Downlink Initial Rank After Handover	NRDUCellPdsch. <i>DLInitRankAfter-Ho</i>	None	It is recommended that Huawei engineers configure this parameter based on network requirements.
DL Link Adaptation Enhancement Switch	NRDUCellPdsch. <i>DLLinkAdaptEnhancementSw</i>	DL_RANK_ADAPT_SW	Select this option.
DL Rank Adaptation SRS PreSINR Threshold	NRDUCellDLRank. <i>DLRankAdaptSrsPresinrThld</i>	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank2 Increase Spectral Eff Thld	NRDUCellDLRank <i>.DLRank2IncSpctEffThld</i>	None	- When downlink single-UE multi-layer power control is disabled (by deselecting the DL_SU_MULTILAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), use the default value. - When downlink single-UE multi-layer power control is enabled (by selecting the DL_SU_MULTILAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), it is recommended that this parameter be configured by Huawei engineers based on network requirements.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank3 Increase Spectral Eff Thld	NRDUCellDLRank <i>.DLRank3IncSpctEffThld</i>	None	• When downlink single-UE multi-layer power control is disabled (by deselecting the DL_SU_MULTILAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), use the default value. • When downlink single-UE multi-layer power control is enabled (by selecting the DL_SU_MULTILAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), it is recommended that this parameter be configured by Huawei engineers based on network requirements.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank3 Decrease Spectral Eff Thld	NRDUCellDLRank .DLRank3DecSpct EffThld	None	- When downlink single-UE multi-layer power control is disabled (by deselecting the DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), use the default value. - When downlink single-UE multi-layer power control is enabled (by selecting the DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), it is recommended that this parameter be configured by Huawei engineers based on network requirements.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank4 Decrease Spectral Eff Thld	NRDUCellDLRank <i>.DLRank4DecSpct EffThld</i>	None	- When downlink single-UE multi-layer power control is disabled (by deselecting the DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), use the default value. - When downlink single-UE multi-layer power control is enabled (by selecting the DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayersSw parameter), it is recommended that this parameter be configured by Huawei engineers based on network requirements.

Parameter Name	Parameter ID	Option	Setting Notes
DL Rank Selection Algorithm Switch	NRDUCelIDlRank .DlRankSelAlgoSw	DL_MULTI_LAYER_PC_ADAPT_SW	You are advised to select this option when downlink single-UE multi-layer power control is enabled (by selecting the DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCelAlgoSwitch.SuMimoMultipleLayerSw parameter).
DL Non-PC Rank Increase Spct Eff Thld	NRDUCelIDlRank .DlNonPcRankIncSpctEffThld	None	Use the default value.
DL Non-PC Rank Decrease Spct Eff Thld	NRDUCelIDlRank .DlNonPcRankDecSpctEffThld	None	Use the default value.
DL Max Rank Limit Layer Gap Thld	NRDUCelIDlRank .DlMaxRankLimitLayerGapThld	None	Use the default value.
Multi-Layer PC High Rank MCS Thld	NRDUCelIDlRank .MultiLayerPcHighRankMcsThld	None	Use the default value.
DL Rank Selection Algorithm Switch	NRDUCelIDlRank .DlRankSelAlgoSw	DL_SMALL_PKT_CEU_RANK_ADAPT_SW	Use the default value.
SU-MIMO Multiple Layers Switch	NRDUCelAlgoSwitch.SuMimoMultipleLayerSw	DL_SU_MULTI_LAYER_PWR_CTRL_SW	Use the default value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Low-frequency NR TDD:

```
//Configuring the downlink initial rank
MOD NRDUCELLPDSCH: NrDuCellId=0, DlnitRank=RANK1;
//Configuring the downlink initial rank after a handover
MOD NRDUCELLPDSCH: NrDuCellId=0, DlnitRankAfterHo=RANK1;
//Turning on the switch for downlink rank adaptation with SRS-based weighting
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptEnhancementSw=DL_RANK_ADAPT_SW-1;
//Configuring the spectral efficiency threshold for downlink rank increase from rank 2
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRank2IncSptEffThld=110;
//Configuring the spectral efficiency threshold for downlink rank increase from rank 3
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRank3IncSptEffThld=110;
//Configuring the spectral efficiency threshold for downlink rank decrease from rank 3
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRank3DecSptEffThld=110;
//Configuring the spectral efficiency threshold for downlink rank decrease from rank 4
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRank4DecSptEffThld=110;
//Configuring the inter-layer difference threshold for limiting the highest downlink rank
MOD NRDUCELLDLRANK: NrDuCellId=0, DlMaxRankLimitLayerGapThld=1000;
//Configuring the SRS PreSINR threshold for triggering downlink rank adaptation
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankAdaptSrsPresinrThld=-2;
//Turning on the switch for downlink single-UE multi-layer power control adaptation
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_MULTI_LAYER_PC_ADAPT_SW-1;
//Turning on the switch for downlink single-UE multi-layer power control
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_PWR_CTRL_SW-1;
//Configuring the high-rank-oriented MCS threshold for multi-layer power control
MOD NRDUCELLDLRANK: NrDuCellId=0, MultiLayerPcHighRankMcsThld=16;
//Configuring the spectral efficiency threshold for downlink rank increase from rank 2 or 3 for UEs on which
downlink single-UE multi-layer power control has not taken effect
MOD NRDUCELLDLRANK: NrDuCellId=0, DlNonPcRankIncSptEffThld=110;
//Configuring the spectral efficiency threshold for downlink rank decrease from rank 3 or 4 for UEs on
which downlink single-UE multi-layer power control has not taken effect
MOD NRDUCELLDLRANK: NrDuCellId=0, DlNonPcRankDecSptEffThld=110;
//Turning on the switch for downlink rank optimization for cell edge UEs performing small-packet services
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SMALL_PKT_CEU_RANK_ADAPT_SW-1;
```

High-frequency NR TDD:

```
//Configuring the downlink initial rank
MOD NRDUCELLPDSCH: NrDuCellId=0, DlnitRank=RANK1;
//Configuring the downlink initial rank after a handover
MOD NRDUCELLPDSCH: NrDuCellId=0, DlnitRankAfterHo=RANK1;
```

Deactivation Command Examples

Low-frequency NR TDD:

```
//Turning off the switch for downlink rank adaptation with SRS-based weighting
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptEnhancementSw=DL_RANK_ADAPT_SW-0;
//Turning off the switch for downlink rank optimization for cell edge UEs performing small-packet services
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=DL_SMALL_PKT_CEU_RANK_ADAPT_SW-0;
```

High-frequency NR TDD: None

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

None

6.4.3 Network Monitoring

The indicators listed in [6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

In NR TDD scenarios, the **N.PDSCH.InitTbDL.Rank1** to **N.PDSCH.InitTbDL.Rank4** counters can be used to observe the changes in the number of times a rank is used in downlink scheduling.

7 MCS Selection

The MCS selection procedure by the gNodeB varies before and after the reception of CSI measurement results from UEs:

- Before the reception of CSI measurement results from UEs: MCS selection is required when a UE newly accesses a cell or is newly handed over to a cell.
 - When a UE newly accesses a cell, the MCS used for downlink transmission is specified by the **NRDUCellPdsch.DlInitMcs** parameter.
 - When a UE is newly handed over to a cell, the MCS used for downlink transmission is specified by the **NRDUCellPdsch.DlInitMcsAfterHo** parameter.
- After the reception of CSI measurement results from UEs: Basic functions and relevant enhancements for MCS selection are involved.
 - Basic functions for MCS selection: include initial MCS selection and MCS adjustment. For details, see [7.1 Basic Functions for MCS Selection](#).
 - Enhanced functions for MCS selection: are described in [Table 7-1](#).

Table 7-1 Enhanced functions for MCS selection

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Downlink maximum modulation order optimization	√	-	7.2 Downlink Maximum Modulation Order Optimization
Downlink post-handover fast decision	√	√	7.3 Downlink Post-Handover Fast Decision

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
MCS index selection for small-packet services	√	√	7.4 MCS Index Selection for Small-Packet Services
Enhanced downlink target IBLER adaptation	-	√	7.5 Enhanced Downlink Target IBLER Adaptation
Target IBLER adaptation based on interference from neighboring cells	√	-	7.6 Target IBLER Adaptation Based on Interference from Neighboring Cells
Post-weight-switching initial MCS adjustment optimization	√	-	7.7 Post-Weight-Switching Initial MCS Adjustment Optimization
Common Control Info Minimum MCS	√	-	7.8 Minimum MCS Index for Common Control Information
Target code rate optimization	√	√	7.9 Target Code Rate Optimization
Downlink MCS index selection optimization for UEs using simplified SRS-based weights	√	-	7.10 Downlink MCS Index Selection Optimization for UEs Using Simplified SRS-based Weights

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Downlink RMSI payload reduction	-	√	7.11 Downlink RMSI Payload Reduction
Downlink Msg4 payload reduction	√	√	7.12 Downlink Msg4 Payload Reduction
Downlink initial MCS adjustment	√	√	7.13 Downlink Initial MCS Adjustment

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

7.1 Basic Functions for MCS Selection

7.1.1 Principles

Basic functions for MCS selection: include initial MCS selection and MCS adjustment.

- Initial MCS selection: After a UE reports CSI measurement results in the serving cell, the base station automatically selects an initial MCS for user-plane data transmission based on the UE-reported channel quality indicator (CQI).
- MCS adjustment: If the UE-reported channel measurement information is inaccurate or the air-interface channel conditions change rapidly, the initial MCS may not reflect the actual channel quality at the time of scheduling and the initial block error rate (IBLER) will not close in on the optimum value. Therefore, after initial MCS selection, the downlink scheduler periodically checks HARQ feedback from the UE. Based on HARQ feedback, the downlink scheduler determines how much the selected MCS deviates from the actual channel quality and adapts the MCS to the channel quality. This allows the IBLER to close in on the optimal value, thereby improving downlink throughput of the system.

MCS adjustment involves the following aspects:

- Determining the MCS adjustment direction: The initial MCS adjustment value is specified by the **NRDUCellPdsch.DlInitialMcsAdjValue** parameter. During MCS adjustment, the scheduler calculates the MCS

adjustment value based on the downlink target IBLER (specified by the **NRDUCellPdsch.DlTargetIbler** parameter).

- When the IBLER is higher than the downlink target IBLER, the MCS adjustment value is adjusted downwards.
- When the IBLER is lower than the downlink target IBLER, the MCS adjustment value is adjusted upwards.
- Determining the MCS adjustment step: The MCS adjustment step is specified by the **NRDUCellPdsch.FixedAmcStepValue** parameter.

7.1.2 Network Analysis

7.1.2.1 Benefits

With the basic functions for MCS selection, the gNodeB adaptively selects an MCS for scheduling based on the UE-reported CQI. The initial MCS may not allow the downlink IBLER to close in on the target value. Therefore, the scheduler needs to adjust the MCS based on HARQ feedback to enable the downlink IBLER to close in on the target value, improving the downlink throughput of the system.

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

Basic functions for MCS selection have no impact on the network.

7.1.3 Requirements

7.1.3.1 Licenses

There are no license requirements for basic functions.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

None

7.1.3.2.2 Mutually Exclusive Functions

None

7.1.3.2.3 Function Impacts

None

7.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.1.3.4 Others

None

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

This function is enabled by default and does not need activation.

Table 7-2 describes the parameters used for function optimization. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Initial MCS	NRDUCellPdsch. <i>DlInitMcs</i>	None	It is recommended that Huawei engineers configure this parameter based on network requirements.
Downlink Initial MCS After Handover	NRDUCellPdsch. <i>DlInitMcsAfterHo</i>	None	It is recommended that Huawei engineers configure this parameter based on network requirements.
Downlink Initial MCS Adjust Value	NRDUCellPdsch. <i>DlInitialMcsAdj-Value</i>	None	Use the recommended value.
Downlink Target IBLER	NRDUCellPdsch. <i>DlTargetIbler</i>	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Fixed AMC Step Value	NRDUCeLLPdsch.F ixedAmcStepValu e	None	Use the default value.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to [7.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Configuring the downlink initial MCS
MOD NRDUCELLPDSCH: NrDuCellId=0, DInitMcs=4;
//Configuring the downlink initial MCS after a handover
MOD NRDUCELLPDSCH: NrDuCellId=0, DInitMcsAfterHo=4;
//Configuring the downlink initial MCS adjustment value
MOD NRDUCELLPDSCH: NrDuCellId=0, DInitialMcsAdjValue=-4;
//Configuring the downlink target IBLER
MOD NRDUCELLPDSCH: NrDuCellId=0, DTargetIbler=10;
//Configuring the initial step for downlink AMC
MOD NRDUCELLPDSCH: NrDuCellId=0, FixedAmcStepValue=20;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

7.1.4.3 Network Monitoring

The indicators listed in [7.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

In addition, the **N.ChMeas.PDSCH.MCS.0** to **N.ChMeas.PDSCH.MCS.31** counters can be used to observe the changes in the number of times an MCS index is selected for scheduling in the downlink, and the **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.31** counters can be used to observe such changes in the uplink.

7.2 Downlink Maximum Modulation Order Optimization

7.2.1 Principles

The downlink maximum modulation order optimization function has been introduced to increase downlink spectral efficiency of UEs. This function is controlled by the **DL_MAX_MODULATION_ORDER_OPT_SW** option of the **NRDUCellDlAmc.DlAmcAlgoSw** parameter. If this option is selected:

- When the modulation order supported by a UE is higher than that indicated by the **supportedModulationOrderDL** field, the downlink maximum modulation order selected for the UE can be higher than that indicated by this field, thereby improving downlink spectral efficiency for the UE.
- When the modulation order supported by a UE is not higher than that indicated by the **supportedModulationOrderDL** field, the downlink maximum modulation order selected for the UE cannot be higher than that indicated by this field.

NOTE

For details about the **supportedModulationOrderDL** field, see the descriptions of the *FeatureSetDownlinkPerCC* IE in section 6.3.3 "UE capability information elements" of 3GPP TS 38.331 V15.4.0.

7.2.2 Network Analysis

7.2.2.1 Benefits

After this function is enabled in low frequency bands, downlink spectral efficiency increases when the modulation order supported by a UE is higher than that indicated by **supportedModulationOrderDL**. In peak rate scenarios, this function also increases the downlink peak rate.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

After downlink maximum modulation order optimization is enabled in low-frequency scenarios, if the modulation order supported by a UE is not higher than

that indicated by the **supportedModulationOrderDL** field, the downlink maximum modulation order of the UE cannot be higher than that indicated by this field. In peak rate scenarios, the downlink peak rate of such UEs may decrease.

7.2.3 Requirements

7.2.3.1 Licenses

There are no license requirements for basic functions.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

None

7.2.3.2.2 Mutually Exclusive Functions

None

7.2.3.2.3 Function Impacts

None

7.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.2.3.4 Others

None

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

[Table 7-3](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellDLAmc. <i>DLAmcAlgoSw</i>	DL_MAX_MODULATION_ORDER_OPT_SW	You are advised to select this option.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for downlink maximum modulation order optimization (recommended when the modulation order supported by a UE is higher than that indicated by the supportedModulationOrderDL field)  
MOD NRDUCELLDLAmc: NrDuCellId=0, DLAmcAlgoSw=DL_MAX_MODULATION_ORDER_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for downlink maximum modulation order optimization  
MOD NRDUCELLDLAmc: NrDuCellId=0, DLAmcAlgoSw=DL_MAX_MODULATION_ORDER_OPT_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

In single-UE peak rate scenarios, if the value of **N.ThpVol.DL.Cell** increases after this function is enabled, this function has taken effect.

7.2.4.3 Network Monitoring

The indicators listed in [7.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.3 Downlink Post-Handover Fast Decision

7.3.1 Principles

Consider a cell whose numbers of transmit and receive antennas are the same as the number of CSI-RS ports, for example, an 8T8R cell with eight CSI-RS ports. The number of CSI-RS ports is specified by the **NRDUCellCsirs.FR1MaxCellCsirsPortNum** parameter in NR TDD in low frequency bands or the **NRDUCellCsirs.FR2MaxCellCsirsPortNum** parameter in NR TDD in high frequency bands. PDSCH weight and rank adaptation is slow within a period of time following the first CQI reporting by a UE after the UE is handed over to a cell or beam. As a result, the selected MCS in downlink scheduling does not match the channel quality. Therefore, the downlink post-handover fast decision function is introduced to quickly determine the PDSCH weights and ranks after a UE is handed over to a cell or beam. This function can be enabled by selecting the **DL_QUICK_JUDGE_AFTER_HO_SW** option of the **NRDUCellPdsch.DLinkAdaptEnhancementSw** parameter.

7.3.2 Network Analysis

7.3.2.1 Benefits

In low-frequency NR TDD scenarios, this function allows for fast selection of PDSCH weights and ranks after a UE is handed over to a cell. This reduces the post-handover performance loss that occurs due to inaccurate selection of PDSCH weights and ranks and improves the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) within a certain period of time after the handover to the cell.

In the current version, after this function is enabled in high-frequency NR TDD scenarios, the performance improvement of fast PDSCH weight decision is not significant and this function needs to be optimized. Therefore, this function is not recommended in high-frequency NR TDD scenarios

7.3.2.2 Impacts

The impacts between this function and other functions are described in [7.3.3.2.3 Function Impacts](#).

In NR TDD, this function may cause fluctuations in the **Downlink Resource Block Utilizing Rate (DU)**, average downlink MCS index, average uplink MCS index, and average rank for downlink scheduling in a cell.

7.3.3 Requirements

7.3.3.1 Licenses

There are no license requirements for basic functions.

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

None

7.3.3.2.2 Mutually Exclusive Functions

None

7.3.3.2.3 Function Impacts

None

7.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.3.4 Others

None

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

Table 7-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Link Adaptation Enhancement Switch	NRDUCellPdsch. DLLinkAdaptEnhancementSw	DL_QUICK_JUDGE_AFTER_HO_SW	Low-frequency NR TDD: Select this option. High-frequency NR TDD: You are advised to deselect this option.

7.3.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink post-handover fast decision  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptEnhancementSw=DL_QUICK_JUDGE_AFTER_HO_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink post-handover fast decision  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptEnhancementSw=DL_QUICK_JUDGE_AFTER_HO_SW-0;
```

7.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.3.4.3 Network Monitoring

The indicators listed in [7.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.4 MCS Index Selection for Small-Packet Services

7.4.1 Principles

If there is a lot of padding in the PDSCH resources that the base station allocates for a UE, the base station automatically selects a low MCS index for data transmission. This reduces the padding proportion, decreases the BLER, and improves transmission reliability. When the **SMALL_PKT_MCS_SELECT_SW** option of the **NRDUCellPdsch.DLPdschAlgoSwitch** parameter is selected and there is a lot of padding in the PDSCH resources that the base station allocates for a UE, the base station does not decrease the MCS index. In this case, the proportion of high MCS indexes increases, the downlink IBLER increases, and the downlink user-perceived rate decreases.

7.4.2 Network Analysis

7.4.2.1 Benefits

After this function takes effect and there is a lot of padding in the PDSCH resources that the base station allocates for a UE, the base station does not decrease the MCS index. In this case, the proportion of high MCS indexes and the downlink IBLER increase, and the average downlink cell throughput **Cell Downlink Average Throughput (DU)** and average downlink UE throughput **User Downlink Average Throughput (DU)** decrease.

7.4.2.2 Impacts

The impacts between this function and other functions are described in [7.4.3.2.3 Function Impacts](#).

The network impacts produced by MCS index selection for small-packet services are described in benefit analysis. For details, see [7.4.2.1 Benefits](#).

7.4.3 Requirements

7.4.3.1 Licenses

There are no license requirements for basic functions.

7.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.4.3.2.1 Prerequisite Functions

None

7.4.3.2.2 Mutually Exclusive Functions

None

7.4.3.2.3 Function Impacts

None

7.4.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.4.3.4 Others

None

7.4.4 Operation and Maintenance

7.4.4.1 Data Configuration

7.4.4.1.1 Data Preparation

[Table 7-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DIPdschAlgoSwitch	SMALL_PKT_MCS_SELECT_SW	Use the default value.

7.4.4.1.2 Using MML Commands

Before using MML commands, refer to [7.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for MCS index selection for small-packet services  
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=SMALL_PKT_MCS_SELECT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for MCS index selection for small-packet services  
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=SMALL_PKT_MCS_SELECT_SW-0;
```

7.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.4.4.3 Network Monitoring

The indicators listed in [7.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.5 Enhanced Downlink Target IBLER Adaptation

7.5.1 Principles

Enhanced downlink target IBLER adaptation is introduced to improve UE performance. This function is controlled by the **DL_ENH_IBLER_TARGET_ADAPT_SW** option of the **NRDUCellDlAmc.DlMcsSelAlgoSw** parameter.

- With this option selected: The gNodeB adaptively selects a downlink target IBLER for a UE based on the MCS and channel fluctuations of the UE.
- With this option deselected: The downlink target IBLER for a UE equals the **NRDUCellPdsch.DlTargetIbler** parameter value.

Enhanced downlink target IBLER adaptation is supported in both high and low frequency bands in NR TDD. In high frequency bands, this function is used only in FWA scenarios. In low frequency bands, this function is mainly used for drive tests of large-packet-service UEs.

7.5.2 Network Analysis

7.5.2.1 Benefits

The benefits of this function in high-frequency NR TDD FWA scenarios are as follows:

- When enhanced downlink target IBLER adaptation takes effect for UEs in a cell, the downlink target IBLER decreases and the proportion of downlink retransmissions decreases in the cell. As a result, the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases.
- When all UEs for which this function takes effect run small-packet services in a cell, this function does not produce benefits because MCS index reduction for small-packet services will be triggered for such UEs.
- When all UEs for which this function takes effect in a cell use high-order MCS with the downlink target IBLER being less than 1%, this function does not produce any benefits.

7.5.2.2 Impacts

The impacts between this function and other functions are described in [7.5.3.2.3 Function Impacts](#).

After this function takes effect in high-frequency FWA scenarios, the average MCS index for downlink scheduling may decrease.

7.5.3 Requirements

7.5.3.1 Licenses

There are no license requirements for basic functions.

7.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.5.3.2.1 Prerequisite Functions

None

7.5.3.2.2 Mutually Exclusive Functions

None

7.5.3.2.3 Function Impacts

None

7.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

7.5.3.4 Others

None

7.5.4 Operation and Maintenance

7.5.4.1 Data Configuration

7.5.4.1.1 Data Preparation

[Table 7-6](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL MCS Selection Algorithm Switch	NRDUCeLLDLAmc. <i>DLmcsSelAlgoSw</i>	DL_ENH_IBLER_TARGET_ADAPT_SW	You are advised to select this option in high-frequency NR TDD FWA scenarios and deselect this option in other scenarios.

7.5.4.1.2 Using MML Commands

Before using MML commands, refer to [7.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling enhanced downlink target IBLER adaptation (only in high frequency bands)
MOD NRDUCeLLDLAMC: NrDuCellId=0, DLmcsSelAlgoSw=DL_ENH_IBLER_TARGET_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling enhanced downlink target IBLER adaptation (only in high frequency bands)
MOD NRDUCeLLDLAMC: NrDuCellId=0, DLmcsSelAlgoSw=DL_ENH_IBLER_TARGET_ADAPT_SW-0;
```

7.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.5.4.2 Activation Verification

If the downlink target IBLER decreases, the proportion of downlink retransmissions decreases in the cell, and the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.5.4.3 Network Monitoring

The indicators listed in [7.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.6 Target IBLER Adaptation Based on Interference from Neighboring Cells

7.6.1 Principles

To improve the performance of UEs in the overlapping area between cells, the target IBLER adaptation based on interference from neighboring cells function is introduced. This function is controlled by the **DL_INTRF_BASED_IBLER_ADAPT_SW** option of the **NRDUCellDlAmc.DlMcsSelAlgoSw** parameter:

- With this option selected: If the strength of interference from neighboring cells is detected to fluctuate for a UE in the overlapping area between cells, the target IBLER for the UE is changed from the value of **NRDUCellPdsch.DlTargetIbler** to that of **NRDUCellDlSch.DlIntrfBasedIblerTarget**.
- With this option deselected: The target IBLER for a UE equals the **NRDUCellPdsch.DlTargetIbler** parameter value.

NOTE

1. Target IBLER adaptation based on interference from neighboring cells can be enabled only when the **INTRA_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ.NeighborNrCellMgmtSw** parameter is selected so that the base station collects statistics about neighboring cells based on the UE-reported event A3 measurement results of intra-frequency neighboring cells.
2. Target IBLER adaptation based on interference from neighboring cells can be enabled only for common cells (with the **NRDUCell.NrDuCellNetworkingMode** parameter set to **NORMAL_CELL**).

7.6.2 Network Analysis

7.6.2.1 Benefits

In low-frequency scenarios, the benefits of this function are as follows:

- When there are UEs performing large-packet services in the overlapping area between cells and the RSRP difference between the serving cell and a neighboring cell is greater than -6 dB for such UEs, the PRB usage and scheduling duty cycle of the serving cell are both greater than 20%, the average rank for scheduling is greater than or equal to 2.5, and the MCS index is less than 17: After target IBLER adaptation based on interference from neighboring cells takes effect, the average downlink MCS index of the cell increases, improving the downlink cell spectral efficiency and the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**). The larger the proportion of UEs performing large-packet services, the greater the benefits provided by this function.

- When all UEs in the overlapping area are running small-packet services: This function does not produce benefits because MCS index reduction for small-packet services will be triggered for such UEs.
- When the interference from neighboring cells to the local cell fluctuates slightly (that is, the number of RBs scheduled in the neighboring cells changes slightly), the downlink cell spectral efficiency and the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) may decrease after this function takes effect.

7.6.2.2 Impacts

The impacts between this function and other functions are described in [7.6.3.2.3 Function Impacts](#).

After this function takes effect in low-frequency scenarios, the downlink IBLER of UEs increases and the downlink CCE usage decreases.

7.6.3 Requirements

7.6.3.1 Licenses

There are no license requirements for basic functions.

7.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR neighboring cell management based on intra-frequency neighboring cell statistics	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	Target IBLER adaptation based on interference from neighboring cells can be enabled only when the INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter is selected.

7.6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Target IBLER adaptation based on interference from neighboring cells cannot be enabled for an NR cell in Hyper Cell networking (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL).
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Target IBLER adaptation based on interference from neighboring cells cannot be enabled for an NR cell in Cell Combination networking (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE).

7.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RB allocation for downlink experience optimization	NRDUCellDLSchRes.DIExpOptRbIncreasePct	<i>Downlink Scheduling</i>	When both target IBLER adaptation based on interference from neighboring cells and RB allocation for downlink experience optimization take effect, target IBLER adaptation based on interference from neighboring cells may not produce gains.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellALgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both "target IBLER adaptation based on interference from neighboring cells" and "energy saving based on flexible frequency-domain scheduling" are in effect, the latter function may cause the former function to deliver no benefits.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellALgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both "target IBLER adaptation based on interference from neighboring cells" and "TTI-level channel shutdown" are in effect, the former function may deliver no benefits.

7.6.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.6.3.4 Others

None

7.6.4 Operation and Maintenance

7.6.4.1 Data Configuration

7.6.4.1.1 Data Preparation

Table 7-7 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL MCS Selection Algorithm Switch	NRDUCellDLAMC. <i>DLMcsSelAlgoSw</i>	DL_INTRF_BASED_IBLER_ADAPT_SW	You are advised to select this option.
Downlink Interference-based IBLER Target	NRDUCellDLSch. <i>DLIntrfBasedIblerTarget</i>	None	Use the default value.

7.6.4.1.2 Using MML Commands

Before using MML commands, refer to [7.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the target IBLER based on interference from neighboring cells
MOD NRDUCELLDLSCH: NrDuCellId=0, DLIntrfBasedIblerTarget=20;
//Enabling target IBLER adaptation based on interference from neighboring cells (with intra-frequency neighboring cell measurement configured in advance)
MOD NRDUCELLDLAMC: NrDuCellId=0, DLMcsSelAlgoSw=DL_INTRF_BASED_IBLER_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling target IBLER adaptation based on interference from neighboring cells
MOD NRDUCELLDLAMC: NrDuCellId=0, DLMcsSelAlgoSw=DL_INTRF_BASED_IBLER_ADAPT_SW-0;
```

7.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.6.4.2 Activation Verification

When there are UEs performing large-packet services in the overlapping area between cells and the RSRP difference between the serving cell and a neighboring cell is greater than -6 dB for such UEs, the PRB usage and scheduling duty cycle of the serving cell are both greater than 20%, the average rank for scheduling is greater than or equal to 2.5, and the MCS index is less than 17: If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.6.4.3 Network Monitoring

The indicators listed in [7.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.7 Post-Weight-Switching Initial MCS Adjustment Optimization

7.7.1 Principles

The post-weight-switching initial MCS adjustment optimization function is introduced in low-frequency TDD scenarios to improve the initial MCS after switching from PMI-based weights to SRS-based weights. This function is controlled by the **SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW** option of the **NRDUCellDLAmc.DlMcsSelAlgoSw** parameter.

- If this option is selected, the initial MCS after switching from PMI-based weights to SRS-based weights can be adjusted based on the initial MCS offset for SRS-based weights (specified by the **NRDUCellDLAmc.SrsWeightInitialMcsOffset** parameter).
- If this option is deselected, the initial MCS after switching from PMI-based weights to SRS-based weights is the same as the latest MCS for PMI-based weights before the switching.

7.7.2 Network Analysis

7.7.2.1 Benefits

After this function takes effect in low-frequency NR TDD scenarios, the initial MCS after switching from PMI-based weights to SRS-based weights increases and the average downlink rank increases, thereby increasing the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink user-perceived rate.

7.7.2.2 Impacts

The impacts between this function and other functions are described in [7.7.3.2.3 Function Impacts](#).

In low-frequency NR TDD, this function increases the initial MCS index after switching from PMI-based weights to SRS-based weights and the average downlink rank, thereby increasing the downlink IBLER.

7.7.3 Requirements

7.7.3.1 Licenses

There are no license requirements for basic functions.

7.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrsSrsAlgoSwitch parameter	<i>Channel Management</i>	Post-weight-switching initial MCS adjustment optimization takes effect only when UE-characteristic-based SRS period adaptation is enabled.

7.7.3.2.2 Mutually Exclusive Functions

None

7.7.3.2.3 Function Impacts

None

7.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.7.3.4 Others

None

7.7.4 Operation and Maintenance

7.7.4.1 Data Configuration

7.7.4.1.1 Data Preparation

Table 7-8 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL MCS Selection Algorithm Switch	NRDUCellDLAmc. DIMcsSelAlgoSw	SRS_WEIGHT_INIT _MCS_ADJ_OPT_S W	You are advised to select this option.
SRS-based Weight Initial MCS Offset	NRDUCellDLAmc. SrsWeightInitial McsOffset	None	Use the default value.

7.7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling post-weight-switching initial MCS adjustment optimization
MOD NRDUCELLDLAMC: NrDuCellId=0, DLMcsSelAlgoSw=SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW-1;
//Configuring the initial MCS offset for SRS-based weights
MOD NRDUCELLDLAMC: NrDuCellId=0, SrsWeightInitialMcsOffset=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling post-weight-switching initial MCS adjustment optimization
MOD NRDUCELLDLAMC: NrDuCellId=0, DLMcsSelAlgoSw=SRS_WEIGHT_INIT_MCS_ADJ_OPT_SW-0;
```

7.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.7.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.7.4.3 Network Monitoring

The indicators listed in [7.7.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.8 Minimum MCS Index for Common Control Information

7.8.1 Principles

The **NRDUCellPdsch.CommonCtrlInfoMinimumMcs** parameter specifies the minimum MCS index for transmitting common control information on the base station side. This function is supported only in low frequency bands, not in high frequency bands.

7.8.2 Network Analysis

7.8.2.1 Benefits

In low-frequency scenarios, a larger value of the minimum MCS index for common control information results in fewer PDSCH RBs used for sending common control information by the gNodeB, more PDSCH RBs that can be allocated by the gNodeB for UE data transmission, and higher average downlink UE throughput (**User Downlink Average Throughput (DU)**). A smaller value results in the opposite effects.

7.8.2.2 Impacts

The impacts between this function and other functions are described in [7.8.3.2.3 Function Impacts](#).

In low-frequency scenarios, inconsistent minimum MCS indexes for common control information across cells cause misalignment of RBs. This results in inter-cell interference and a decrease in downlink UE throughput.

7.8.3 Requirements

7.8.3.1 Licenses

There are no license requirements for basic functions.

7.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.8.3.2.1 Prerequisite Functions

None

7.8.3.2.2 Mutually Exclusive Functions

None

7.8.3.2.3 Function Impacts

None

7.8.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.8.3.4 Others

None

7.8.4 Operation and Maintenance

7.8.4.1 Data Configuration

7.8.4.1.1 Data Preparation

[Table 7-9](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Common Control Info Minimum MCS	NRDUCellPdsch.CommonCtrlInfoMinimumMcs	None	Use the default value.

7.8.4.1.2 Using MML Commands

Before using MML commands, refer to [7.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the minimum MCS index for common control information
MOD NRDUCELLPDSCH: NrDuCellId=0, CommonCtrlInfoMinimumMcs=0;
```

Deactivation Command Examples

Deactivation command examples are not involved.

7.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.8.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) changes after this function is enabled, this function has taken effect.

7.8.4.3 Network Monitoring

The indicators listed in [7.8.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.9 Target Code Rate Optimization

7.9.1 Principles

Through **NRDUCellDisch.PagingAndRarTbScalFactor**, the base station in scheduling can reduce the target code rate for paging messages and RAR messages to improve the coverage. The resultant code rate is 1/2 or 1/4 of the code rate corresponding to the MCS used for scheduling.

7.9.2 Network Analysis

7.9.2.1 Benefits

With target code rate reduction for paging messages and RAR messages, the number of UEs paged over the Uu interface (**N.Paging.UU.Ue.Num**) increases, thereby increasing the number of online UEs in a cell.

7.9.2.2 Impacts

The impacts between this function and other functions are described in [7.9.3.2.3 Function Impacts](#).

When the target code rate for paging messages and RAR messages is reduced, these messages occupy more RBs, decreasing RBs that can be allocated for the PDSCH. The average downlink cell throughput and average downlink user-perceived rate may decrease in low-frequency heavily loaded cells, but there is no impact on high-frequency cells. In addition, after the code rate reduction for paging messages, the random access success rate may decrease because more UEs initiate random access but the uplink coverage is limited.

7.9.3 Requirements

7.9.3.1 Licenses

There are no license requirements for basic functions.

7.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.9.3.2.1 Prerequisite Functions

None

7.9.3.2.2 Mutually Exclusive Functions

None

7.9.3.2.3 Function Impacts

None

7.9.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.9.3.4 Others

None

7.9.4 Operation and Maintenance

7.9.4.1 Data Configuration

7.9.4.1.1 Data Preparation

[Table 7-10](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Paging and RAR TB Scaling Factor	NRDUCellDISch.PagingAndRarTbscalFactor	None	Use the default value.

7.9.4.1.2 Using MML Commands

Before using MML commands, refer to [7.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the target code rate scaling factor (1/2 used as an example) for paging messages and RAR messages  
MOD NRDUCELLDLSCH: NrDuCellId=0, PagingAndRarTbScalFactor=SCALING_FACTOR_1_2;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling target code rate optimization  
MOD NRDUCELLDLSCH: NrDuCellId=0, PagingAndRarTbScalFactor=SCALING_DISABLE;
```

7.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.9.4.2 Activation Verification

If the number of paged UEs over the Uu interface (**N.Paging.UU.Ue.Num**) increases after this function is enabled, this function has taken effect.

7.9.4.3 Network Monitoring

The indicators listed in [7.9.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.10 Downlink MCS Index Selection Optimization for UEs Using Simplified SRS-based Weights

7.10.1 Principles

In low-frequency TDD scenarios, for UEs using simplified SRS-based weights, beamforming gains are not considered during MCS selection for data transmission. This compromises performance of these UEs. Downlink MCS index selection optimization for UEs using simplified SRS-based weights is introduced to increase the MCS indexes for such UEs. This function is controlled by the

SRS_WEIGHT_SIMPLE_BFGAIN_SW option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter.

- If this option is selected, beamforming gain compensation is considered in MCS index selection for UEs using simplified SRS-based weights. This increases the MCS indexes selected for these UEs.
- If this option is deselected, beamforming gain compensation is not considered in MCS index selection for UEs using simplified SRS-based weights.

Simplified SRS-based weights is used for small-packet UEs when UE-characteristic-based SRS period adaptation takes effect and there are a large number of UEs in a cell. Such small-packet UEs are also referred to as UEs using simplified SRS-based weights. For details about UE-characteristic-based SRS period adaptation, see *Channel Management*.

7.10.2 Network Analysis

7.10.2.1 Benefits

In low-frequency NR TDD scenarios, this function increases the MCS indexes used for downlink transmission for UEs using simplified SRS-based weights, and increases the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**).

7.10.2.2 Impacts

The impacts between this function and other functions are described in [7.10.3.2.3 Function Impacts](#).

In low-frequency NR TDD, this function increases the MCS indexes used for downlink transmission for UEs using simplified SRS-based weights. As a result, the downlink IBLER increases. Downlink IBLER = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB) \times 100\%$

7.10.3 Requirements

7.10.3.1 Licenses

There are no license requirements for basic functions.

7.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.10.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	Downlink MCS index selection optimization for UEs using simplified SRS-based weights can take effect only when UE-characteristic-based SRS period adaptation takes effect and there are a large number of UEs in a cell.

7.10.3.2.2 Mutually Exclusive Functions

None

7.10.3.2.3 Function Impacts

None

7.10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.10.3.4 Others

None

7.10.4 Operation and Maintenance

7.10.4.1 Data Configuration

7.10.4.1.1 Data Preparation

Table 7-11 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-11 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL MCS Selection Algorithm Switch	NRDUCellDLAmc. DLmcsSelAlgoSw	SRS_WEIGHT_SIM PLE_BFGAIN_SW	In low-frequency NR TDD scenarios, you are advised to select this option.

7.10.4.1.2 Using MML Commands

Before using MML commands, refer to [7.10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink MCS index selection optimization for UEs using simplified SRS-based weights  
MOD NRDUCELLDLAMC: NrDuCellId=0, DLmcsSelAlgoSw=SRS_WEIGHT_SIMPLE_BFGAIN_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink MCS index selection optimization for UEs using simplified SRS-based weights  
MOD NRDUCELLDLAMC: NrDuCellId=0, DLmcsSelAlgoSw=SRS_WEIGHT_SIMPLE_BFGAIN_SW-0;
```

7.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.10.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.10.4.3 Network Monitoring

The indicators listed in [7.10.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.11 Downlink RMSI Payload Reduction

7.11.1 Principles

During initial access, a UE obtains the scheduling information of SIB1 through control-resource set (CORESET) 0 to complete initial access. Although a lower MCS index used by the gNodeB to schedule common control messages leads to better UE demodulation performance, common control messages must be scheduled within the frequency-domain resources corresponding to CORESET 0.

To improve the demodulation performance of common control messages on the UE side, the downlink remaining minimum system information (RMSI) payload reduction function is introduced. This function is controlled by the **RMSI_PAYLOAD_REDUCE_SW** option of the **NRDUCellDLSch.DLSchAlgoSwitch** parameter. When this function takes effect, the number of bits occupied by RMSI in downlink common control messages and the MCS index for RMSI scheduling decrease, which improves RMSI demodulation performance on the UE side. This function is supported only in high-frequency NR TDD cells.

7.11.2 Network Analysis

7.11.2.1 Benefits

In SA networking or NSA and SA hybrid networking, after this function is enabled, the number of RMSI-occupied bits and the MCS index for RMSI scheduling decrease, and the number of random access requests received by the gNodeB increases. In other scenarios, enabling this function has no impact.

7.11.2.2 Impacts

The impacts between this function and other functions are described in [7.11.3.2.3 Function Impacts](#).

In SA networking or NSA and SA hybrid networking, after this function is enabled, Msg4 is scheduled based on the default PDSCH time-domain configuration table. This may reduce the time-domain resources for Msg4 transmission. In this case, the MCS index for Msg4 scheduling may increase to ensure normal scheduling of such messages. Therefore, it is good practice to enable downlink RMSI payload reduction and downlink Msg4 payload reduction together. In other scenarios, enabling this function has no impact.

7.11.3 Requirements

7.11.3.1 Licenses

There are no license requirements for basic functions.

7.11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.11.3.2.1 Prerequisite Functions

None

7.11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	Downlink RMSI payload reduction cannot be enabled in the wide-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104(Wide Beam Scenario)).

7.11.3.2.3 Function Impacts

None

7.11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

7.11.3.4 Others

None

7.11.4 Operation and Maintenance

7.11.4.1 Data Configuration

7.11.4.1.1 Data Preparation

[Table 7-12](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCellDISch. DISchAlgoSwitch	RMSI_PAYLOAD_REDUCE_SW	In SA networking or NSA and SA hybrid networking, you are advised to select this option together with the MSG4_PAYLOAD_REDUCE_SW option of the NRDUCellDISch. DISchAlgoSwitch parameter.

7.11.4.1.2 Using MML Commands

Before using MML commands, refer to [7.11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink RMSI payload reduction  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=RMSI_PAYLOAD_REDUCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink RMSI payload reduction  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=RMSI_PAYLOAD_REDUCE_SW-0;
```

7.11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.11.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.11.4.3 Network Monitoring

The indicators listed in [7.11.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.12 Downlink Msg4 Payload Reduction

7.12.1 Principles

During initial access, a UE obtains the scheduling information of SIB1 through control-resource set (CORESET) 0 to complete initial access. Although a lower MCS index used by the gNodeB to schedule common control messages leads to better UE demodulation performance, common control messages must be scheduled within the frequency-domain resources corresponding to CORESET 0.

To improve the demodulation performance of common control messages on the UE side, the downlink Msg4 payload reduction function is introduced. This function is controlled by the **MSG4_PAYLOAD_REDUCE_SW** option of the **NRDUCELLDLSCH.DISchAlgoSwitch** parameter. When this function takes effect, the number of bits occupied by Msg4 in downlink common control messages and the MCS index for Msg4 scheduling decrease, which improves the performance of UEs to demodulate Msg4.

7.12.2 Network Analysis

7.12.2.1 Benefits

In SA networking or NSA and SA hybrid networking, after this function is enabled, the number of Msg4-occupied bits and the MCS index for Msg4 scheduling decrease, and the **RRC Setup Success Rate (CU)** increases. In other scenarios, enabling this function has no impact.

7.12.2.2 Impacts

The impacts between this function and other functions are described in [7.12.3.2.3 Function Impacts](#).

In SA networking or NSA and SA hybrid networking, after this function is enabled, the search space period for paging messages is shortened, and UE power consumption may increase. In other scenarios, enabling this function has no impact.

7.12.3 Requirements

7.12.3.1 Licenses

There are no license requirements for basic functions.

7.12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.12.3.2.1 Prerequisite Functions

None

7.12.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH resource reduction for heavy-traffic UEs in single-beam scenarios	NRDUCellPucch.PucchResrcRednMthd	<i>Channel Management</i>	In the single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwav-Beam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)), if NRDUCellPucch.PucchResrcRednMthd is set to MASS_FLOW , downlink Msg4 payload reduction cannot be enabled.
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	

7.12.3.2.3 Function Impacts

None

7.12.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.12.3.4 Others

None

7.12.4 Operation and Maintenance

7.12.4.1 Data Configuration

7.12.4.1.1 Data Preparation

[Table 7-13](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-13 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCeLLDLSch. <i>DISchAlgoSwitch</i>	MSG4_PAYLOAD_REDUCE_SW	In SA networking or NSA and SA hybrid networking, you are advised to select this option.

7.12.4.1.2 Using MML Commands

Before using MML commands, refer to [7.12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink Msg4 payload reduction  
MOD NRDUCeLLDLSCH: NrDuCellId=0, DISchAlgoSwitch=MSG4_PAYLOAD_REDUCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink Msg4 payload reduction  
MOD NRDUCeLLDLSCH: NrDuCellId=0, DISchAlgoSwitch=MSG4_PAYLOAD_REDUCE_SW-0;
```

7.12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.12.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

7.12.4.3 Network Monitoring

The indicators listed in [7.12.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

7.13 Downlink Initial MCS Adjustment

7.13.1 Principles

The gNodeB reselects an MCS for a UE after the UE accesses or is handed over to a cell. However, the selected MCS may be inaccurate. To address this issue, downlink initial MCS adjustment is introduced. With this function, the gNodeB quickly adjusts the MCS based on HARQ feedback before the first CQI reporting. This function is controlled by the **DL_INIT_MCS_ADJ_SW** option of the **NRDUCellPdsch.DLinkAdaptAlgoSwitch** parameter. If this option is selected, the gNodeB adjusts the MCS before the first CQI reporting, thereby increasing the downlink UE rate. If this option is deselected, the gNodeB does not adjust the MCS before the first CQI reporting, which may result in an inaccurate initial MCS selection. If this occurs, the downlink UE rate decreases.

7.13.2 Network Analysis

7.13.2.1 Benefits

With this function, the gNodeB adjusts the MCS before the first CQI reporting, improving the MCS selection accuracy. This in return increases the downlink user-perceived rate and **User Downlink Average Throughput (DU)**.

7.13.2.2 Impacts

The impacts between this function and other functions are described in [7.13.3.2.3 Function Impacts](#).

Fast MCS adjustment may result in a high MCS index. A high MCS index causes consecutive bit errors, which may decrease the downlink UE rate and even cause service drops in extreme scenarios.

This function may cause fluctuations in the values of indicators, such as the Average DownLink MCS, downlink IBLER, and average downlink rank.

7.13.3 Requirements

7.13.3.1 Licenses

There are no license requirements for basic functions.

7.13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.13.3.2.1 Prerequisite Functions

None

7.13.3.2.2 Mutually Exclusive Functions

None

7.13.3.2.3 Function Impacts

None

7.13.3.3 Hardware

For details, see [4.3.3 Hardware](#).

7.13.3.4 Others

None

7.13.4 Operation and Maintenance

7.13.4.1 Data Configuration

7.13.4.1.1 Data Preparation

Table 7-14 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-14 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Link Adaptation Algorithm Switch	NRDUCellPdsch.DLLinkAdaptAlgoSwitch	DL_INIT_MCS_ADJ_SW	You are advised to select this option.

7.13.4.1.2 Using MML Commands

Before using MML commands, refer to [7.13.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink initial MCS adjustment  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptAlgoSwitch=DL_INIT_MCS_ADJ_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink initial MCS adjustment  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptAlgoSwitch=DL_INIT_MCS_ADJ_SW-0;
```

7.13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.13.4.2 Activation Verification

If the **Cell Downlink Average Throughput (DU)** increases after this function is enabled, this function has taken effect.

7.13.4.3 Network Monitoring

The indicators listed in [7.13.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8 PDSCH Time-Frequency Resource Allocation

The PDSCH time-frequency resources include both time- and frequency-domain resources, and allocating them means setting appropriate resource sizes and positions to UEs.

- PDSCH frequency-domain resource allocation

Section 5.1.2.2 "Resource allocation in frequency domain" in 3GPP TS 38.214 V15.4.0 defines two resource allocation schemes: type 0 and type 1. In type 0, contiguous or non-contiguous resource block groups (RBGs) can be allocated. In type 1, only contiguous RBs can be allocated.

 - In the current version, resource allocation in type 0 is the default for both UE-level control-plane information and user-plane information. [Figure 8-1](#) and [Figure 8-2](#) show non-contiguous and contiguous resource allocation in type 0, respectively. This example assumes that one RBG contains four RBs and the scheduler allocates one RBG to UE 1, two RBGs to UE 2, and one RBG to UE 3.

Figure 8-1 Non-contiguous resource allocation in type 0

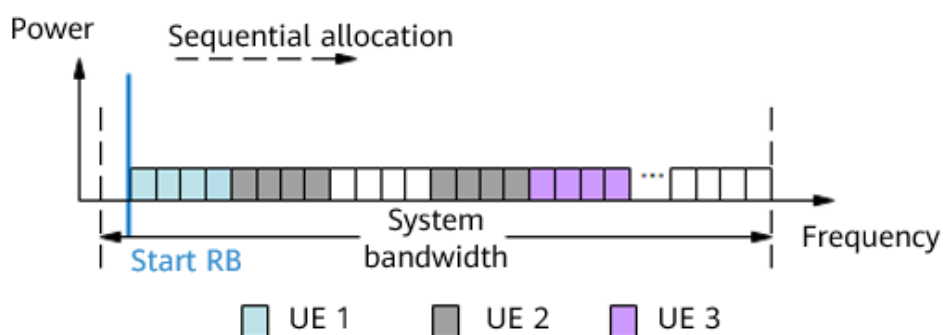
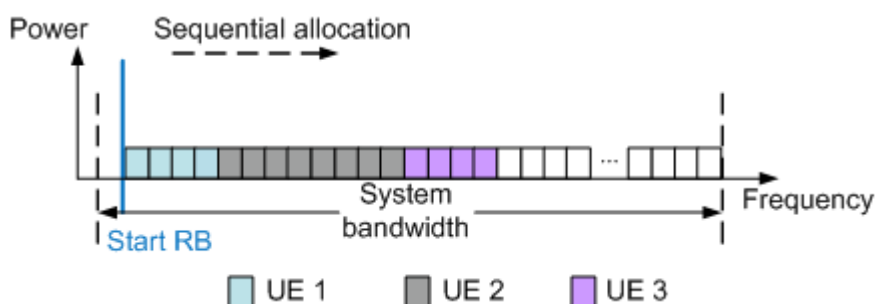
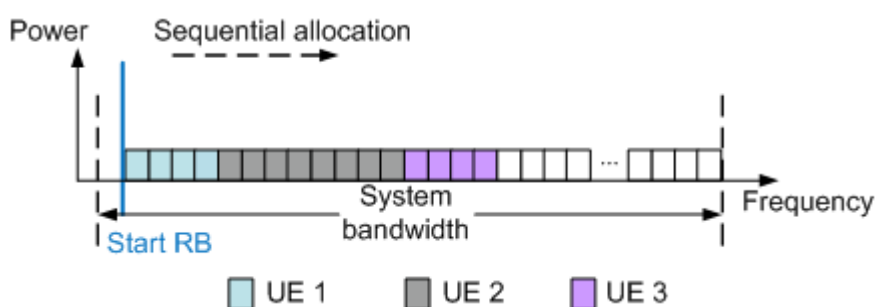


Figure 8-2 Contiguous resource allocation in type 0**NOTE**

The number of RBs contained in an RBG varies with the bandwidth. For details, see Table 5.3.2-1 Transmission bandwidth configuration NRB for FR1 in 3GPP TS 38.104 V15.5.0 and Table 5.1.2.2.1-1 Nominal RBG size P in 3GPP TS 38.214 V15.4.0. 3GPP specifications define two RBG size configurations: configuration 1 and configuration 2. Huawei uses configuration 1.

- In the current version, contiguous resource allocation in type 1 is applied to common control information. As shown in [Figure 8-3](#), the scheduler allocates four RBs to UE 1, eight RBs to UE 2, and four RBs to UE 3.

Figure 8-3 Contiguous resource allocation in type 1

- PDSCH time-domain resource allocation

The scheduler needs to indicate the scheduled slot and both the start and end positions of the scheduled OFDM symbols in the slot to a UE. PDSCH time-domain resource allocation is classified into type A and type B, depending on the combination of the start symbol position and scheduling length. For details, see section 5.1.2.1 "Resource allocation in time domain" in 3GPP TS 38.214 V15.4.0.

During downlink data transmission, the scheduler selects a more appropriate resource allocation mode based on scheduling information. [Table 8-1](#) lists optimization functions.

Table 8-1 Optimization functions for PDSCH time-frequency resource allocation

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Downlink type 1 resource allocation for UEs performing small-packet services	√	-	8.1 Downlink Type 1 Resource Allocation for UEs Performing Small-packet Services
Downlink supplementary type 1 scheduling for UEs performing small-packet services	√	-	8.2 Downlink Supplementary Type 1 Scheduling for UEs Performing Small-Packet Services
Downlink TBS calculation optimization and downlink RB estimation optimization	√	√	8.3 Downlink TBS Calculation Optimization and Downlink RB Estimation Optimization
RB allocation for downlink experience optimization	√	-	8.4 RB Allocation for Downlink Experience Optimization
Downlink tail packet scheduling optimization	-	√	8.5 Downlink Tail Packet Scheduling Optimization
Sparsified RMSI scheduling policy	-	√	8.7 Sparsified RMSI Scheduling Policy

Function Name	Low-Frequency NR TDD	High-Frequency NR TDD	Reference
Downlink interference randomization-based scheduling	√	-	8.8 Downlink Interference Randomization-based Scheduling
PDSCH resource mapping	√	√	8.9 PDSCH Resource Mapping
PDSCH PTRS allocation	-	√	8.10 PDSCH PTRS Allocation
DCCH type 1 scheduling	√	-	8.11 DCCH Type 1 Scheduling
Optimized scheduling of tail RBGs	√	-	8.12 Optimized Scheduling of Tail RBGs

NOTE

In the preceding table, "√" indicates that the function is supported in the specified RAT, while "-" indicates that the function is not supported.

8.1 Downlink Type 1 Resource Allocation for UEs Performing Small-packet Services

8.1.1 Principles

Downlink type 1 resource allocation for UEs performing small-packet services is introduced in low-frequency scenarios and is controlled by the **SMALL_PKT_TYPE1_SCH_SW** option of the **NRDUCellPdsch.DLPdschAlgoSwitch** parameter. After this function takes effect, the resource allocation granularity of four RBs can be used for UEs performing small-packet services when the UEs support PDSCH type 0 or 1 dynamic indication (the capability field is **dynamicSwitchRA-Type0-1-PDSCH**). This function increases the average downlink spectral efficiency and average downlink cell throughput without significantly affecting the demodulation performance of UEs performing small-packet services. It is recommended that this function be enabled in heavy-load scenarios. This function is supported only in low frequency bands, not in high frequency bands.

8.1.2 Network Analysis

8.1.2.1 Benefits

After this function takes effect in low-frequency heavy-load scenarios, the number of scheduled UEs in the downlink may increase, the average MCS index for downlink scheduling increases, and the average downlink cell throughput and average downlink user-perceived rate increase.

8.1.2.2 Impacts

The impacts between this function and other functions are described in [8.2.3.2.3 Function Impacts](#).

In the current version, it is good practice to disable downlink type 1 resource allocation for UEs performing small-packet services. The following impacts exist after this function takes effect in low-frequency scenarios: The downlink IBLER increases, the downlink PRB usage decreases, and the downlink equivalent CCE usage may increase. If this function is enabled for heavily loaded cells, the probability of uplink CCE allocation failures may increase. As a result, the average uplink cell throughput and average uplink user-perceived rate decrease. If this function is enabled for lightly loaded cells, the average downlink cell throughput decreases and the average downlink user-perceived rate may decrease.

8.1.3 Requirements

8.1.3.1 Licenses

There are no license requirements for basic functions.

8.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.1.3.2.1 Prerequisite Functions

None

8.1.3.2.2 Mutually Exclusive Functions

None

8.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When "PDSCH MU-MIMO" is enabled, "downlink type 1 resource allocation for UEs performing small-packet services" does not take effect for MU-MIMO UEs.

8.1.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.1.3.4 Others

None

8.1.4 Operation and Maintenance

8.1.4.1 Data Configuration

8.1.4.1.1 Data Preparation

[Table 8-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DlPdschAlgoSwitch	SMALL_PKT_TYPE1_SCH_SW	When the downlink load is heavy (the downlink PRB usage is greater than 80%), the downlink equivalent CCE usage of the slot where the uplink DCI is located is less than 70%, and the downlink CCE allocation failure rate is less than 30%, you are advised to select this option. In other scenarios, you are advised to deselect this option.

8.1.4.1.2 Using MML Commands

Before using MML commands, refer to [8.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink type 1 resource allocation for UEs performing small-packet services  
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=SMALL_PKT_TYPE1_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink type 1 resource allocation for UEs performing small-packet services  
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=SMALL_PKT_TYPE1_SCH_SW-0;
```

8.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.1.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

8.1.4.3 Network Monitoring

The indicators listed in [8.1.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.2 Downlink Supplementary Type 1 Scheduling for UEs Performing Small-Packet Services

8.2.1 Principles

Downlink supplementary type 1 scheduling for UEs performing small-packet services is introduced in low frequency bands. This function is controlled by the **SMALL_PKT_TYPE1_SUPPL_SCH_SW** option of the **NRDUCellDISchRes.DISchResAlgoSw** parameter. Only when downlink type 1 resource allocation for UEs performing small-packet services is enabled, can this function take effect and utilize the resources that remain after downlink type 1 resource allocation. This increases the cell PRB usage in heavy-load scenarios. This function is supported only in low frequency bands, not in high frequency bands.

8.2.2 Network Analysis

8.2.2.1 Benefits

After downlink supplementary type 1 scheduling for UEs performing small-packet services takes effect in low-frequency heavy-load scenarios, the cell PRB usage, downlink CCE load, number of UEs scheduled in the downlink, and average downlink cell throughput increase.

8.2.2.2 Impacts

The impacts between this function and other functions are described in [8.2.3.2.3 Function Impacts](#).

After this function takes effect in low-frequency scenarios, the average downlink UE throughput may decrease.

8.2.3 Requirements

8.2.3.1 Licenses

There are no license requirements for basic functions.

8.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink type 1 resource allocation for UEs performing small-packet services	SMALL_PKT_TYPE1_SCH_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	The switch for downlink supplementary type 1 scheduling of UEs performing small-packet services can be turned on only when the switch for downlink type 1 resource allocation for such UEs is on.

8.2.3.2.2 Mutually Exclusive Functions

None

8.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	FWA RedCap downlink type 1 scheduling	FWA_TYPE1_4RB_SCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	After supplementary scheduling in the case of type-1 resource allocation for small-packet services takes effect, FWA RedCap downlink type 1 scheduling does not take effect for UEs that meet the conditions for type-1 resource allocation for small-packet services.

8.2.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.2.3.4 Others

None

8.2.4 Operation and Maintenance

8.2.4.1 Data Configuration

8.2.4.1.1 Data Preparation

Table 8-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellDLSchRes.DLSchResAlgoSw	SMALL_PKT_TYPE1_SUPPL_SCH_SW	You are advised to select this option when the switch for downlink type 1 resource allocation for UEs performing small-packet services is on and the downlink load is extremely heavy (the downlink PRB usage is greater than 95%). In other scenarios, you are advised to deselect this option.

8.2.4.1.2 Using MML Commands

Before using MML commands, refer to [8.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink supplementary type 1 scheduling for UEs performing small-packet services
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DLSchResAlgoSw=SMALL_PKT_TYPE1_SUPPL_SCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink supplementary type 1 scheduling for UEs performing small-packet services
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DLSchResAlgoSw=SMALL_PKT_TYPE1_SUPPL_SCH_SW-0;
```

8.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.2.4.2 Activation Verification

If the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

8.2.4.3 Network Monitoring

The indicators listed in [8.2.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.3 Downlink TBS Calculation Optimization and Downlink RB Estimation Optimization

8.3.1 Principles

When downlink TBS calculation optimization (controlled by the **TBS_CALC_OPT_SW** option of the **NRDUCellPdsch.DIPdschAlgoSwitch** parameter) and downlink RB estimation optimization (controlled by the **RB_ESTIMATE_OPT_SW** option of the **NRDUCellPdsch.DIPdschAlgoSwitch** parameter) are enabled, the scheduler precisely allocates RBs to UEs. To do so, the scheduler uses information about the data to be scheduled as well as RBs and REs available in the cell bandwidth. This speeds up downlink UE data transmission and improves the downlink user-perceived rate.

8.3.2 Network Analysis

8.3.2.1 Benefits

After the functions are enabled, RB allocation is more accurate and the downlink user-perceived rate increases.

8.3.2.2 Impacts

The impacts between this function and other functions are described in [8.3.3.2.3 Function Impacts](#).

After the functions are enabled, indicators related to downlink scheduling (such as the average downlink MCS index and average downlink rank) and downlink PRB usage change. In addition, the **User Downlink Average Throughput (DU)** may decrease because RB allocation becomes more accurate, fewer data packets for UEs are segmented, and more data is scheduled in the last slot during which the downlink buffer becomes empty.

8.3.3 Requirements

8.3.3.1 Licenses

There are no license requirements for basic functions.

8.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.3.2.1 Prerequisite Functions

None

8.3.3.2.2 Mutually Exclusive Functions

None

8.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Data convergence in symbols	NRDUCellPowerSaving.SymbolCompressLength	<i>Energy Conservation and Emission Reduction</i>	If "data convergence in symbols" is in effect but "downlink RB estimation optimization" does not take effect, the user-perceived rate may decrease. You are advised to enable "downlink RB estimation optimization" when "data convergence in symbols" is in effect.
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SWITCH option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If "energy saving based on flexible frequency-domain scheduling" is in effect but downlink RB estimation optimization does not take effect, the user-perceived rate may decrease. You are advised to enable downlink RB estimation optimization when "energy saving based on flexible frequency-domain scheduling" is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If TTI-level channel shutdown is in effect but downlink RB estimation optimization does not take effect, the user-perceived rate may decrease. You are advised to enable downlink RB estimation optimization when TTI-level channel shutdown is in effect.

8.3.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.3.3.4 Others

None

8.3.4 Operation and Maintenance

8.3.4.1 Data Configuration

8.3.4.1.1 Data Preparation

[Table 8-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DIPdschAlgoSwitch	TBS_CALC_OPT_SW	You are advised to select this option.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DIPdschAlgoSwitch	RB_ESTIMATE_OPT_SW	You are advised to select this option.

8.3.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for downlink TBS calculation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=TBS_CALC_OPT_SW-1;
//Turning on the switch for downlink RB estimation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for downlink TBS calculation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=TBS_CALC_OPT_SW-0;
//Turning off the switch for downlink RB estimation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=RB_ESTIMATE_OPT_SW-0;
```

8.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.3.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) changes after this function is enabled, this function has taken effect.

8.3.4.3 Network Monitoring

The indicators listed in [8.3.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.4 RB Allocation for Downlink Experience Optimization

8.4.1 Principles

To improve downlink user experience, the RB allocation for downlink experience optimization function is introduced. When the air interface resources in a slot are not limited, appropriate UEs are selected and the number of RBs allocated to selected UEs will be increased by the percentage specified by the **NRDUCellDLSchRes.DIExpOptRbIncreasePct** parameter. When the **NRDUCellDLSchRes.DIExpOptRbIncreasePct** parameter is set to **0**, this function

does not take effect. This function is supported only in low frequency bands, not in high frequency bands.

8.4.2 Network Analysis

8.4.2.1 Benefits

When the downlink PRB usage indicates light load in low-frequency scenarios, the downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) increase.

8.4.2.2 Impacts

The impacts between this function and other functions are described in [8.4.3.2.3 Function Impacts](#).

In low-frequency networks, this function has the following impacts:

- When the downlink PRB usage indicates heavy load, the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** decrease.
- A larger increase percentage indicated by **NRDUCellDISchRes.DIExpOptRbIncreasePct** results in a lower average MCS index for downlink scheduling, a lower downlink BLER, and a higher downlink PRB usage, but stronger inter-cell interference.

8.4.3 Requirements

8.4.3.1 Licenses

There are no license requirements for basic functions.

8.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.4.3.2.1 Prerequisite Functions

None

8.4.3.2.2 Mutually Exclusive Functions

None

8.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SWITCH.AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i>	After RB allocation for downlink experience optimization takes effect, the downlink PRB usage increases and the number of scenarios where downlink interference randomization-based scheduling can take effect decreases.
Low-frequency TDD	Target IBLER adaptation based on interference from neighboring cells	DL_INTRF_BASED_IBLER_ADAPT_SWITCH option of the NRDUCellDLAmc.DIMcsSelAlgoSw parameter	<i>Downlink Scheduling</i>	When both target IBLER adaptation based on interference from neighboring cells and RB allocation for downlink experience optimization take effect, target IBLER adaptation based on interference from neighboring cells may not produce gains.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellALgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both RB allocation for downlink experience optimization and intelligent carrier shutdown take effect, an exit from intelligent carrier shutdown may be performed earlier than expected.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellALgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both RB allocation for downlink experience optimization and RF channel intelligent shutdown take effect, an exit from RF channel intelligent shutdown may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both RB allocation for downlink experience optimization and energy saving based on flexible frequency-domain scheduling take effect, an exit from energy saving based on flexible frequency-domain scheduling may be performed earlier than expected.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCH_DULE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both RB allocation for downlink experience optimization and power saving scheduling take effect, an exit from power saving scheduling may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: - Intra-base-station multi-carrier: INTRA_GNB_MULTICARRIER_SD_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter - Inter-base-station multi-carrier: INTER_GNB_MULTICARRIER_SD_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both "RB allocation for downlink experience optimization" and "fast carrier shutdown" are in effect, an exit from fast carrier shutdown may occur earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
		<i>xtSwitch</i> parameter Basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both RB allocation for downlink experience optimization and RF module fast deep dormancy take effect, an exit from RF module fast deep dormancy may be performed earlier than expected.

8.4.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.4.3.4 Others

None

8.4.4 Operation and Maintenance

8.4.4.1 Data Configuration

8.4.4.1.1 Data Preparation

[Table 8-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
RB Increase Percentage for DL Exp Opt	NRDUCellDLSchRes.DIExpOptRbIncreasePct	None	You are advised to set this parameter to 20 when the downlink PRB usage indicates light load and to 0 when the downlink PRB usage indicates heavy load.

8.4.4.1.2 Using MML Commands

Before using MML commands, refer to [8.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling RB allocation for downlink experience optimization (value 20 used as an example)
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DIExpOptRbIncreasePct=20;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling RB allocation for downlink experience optimization
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DIExpOptRbIncreasePct=0;
```

8.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.4.2 Activation Verification

When the downlink PRB usage indicates light load, if the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) increases after this function is enabled, this function has taken effect.

8.4.4.3 Network Monitoring

The indicators listed in [8.4.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.5 Downlink Tail Packet Scheduling Optimization

8.5.1 Principles

MCS index reduction due to small-packet services or rate matching for CSI-RS/TRS slots may cause packet truncation in the event of excess MCS index reduction. For this reason, the downlink tail packet scheduling optimization function is introduced. This function is controlled by the **TAIL_PKT_SCH_OPT_SW** option of the **NRDUCellPdsch.DLPdschAlgoSwitch** parameter. When this option is selected, the scheduler appropriately adjusts the MCS for UEs enabled with MCS index reduction due to small-packet services or rate matching, preventing UE data packet truncation and improving the downlink user-perceived rate. This function is supported in high-frequency NR TDD scenarios.

8.5.2 Network Analysis

8.5.2.1 Benefits

After this function takes effect, truncation of UE data packets can be effectively reduced, thereby improving the average downlink user-perceived rate.

8.5.2.2 Impacts

The impacts between this function and other functions are described in [8.5.3.2.3 Function Impacts](#).

8.5.3 Requirements

8.5.3.1 Licenses

There are no license requirements for basic functions.

8.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.5.3.2.1 Prerequisite Functions

None

8.5.3.2.2 Mutually Exclusive Functions

None

8.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	None	None	None	None

8.5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.5.3.4 Others

None

8.5.4 Operation and Maintenance

8.5.4.1 Data Configuration

8.5.4.1.1 Data Preparation

[Table 8-6](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. DlPdschAlgoSwitch	TAIL_PKT_SCH_OP T_SW	You are advised to select this option.

8.5.4.1.2 Using MML Commands

Before using MML commands, refer to [8.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink tail packet scheduling optimization  
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=TAIL_PKT_SCH_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink tail packet scheduling optimization  
MOD NRDUCELLPDSCH: NrDuCellId=0, DiPdschAlgoSwitch=TAIL_PKT_SCH_OPT_SW-0;
```

8.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.5.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) changes after this function is enabled, this function has taken effect.

8.5.4.3 Network Monitoring

The indicators listed in [8.5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.6 Downlink Precise RB Estimation

8.6.1 Principles

Downlink precise RB estimation is introduced to meet users' requirements for high-speed downlink experience and achieve optimal resource utilization. This function is controlled by the **PRECISE_DL_RB_ESTIMATE_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter. With this function, the downlink RB

estimation time is delayed, which is closer to the actual scheduling time. This improves RB allocation accuracy, thereby increasing the downlink user-perceived rate.

8.6.2 Network Analysis

8.6.2.1 Benefits

This function enables the scheduler to allocate RBs more accurately, thereby increasing the **User Downlink Average Throughput (DU)**.

8.6.2.2 Impacts

The impacts between this function and other functions are described in [8.6.3.2.3 Function Impacts](#).

8.6.3 Requirements

8.6.3.1 Licenses

There are no license requirements for basic functions.

8.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.6.3.2.1 Prerequisite Functions

None

8.6.3.2.2 Mutually Exclusive Functions

None

8.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When enabled together with downlink precise RB estimation, NR inter-carrier dynamic power sharing may produce decreased benefits.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When enabled together with downlink precise RB estimation, NR multi-band power adaptive scheduling may produce reduced benefits.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When enabled together with downlink precise RB estimation, interference avoidance may produce increased benefits.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When enabled together with downlink precise RB estimation, CA Performance Boost Based on Multi-dimensional Optimization may produce increased benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When enabled together with downlink precise RB estimation, Co-MM may produce increased benefits.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CarrierEnhancedAllocationSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When enabled together with downlink precise RB estimation, Single DCI may produce increased benefits.
Low-frequency TDD	Enhanced Co-MM	INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When enabled together with downlink precise RB estimation, enhanced Co-MM may produce increased benefits.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAllocationSwitch.CompSwitch parameter	<i>CoMP</i>	When enabled together with downlink precise RB estimation, intra-base-station DL CoMP may produce increased benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetWorkingMode set to HYPER_CELL_COMBINE_MODE NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	When enabled together with downlink precise RB estimation, Cell Combination may produce increased benefits.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When enabled together with downlink precise RB estimation, coordinated interference management may produce reduced benefits.

8.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- TDD
All NR-capable main control boards as well as NR TDD-capable UBBPg and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.6.3.4 Others

None

8.6.4 Operation and Maintenance

8.6.4.1 Data Configuration

8.6.4.1.1 Data Preparation

[Table 8-7](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Switch	NRDUCellDLSch. DISchAlgoSwitch	PRECISE_DL_RB_E STIMATE_SW	It is recommended that this option be selected when RB resource estimation is involved.

8.6.4.1.2 Using MML Commands

Before using MML commands, refer to [8.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink precise RB estimation
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=PRECISE_DL_RB_ESTIMATE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink precise RB estimation  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=PRECISE_DL_RB_ESTIMATE_SW-0;
```

8.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.6.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) changes after this function is enabled, this function has taken effect.

8.6.4.3 Network Monitoring

The indicators listed in [8.6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

8.7 Sparsified RMSI Scheduling Policy

8.7.1 Principles

In high-frequency scenarios, when the sparsified RMSI scheduling policy function is enabled (with **NRDUCellDLSch.RmsiSparsifiedSchPolicy** set to a value other than **OFF**), the gNodeB uses different scheduling policies based on the parameter settings to implement sparsified RMSI scheduling. This can reduce the use of time-domain resources for RMSI transmission and increase time-domain resources available for the PDSCH, further improving user experience. It is recommended that this function be enabled in SA networking or NSA and SA hybrid networking.

8.7.2 Network Analysis

8.7.2.1 Benefits

In high-frequency NR TDD scenarios, this function increases time-domain resources available for the PDSCH, thereby increasing the average downlink UE throughput (**User Downlink Average Throughput (DU)**).

8.7.2.2 Impacts

The impacts between this function and other functions are described in [8.7.3.2.3 Function Impacts](#).

In high-frequency NR TDD, this function decreases the number of RMSI transmissions and the PDCCH CCE aggregation level. Therefore, the PDCCH CCE usage (**N.CCE.Used.Avg/N.CCE.Avail.Avg**) may fluctuate. In addition, this function increases the number of scheduled slots, and therefore the **Cell Downlink Average Throughput (DU)** may decrease and the following indicators decrease:

- First packet transmission delay, which is calculated using the following formula: $(\text{N.Traffic.DL.RlcFirstPktDelay.Time} / \text{N.Qos.DL.RlcFirstPktDelay.Num}) / 1000$
- **N.User.RRCCConn.Active.DL.Avg**

8.7.3 Requirements

8.7.3.1 Licenses

There are no license requirements for basic functions.

8.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Slot configuration	NRDUCell.SlotAssignment	<i>Channel Management</i>	The slot configuration must be 4_1_DDDSU when the sparsified RMSI scheduling policy function is to be enabled.
High-frequency TDD	SIB1 period	NRDUCell.Sib1Period	<i>5G Networking and Signaling</i>	The value of the parameter for the sparsified RMSI scheduling policy function, NRDUCellDL.Sch.RmsiSparsifiedSchPolicy , is in the format of XMS_YBEAM . <i>X</i> must be greater than or equal to the SIB1 period.

8.7.3.2.2 Mutually Exclusive Functions

None

8.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Single-beam scenario of mmWave 3D coverage pattern	NRDUCellTrpMmwaveBeam.CoverageScenario set to SCENARIO_103	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwaveBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)), the sparsified RMSI scheduling policy function does not take effect.
High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SWITCH option of the NRDUCellAllgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If both dynamic adjustment of the RMSI broadcast interval and the sparsified RMSI scheduling policy are in effect, the broadcast interval takes the greater one of the interval values involved in the two functions.

8.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

8.7.3.4 Others

None

8.7.4 Operation and Maintenance

8.7.4.1 Data Configuration

8.7.4.1.1 Data Preparation

Table 8-8 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Sparsified RMSI scheduling policy	NRDUCellDLSch.RmsiSparsifiedSchPolicy	None	• In high-frequency NR TDD FWA scenarios, the value 40MS_2BEAM is recommended. • In high-frequency NR TDD eMBB scenarios, the value 80MS_2BEAM is recommended.

8.7.4.1.2 Using MML Commands

Before using MML commands, refer to **8.7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling sparsified RMSI scheduling policy (using high-frequency NR TDD FWA scenarios as an example)
MOD NRDUCELLDLSCH: NrDuCellId=0, RmsiSparsifiedSchPolicy=40MS_2BEAM;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling sparsified RMSI scheduling policy (using high-frequency NR TDD FWA scenarios as an example)
MOD NRDUCELLDLSCH: NrDuCellId=0, RmsiSparsifiedSchPolicy=OFF;
```

8.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.7.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) changes after this function is enabled, this function has taken effect.

8.7.4.3 Network Monitoring

For details, see [8.7.2 Network Analysis](#).

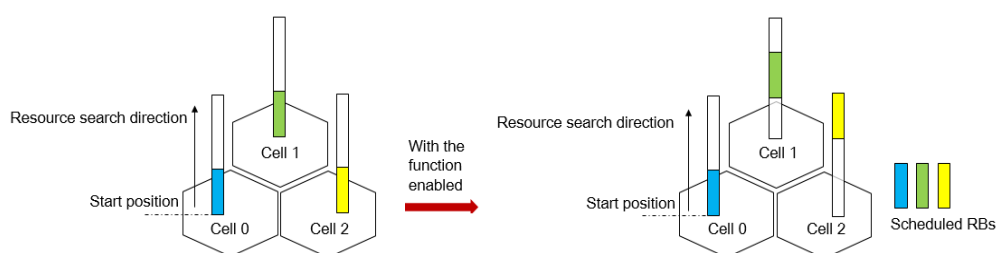
8.8 Downlink Interference Randomization-based Scheduling

8.8.1 Principles

Downlink interference randomization-based scheduling staggers resources allocated to adjacent cells as far away from each other as possible in multi-cell networking scenarios. This reduces downlink interference from neighboring cells to UEs in the local cell and improves downlink system capacity and cell edge UE throughput when the network is moderately or lightly loaded. This function can be enabled by selecting the **DL_INTRF_RANDOM_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. This function is supported only in low frequency bands, not in high frequency bands.

Figure 8-4 shows how downlink interference randomization-based scheduling works.

Figure 8-4 Downlink interference randomization-based scheduling



Before this function is enabled, the RB allocation start positions are not staggered between adjacent cells. Instead, downlink RBs are scheduled at the same start position in the frequency band. The probability of RB position overlapping is high and UEs in the local cell experience a large amount of downlink interference from neighboring cells.

After this function is enabled, the RB allocation start positions are staggered between adjacent cells based on the physical cell identifier (PCI). Downlink RBs are scheduled from different start positions in the frequency band, reducing downlink interference from neighboring cells to UEs in the local cell.

8.8.2 Network Analysis

8.8.2.1 Benefits

When the cell load is less than or equal to 30%, the throughput for UEs at the cell edge can increase by a maximum of 5%. When the cell load is light, there is a high probability that different cells occupy different time-domain resources. In this case, enabling this function may deliver no benefits. A smaller inter-site distance indicates stronger interference from neighboring cells and higher benefits delivered by this function.

When the cell load is greater than 30%, this function may deliver no benefits. However, it will never generate negative results.

NOTE

Cell load = $N.PRB.DL.Used.Avg / N.PRB.DL.Avail.Avg$

8.8.2.2 Impacts

The impacts between this function and other functions are described in [8.8.3.2.3 Function Impacts](#).

8.8.3 Requirements

8.8.3.1 Licenses

There are no license requirements for basic functions.

8.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.8.3.2.1 Prerequisite Functions

None

8.8.3.2.2 Mutually Exclusive Functions

None

8.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDUCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	For hyper cells, downlink interference randomization-based scheduling is supported only between TRPs within a hyper cell.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	After the switch for low latency and high reliability is turned on (by selecting the HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter), downlink interference randomization-based scheduling does not take effect when independent resource reservation in non-slicing scenarios applies and NRDUCellRes.DIMinRbStartPos is set to 0 .
Low-frequency TDD	Interference avoidance for indoor cells	INDOOR_INTERF_AVOID_SW option of the NRDUCellAlgoSwitch.IndoorOutdoorCooperationSw parameter	<i>3D Networking Experience Improvement</i>	When enabled together with downlink interference randomization-based scheduling, interference avoidance for indoor cells is used for PDSCH scheduling.
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	When both the NR DU cell resource management between network slices function and the downlink interference randomization-based scheduling function are enabled, and both the minimum RB quota and its start position are configured, downlink interference randomization-based scheduling is not performed within the minimum RB quota.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink multi-BWP interference avoidance	DL_BWP_HYBRID_INTRF_RAN_DOM_SW option of the NRDUCellIDSc h.DlSchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	If downlink multi-BWP interference avoidance is not in effect, resources on the entire cell spectrum can be used to randomize interference between cells on BWP1. If downlink multi-BWP interference avoidance is in effect, only resources on the non-BWP2 spectrum can be used for the same purpose, which may reduce the benefits produced by downlink interference randomization-based scheduling.
Low-frequency TDD	Secondary harmonic interference avoidance	HARMONIC_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.NsaDca lgoSwitch parameter	<i>EPC-based NSA Basics</i>	When enabled together with downlink interference randomization-based scheduling, NR frequency-domain avoidance for secondary harmonic interference avoidance does not take effect.
Low-frequency TDD	Start RB avoidance	START_RB_AVOID_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	When both start RB avoidance and downlink interference randomization-based scheduling are enabled, their benefits for non-RedCap UEs decrease. This occurs when the downlink RBs scheduled for non-RedCap UEs do not overlap with the low frequency band (20 MHz) and non-RedCap UEs are scheduled to use frequency resources other than those on the BWP1 activated for RedCap UEs. Consequently, the benefits produced by avoidance of RedCap downlink scheduling vary across cells with different PCIs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Avoidance of RedCap downlink scheduling	AVOID_REDCA P_DL_SCH_SW option of the NRDUCellRedCap.RedCapSchedAlgoSwitch parameter	<i>RedCap</i>	When both avoidance of RedCap downlink scheduling and downlink interference randomization-based scheduling are enabled, their benefits for non-RedCap UEs decrease. This occurs when the downlink RBs scheduled for non-RedCap UEs do not overlap with those scheduled for RedCap UEs and non-RedCap UEs are scheduled to use frequency resources other than those on the BWP1 activated for RedCap UEs. Consequently, the benefits produced by avoidance of RedCap downlink scheduling vary across cells with different PCIs.

8.8.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.8.3.4 Others

None

8.8.4 Operation and Maintenance

8.8.4.1 Data Configuration

8.8.4.1.1 Data Preparation

[Table 8-9](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 8-9 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	DL_INTRF_RANDOM_SW	Select this option when the cell load is 30% or less.

8.8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink interference randomization-based scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_INTRF_RANDOM_SW-1;
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink interference randomization-based scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_INTRF_RANDOM_SW-0;
```

8.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.8.4.2 Activation Verification

If the average downlink UE throughput (**User Downlink Average Throughput (DU)**) increases after downlink interference randomization-based scheduling is enabled, this function has taken effect.

8.8.4.3 Network Monitoring

Monitor **User Downlink Average Throughput (DU)** to obtain the average downlink UE throughput.

8.9 PDSCH Resource Mapping

8.9.1 Principles

Puncturing of other channels or signals can be performed for downlink data channels in resource mapping mode, thereby increasing the available RBs/REs on the PDSCH or mitigating neighboring cell interference. For details, see section 5.1.4 "PDSCH resource mapping" in 3GPP TS 38.214 V15.4.0. Downlink scheduling supports PDSCH resource mapping, which is controlled by the **NRDUCellPdsch.RateMatchSwitch** parameter.

In low-frequency NR TDD, the PDSCH resource mapping function allows puncturing of the CSI-RS, TRS, PDCCH, SSB, and DC component interference signal for the PDSCH, by using the respective subfunctions CSI-RS rate matching, TRS rate matching, PDCCH rate matching, SSB rate matching, and DC component rate matching.

In high-frequency NR TDD, PDSCH resource mapping allows the CSI-RS-based and PDCCH-based puncturing on the PDSCH, by using the respective subfunctions CSI-RS rate matching and PDCCH rate matching.

NOTE

The CSI-RS mentioned in this chapter is specifically referred to as the reference signal for channel measurement (CSI-RS for CM) in the downlink. For details about CSI-RS for CM, see *Channel Management*.

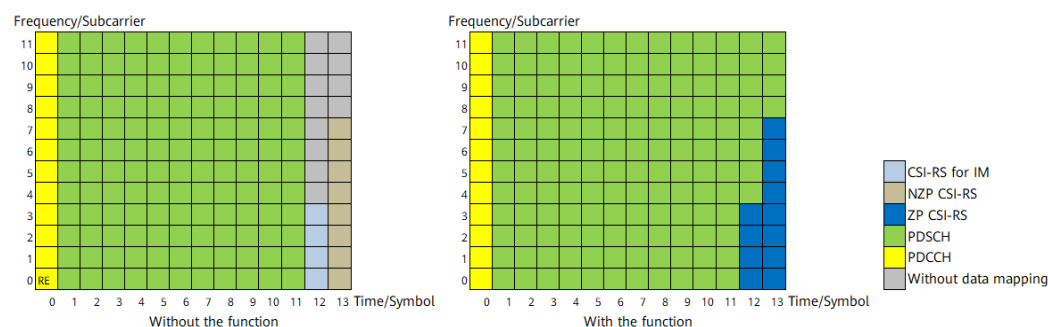
8.9.1.1 CSI-RS Rate Matching

This function supports mapping PDSCH and CSI-RS resources onto the same OFDM symbols and is enabled using the **CSIRS_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter.

On a time-domain OFDM symbol, CSI-RS for IM or NZP CSI-RS occasionally does not fully occupy the corresponding frequency-domain resources. In accordance with specifications, ZP CSI-RS is used to inform UEs of the REs that are not mapped onto any data, as shown in [Figure 8-5](#).

Before the function is enabled, the REs onto which no data is mapped are wasted. After this function is enabled, the gNodeB uses ZP CSI-RS resources to inform UEs of the remaining frequency-domain resources for mapping data, thereby increasing the number of available REs of the UEs.

Figure 8-5 Rate matching pattern



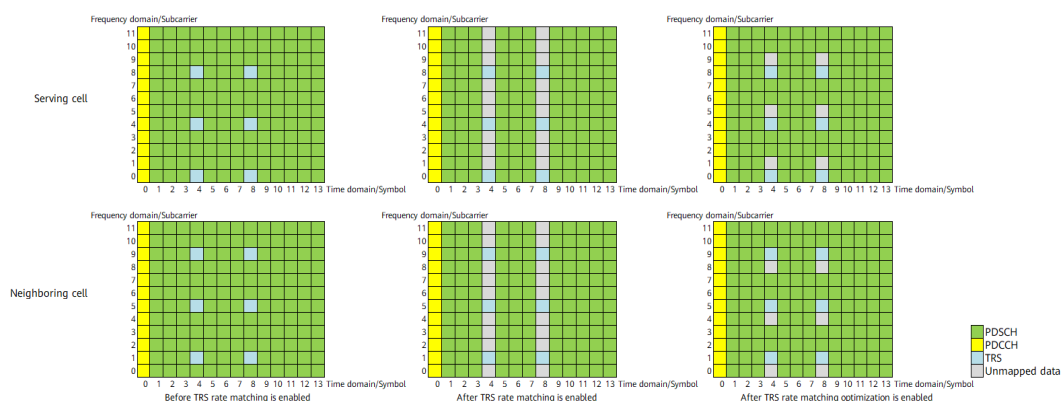
8.9.1.2 TRS Rate Matching

PDSCH resource mapping supports TRS-based puncturing on the PDSCH, thereby mitigating interference from neighboring cells or increasing available REs on the PDSCH.

TRS rate matching is controlled by the **TRS_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter and is enabled by default. This function ensures that a neighboring cell's PDSCH and the serving cell's TRS do not interfere with each other by avoiding mapping the PDSCH to the REs on the OFDM symbols that carry TRS resources in the serving and neighboring cells, thereby increasing the SINR.

In low-frequency NR TDD, after TRS rate matching is enabled, no data is mapped onto TRS symbols, leading to a low frequency-domain resource usage. To improve the frequency-domain resource usage in peak rate or drive test scenarios, TRS rate matching optimization is introduced. This function is controlled by the **TRS_RATEMATCH_OPT_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter and can be enabled only when the TRS rate matching switch is on. Assume that TRS rate matching is enabled. If TRS rate matching optimization is not enabled, no data is mapped onto frequency-domain positions corresponding to TRS symbols in the serving and neighboring cells; if TRS rate matching optimization is enabled, data is mapped onto certain frequency-domain positions corresponding to some TRS slots in the serving and neighboring cells, thereby increasing the average downlink cell throughput and average downlink UE throughput. [Figure 8-6](#) illustrates this comparison.

Figure 8-6 Comparison before and after enabling TRS rate matching and TRS rate matching optimization



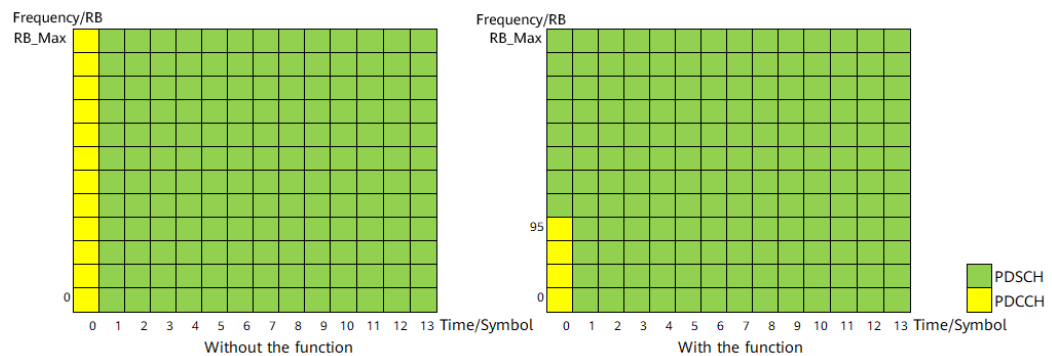
8.9.1.3 PDCCH Rate Matching

Section 5.1.4 "PDSCH resource mapping" in 3GPP TS 38.214 V15.4.0 defines two PDCCH rate matching modes, one where the gNodeB configures the rate matching pattern for UEs ([Rate-Matching-Pattern-Configuration-based PDCCH Rate Matching](#)) and one where it does not ([Rate-Matching-Pattern-Configuration-free PDCCH Rate Matching](#)). In NR TDD scenarios, both rate-matching-pattern-configuration-based and configuration-free PDCCH rate matching functions are supported.

Rate-Matching-Pattern-Configuration-based PDCCH Rate Matching

This function supports mapping PDSCH and PDCCH resources onto the same OFDM symbols and is enabled using the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter. The PDCCH-occupied frequency-domain resource range mapped onto OFDM symbols can be configured by using the **NRDUCellPdcch.OccupiedRbNum** parameter. **Figure 8-7** shows the effect of PDCCH rate matching when **NRDUCellPdcch.OccupiedRbNum** is set to **8** (corresponding to an actual configuration of 96 RBs). **RB_Max** indicates the maximum number of RBs within the system bandwidth.

Figure 8-7 Rate matching pattern



The configuration of the **NRDUCellPdcch.OccupiedRbNum** parameter works as follows:

- When **NRDUCellPdcch.OccupiedRbNum** is set to **0** or the frequency-domain range specified by this parameter is greater than or equal to the number of RBs corresponding to the downlink system bandwidth specified by the **NRDUCell.DlBandwidth** parameter, the PDCCH occupies a full bandwidth and the rate-matching-pattern-configuration-based PDCCH rate matching function does not take effect.
- When **NRDUCellPdcch.OccupiedRbNum** is set to a value greater than **0** and is smaller than the number of RBs corresponding to the downlink system bandwidth specified by the **NRDUCell.DlBandwidth** parameter:
 - A smaller value of this parameter allows the PDCCH to use fewer frequency-domain resources in a cell. This leads to more available resources for the PDSCH and greater PDSCH capacity improvement.
 - A larger value of this parameter allows the PDCCH to use more frequency-domain resources in a cell. This leads to fewer available resources for the PDSCH and smaller PDSCH capacity improvement.

NOTE

When the **NRDUCellPdcch.OccupiedRbNum** parameter is set to a value less than **8** (corresponding to an actual configuration of 96 RBs), there is a risk that UEs cannot access the network due to insufficient PDCCH resources. The value of this parameter must be adapted to scenarios.

When the number of required PDCCH RBs is greater than the number of RBs configured using the **NRDUCellPdcch.OccupiedRbNum** parameter, the number of scheduling times may decrease.

To support downlink control information (DCI) triggered by uplink and downlink scheduling and prevent UE access failures or service drops caused by

scheduling failures, the **NRDUCellPdcch.OccupiedRbNum** parameter needs to be configured based on UE distribution in the cell.

- When **NRDUCellPdcch.OccupiedSymbolNum** is set to **1SYM**:
 - If a cell has UEs at the cell edge or with poor channel conditions (regardless of whether there are UEs near the cell center, at a medium distance from the cell center, with common channel conditions, or with good channel conditions), it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **8**, which corresponds to an actual configuration of 96 RBs.
 - If a cell does not have UEs at the cell edge or with poor channel conditions, but has UEs at a medium distance from the cell center or with common channel conditions (regardless of whether there are UEs near the cell center or with good channel conditions), it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **4**, which corresponds to an actual configuration of 48 RBs.
 - If a cell does not have UEs at the cell edge, at a medium distance from the cell center, with poor channel conditions, or with common channel conditions, it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **2**, which corresponds to an actual configuration of 24 RBs.

NOTE

When **NRDUCellPdcch.OccupiedSymbolNum** is set to **1SYM**, **NRDUCellPdcch.OccupiedRbNum** cannot be set to **1**, which corresponds to an actual configuration of 12 RBs. Otherwise, UEs near the cell center cannot be scheduled.

When the UL and DL decoupling function is enabled (by setting the **NRDUCellAlgoSwitch.UlDlDecouplingSwitch** parameter to **ON** for an NR TDD cell), it is recommended that the **NRDUCellPdcch.OccupiedRbNum** parameter be configured as follows to ensure sufficient CCE resources for SUL UEs:

- If a cell has UEs at the cell edge or with poor channel conditions (regardless of whether there are UEs near the cell center, at a medium distance from the cell center, with common channel conditions, or with good channel conditions), it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **0**, which indicates that the PDCCH occupies a full bandwidth.
- If a cell does not have UEs at the cell edge or with poor channel conditions, but has UEs at a medium distance from the cell center or with common channel conditions (regardless of whether there are UEs near the cell center or with good channel conditions), it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **8**, which corresponds to an actual configuration of 96 RBs.
- If a cell has only UEs near the cell center, but no UEs at the cell edge, at a medium distance from the cell center, with poor channel conditions, or with common channel conditions, it is recommended that **NRDUCellPdcch.OccupiedRbNum** be set to **4**, which corresponds to an actual configuration of 48 RBs.

In low-frequency scenarios, this function applies to single-UE peak rate scenarios and peak rate scenarios involving two to four UEs.

In high-frequency scenarios, this function applies to single-UE peak rate scenarios.

Due to the limited processing capability of 5G C-band CPE V100R002C0X, the rate-matching-pattern-configuration-based PDCCH rate matching function cannot take effect under certain combinations of cell bandwidths and cell center frequencies. In such cases, the cell center frequency can be increased or decreased by an odd number of RBs for the rate-matching-pattern-configuration-based PDCCH rate matching function to take effect. For example, for 5G C-band CPE V100R002C0X in a cell with the C-band 100 MHz bandwidth, the rate-matching-pattern-configuration-based PDCCH rate matching function does not take effect when the cell center frequency is set to **635500** but takes effect when the cell center frequency is set to **635524**.

To solve the compatibility issues between certain UEs and the rate-matching-pattern-configuration-based PDCCH rate matching function, the PDCCH rate matching UE compatibility rectification function is introduced. This function is controlled by the **PDCCH_RATEMATCH_COMPATIBLE_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter. This function is supported only in low-frequency NR TDD cells. After this function is enabled, the gNodeB modifies the PDCCH rate matching pattern configuration of UEs to resolve UE compatibility issues.

Rate-Matching-Pattern-Configuration-free PDCCH Rate Matching

The rate-matching-pattern-configuration-free PDCCH rate matching function does not require the gNodeB to configure the rate matching pattern for UEs, and UEs can proactively detect the PDSCH information on PDCCH symbols. This function is enabled using the **PDCCH_NO_PATTERN_RATE_MATCH_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter.

In low-frequency NR TDD:

- The switch for rate-matching-pattern-configuration-free PDCCH rate matching can be turned on only when the switch for rate-matching-pattern-configuration-based PDCCH rate matching has been turned on. When these two PDCCH rate matching switches are both turned on, rate-matching-pattern-configuration-free PDCCH rate matching preferentially takes effect.
- To increase the probability of rate-matching-pattern-configuration-free PDCCH rate matching taking effect, the pattern-free PDCCH rate matching activation rate increase function is introduced. This function is controlled by the **PDCCH_NO_PAT_RM_RATIO_INC_SW** option of the **NRDUCellServExp.SpdTurboAlgoSwitch** parameter. This function is recommended only in single-UE downlink peak rate scenarios.

In high-frequency NR TDD, it is recommended that this function be enabled only in single-UE downlink peak rate scenarios. In addition, rate-matching-pattern-configuration-based and configuration-free PDCCH rate matching functions cannot be both enabled.

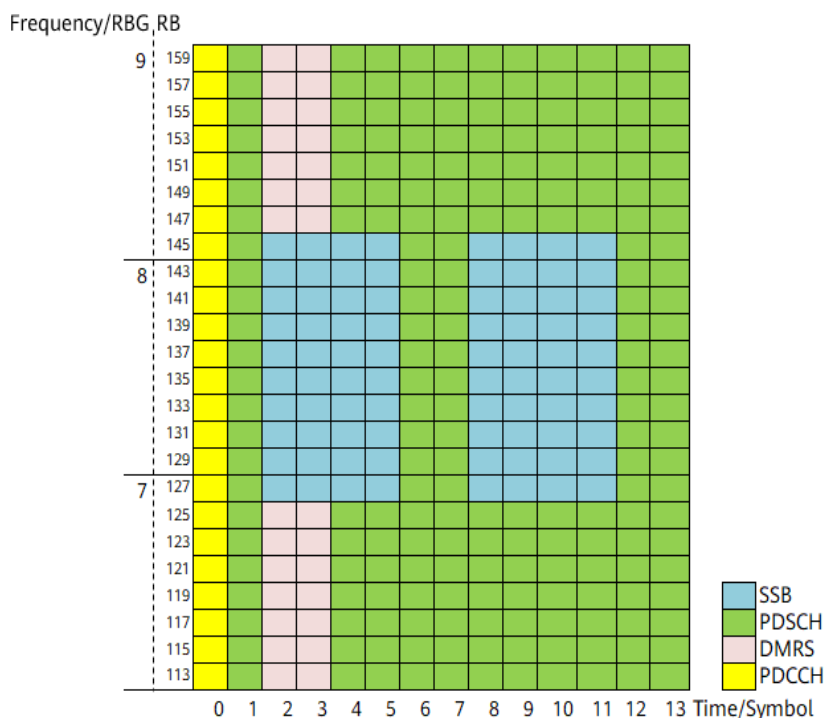
8.9.1.4 SSB Rate Matching

As stipulated in specifications, PDSCH and SSB resources can be mapped onto the same OFDM symbols.

Figure 8-8 shows the rate matching pattern with the 100-MHz NR TDD system bandwidth and two SSB beams (one SSB beam spans symbol 2 to symbol 5 and the other spans symbol 8 to symbol 11).

Figure 8-8 SSB rate matching pattern in TDD (defined in specifications)

Assume that the system bandwidth is 100 MHz.



When the SSB conflicts with the PDSCH DMRS, in the corresponding SSB frequency-domain position, the PDSCH on OFDM symbols has no PDSCH DMRS, which will cause a PDSCH demodulation failure. To avoid such demodulation failures, an RBG that contains SSB is not scheduled. In this context, the SSB rate matching function is introduced to enable the gNodeB to signal to UEs the RBs that cannot carry PDSCH data. This function is enabled using the **SSB_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter and applies to single-UE peak rate scenarios.

NOTE

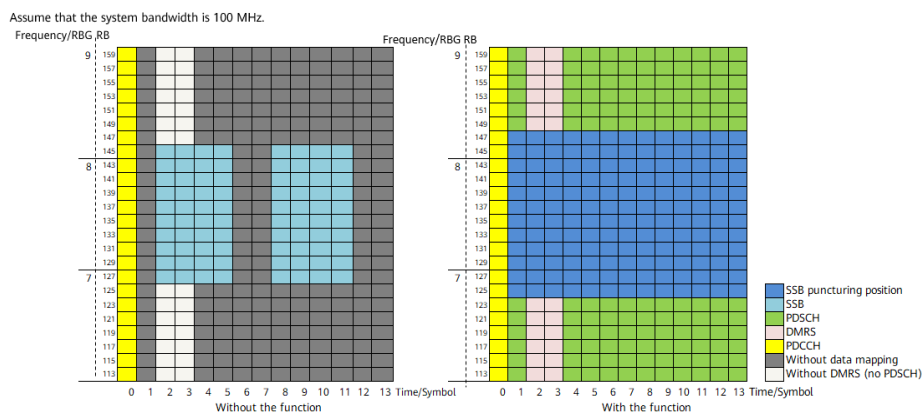
The SSB is present only in some slots, with the SSB period configurable in the time domain.

NR TDD Scenario

Figure 8-9 shows the rate matching pattern with the C-band channel bandwidth of 100 MHz in NR TDD. Before this function is enabled, assuming that the SSB occupies 20 RBs in the center of the cell bandwidth, the SSB occupies three RBGs (based on the default frequency-domain position of the SSB) when scheduling is performed at the RBG level. In this case, no data is mapped onto the 48 RBs in the corresponding frequency domain. After this function is enabled, puncturing on full symbols is supported for SSB rate matching, that is, the gNodeB informs in rate matching mode the UEs of the SSB frequency-domain position (20 or 24 RBs, after outer expansion at the granularity of four RBs on the full bandwidth) and full

symbols (symbol 1 to symbol 13) in the SSB time domain so that the UEs can map data onto the remaining 28 or 24 RBs.

Figure 8-9 Rate matching pattern (TDD)



8.9.1.5 DC Component Rate Matching

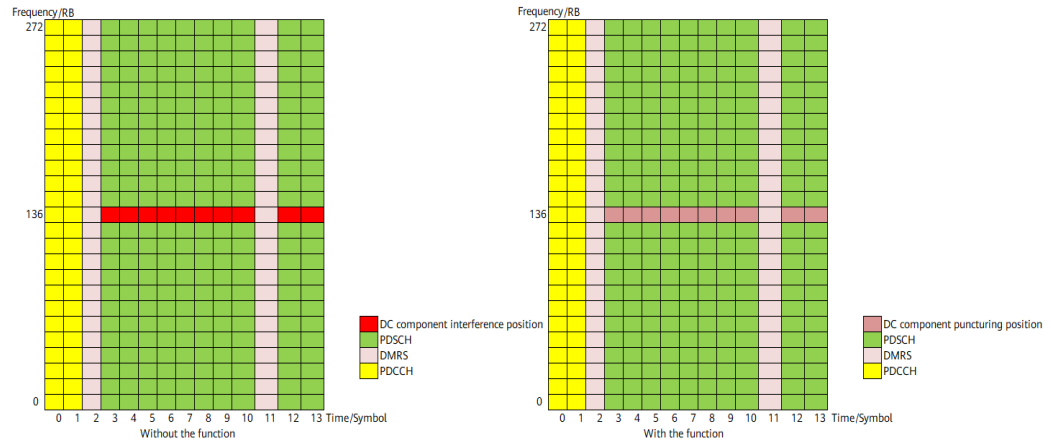
Direct current (DC) component interference signals are generated during signal conversion when a UE receives data. Under the same channel conditions, the probability of failures to demodulate REs at the center of the frequency band of the cell where the UE is located is higher than that to demodulate REs at other positions of the frequency band. As a result, the downlink BLER increases and the MCS index decreases, affecting the downlink user-perceived rate. In this context, the DC component rate matching function is introduced.

The DC component rate matching function is enabled using the **DC_COMPONENT_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter.

DC component rate matching uses the RB-level rate matching mode defined in section 5.1.4.1 "PDSCH resource mapping with RB symbol level granularity" of 3GPP TS 38.214 V15.5.0. **Figure 8-10** uses the 100-MHz bandwidth as an example to explain this function in NR TDD. Without this function, all RBs in the frequency domain can be allocated to UEs. However, interference from the DC component compromises performance of RB demodulation in the corresponding frequency domain. With this function, the gNodeB uses the rate matching mechanism to inform UEs of the frequency-domain position and all time-domain OFDM symbols (specific to the PDSCH) of the DC component. No PDSCH data will then be mapped at this position. The DC component occupies one or two RBs. All RBs, except those occupied by the DC component, can carry PDSCH data. This function is supported only in low frequency bands, not in high frequency bands.

Figure 8-10 Rate matching pattern in NR TDD

Assume that the system bandwidth is 100 MHz.



8.9.2 Network Analysis

8.9.2.1 Benefits

- CSI-RS rate matching

In single-UE peak and single-UE drive test scenarios, the CSI-RS occupies few frequency-domain resources. The remaining RE resources on CSI-RS symbols can be used for PDSCH data transmission, effectively improving the throughput. In addition, interference-measuring CSI-RSs of a cell are aligned with PDSCH transmission data of a neighboring cell in the time and frequency domains, ensuring that the cell can measure the interference of the neighboring cell.

- TRS rate matching

- TRS rate matching

In contiguous networking, this function mitigates mutual interference between the neighboring cell's PDSCH data and the serving cell's TRS. This helps reduce data jitter and improve throughput for CEUs. The improvement is illustrated below: The downlink throughput of some CEUs can increase by about 20% at most in a scenario where the NR TDD system bandwidth is 100 MHz.

- TRS rate matching optimization

In peak rate or drive test scenarios, after TRS rate matching function and TRS rate matching optimization are enabled, REs corresponding to some TRS symbols can be used for PDSCH transmission, increasing the **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.

- PDCCH rate matching

Rate-matching-pattern-configuration-based PDCCH rate matching:

When this function is enabled for a cell in suitable scenarios, the PDCCH occupies few frequency-domain resources. The remaining RBs in PDCCH symbols can be used for PDSCH data transmission, effectively improving the throughput. The improvement is illustrated below: The downlink throughput

can increase by about 6% at most in a scenario where the NR TDD system bandwidth is 100 MHz.

After the PDCCH rate matching UE compatibility rectification function takes effect, the UE access failure rate and cell service drop rate caused by UE compatibility issues may decrease.

Rate-matching-pattern-configuration-free PDCCH rate matching:

When this function takes effect, the remaining RBs in PDCCH symbols are used for PDSCH data transmission, which effectively increases the throughput. The improvement is illustrated below: In a single-UE peak rate scenario in high-frequency NR TDD networking with a channel bandwidth of 100 MHz, the downlink throughput can increase by about 5%–6% at most. In a single-UE peak rate scenario in low-frequency NR TDD networking with a channel bandwidth of 100 MHz, the downlink throughput can increase by about 2%–5% at most. If pattern-free PDCCH rate matching activation rate increase is also enabled, the downlink throughput can further increase by 1%–2% at most.

- SSB rate matching

After this function takes effect, the number of available PDSCH RBs for UEs increases, effectively improving the downlink cell capacity and single-user-perceived downlink rate. A shorter SSB period indicates higher gains. The improvement is illustrated below: The downlink throughput can increase by about 1% at most in a scenario where (1) the NR TDD system bandwidth is 100 MHz, (2) the SSB spans three RBGs in the frequency domain, and (3) the SSB period is 20 ms, and (4) eight SSB beams are configured.

- DC component rate matching

In single-UE scenarios, if the DC component affects the UE demodulation performance, this function reduces the downlink BLER and enables the gNodeB to select a more accurate MCS index for the UE, thereby effectively increasing the downlink UE throughput.

8.9.2.2 Impacts

The impacts between this function and other functions are described in [8.9.3.2.3 Function Impacts](#).

- CSI-RS rate matching

None

- TRS rate matching

- TRS rate matching

This function decreases available PDSCH REs. As a result, for isolated cells, the downlink UE throughput and cell capacity decrease.

- TRS rate matching optimization

None

- PDCCH rate matching

- Rate-matching-pattern-configuration-based PDCCH rate matching

Enabling this function may cause PDCCH resource insufficiency.

Enabling this function may decrease the MCS index, resulting in decreased or even no benefits. If the number of RBs configured for

CORESET 1 (UE-specific CORESET) accounts for an overly large proportion of the total number of RBs on the UE BWP, the downlink UE throughput and cell capacity may even decrease.

In high-frequency NR TDD, enabling this function will cause cell setup failures when the gNodeB is configured with a total of more than eight cells with the 100 MHz bandwidth or more than four cells with the 200 MHz bandwidth.

When the maximum DCI power offset in the common search space (specified by the **NRDUCellChnPwr.MaxCommonDciPwrOffset** parameter) is greater than 0, enabling this function may cause PDCCH power insufficiency.

In low frequency bands, enabling this function increases the downlink BLER for some UEs that support **multipleCORESET** or only **rateMatchingResrcSetSemi-Static**.

- Rate-matching-pattern-configuration-free PDCCH rate matching
Enabling this function may cause the MCS index to decrease. This will reduce positive impact or render none at all.

In low frequency bands, enabling this function increases the downlink BLER for some UEs that support **multipleCORESET** or only **rateMatchingResrcSetSemi-Static**.

In low-frequency TDD, if pattern-free PDCCH rate matching activation rate increase is also enabled, the data transmission delay of UEs performing small-packet services may increase.

- SSB rate matching

None

After this function is enabled, UEs may not demodulate the DCI when the SSB rate matching pattern conflicts with PDCCH resources. As a result, the UE access delay or access failure rate increases.

- DC component rate matching

If there is more than one UE to be scheduled in a cell, this function offers little positive impact or may even render negative results.

NOTE

After the PDCCH rate matching, SSB rate matching, or DC component rate matching function is enabled, the function activation may fail when the number of rate matching patterns that can be delivered by the base station through RRC signaling has reached the upper limit specified in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V16.3.1.

8.9.3 Requirements

8.9.3.1 Licenses

There are no license requirements for basic functions.

8.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.9.3.2.1 Prerequisite Functions

- CSI-RS rate matching
 - None
- TRS rate matching
 - TRS rate matching
 - None
 - TRS rate matching optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS rate matching	TRS_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	TRS rate matching optimization can be enabled only when TRS rate matching is enabled.

- PDCCH rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH rate matching	• PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter • PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Downlink Scheduling</i>	In low-frequency NR TDD networks, rate-matching-pattern-configuration-free PDCCH rate matching can be enabled only when rate-matching-pattern-configuration-based PDCCH rate matching is enabled.
Low-frequency TDD	PDCCH rate matching	PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Pattern-free PDCCH rate matching activation rate increase can be enabled only when the switch for rate-matching-pattern-configuration-free PDCCH rate matching is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	NRDUCellPdcch.OccupiedRbNum must be set to a value greater than 0 and less than 4 for a cell if all the following conditions are met: (1) NRDUCell.FrequencyBand is set to N28 , (2) NRDUCell.DlBandwidth and NRDUCell.UlBandwidth are set to CELL_BW_40M , and (3) rate-matching-pattern-configuration-based PDCCH rate matching is enabled for the cell.

- SSB rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink Additional DMRS Position	NRDUCellPdsch.DlAdditionalDmrPos	<i>Downlink Scheduling</i>	For an NR SDL cell, the NRDUCellPdsch.DlAdditionalDmrPos parameter must be set to NOT_CONFIG if the LTE_NR_SDL_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is deselected and SSB rate matching is enabled for the cell.

RAT	Function Name	Function Switch	Reference	Description
FDD	Configurable time-domain position in the single SSB beam case	None	<i>Channel Management</i>	When the SSB rate matching function is enabled (by selecting the SSB_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter), the NRDUCell.SsbTimePos parameter (for specifying the time-domain position in the single SSB beam case) can only be set to DEFAULT , SSB1 , or SSB3 .

- DC component rate matching
None

8.9.3.2.2 Mutually Exclusive Functions

- CSI-RS rate matching
None
- TRS rate matching
 - TRS rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Narrowband TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	• Narrowband TRS rate matching must be disabled when TRS rate matching is enabled. • For a low-frequency NR cell, if NRDUCell.NrDuCellWorkingMode is set to HYPER_CELL or HYPER_CELL_COMBINE_MODE, TRS rate matching and narrowband TRS rate matching cannot be both disabled.

- TRS rate matching optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB DL Joint Transmission Switch	INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	None	Inter-base-station downlink joint transmission and TRS rate matching optimization are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Intra-base-station DL CoMP and TRS rate matching optimization are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Downlink inter-gNodeB CA	INTER_GNB_ODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Downlink inter-gNodeB CA and TRS rate matching optimization are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	TRS period of 10 ms	NRDUCellCsirs.TrsPeriod set to MS10	<i>Channel Management</i>	When the TRS period is set to 10 ms, the TRS rate matching optimization function cannot be enabled.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_CO_MM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM and TRS rate matching optimization cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-TRS measurement	MULTI_TRS_MULTI_SLOT_MEAS_SW option of the NRDUCell HighSpeedMob.HighSpeedAlgorithmOptSw parameter	<i>High Speed Mobility</i>	TRS rate matching optimization cannot be enabled when multi-TRS measurement is enabled.
Low-frequency TDD	Decoupling of joint transmission from TRS rate matching	JT_TRS_RM_DECOUPLING_SW option of the NRDUCell ServExp.SpdTurboAlgorithmSwitch parameter	None	"Decoupling of joint transmission from TRS rate matching" and "TRS rate matching optimization" cannot be both enabled.

- PDCCH rate matching
 - Rate-matching-pattern-configuration-based PDCCH rate matching:
Mutually exclusive relationships with basic functions under RAN functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	In high-frequency NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when either of the following conditions is met: • NRDUCellPdcch.OccupiedSymbolNum is set to 2SYM (indicating two UE-specific PDCCH symbols). • NRDUCellCoreset.CommonCtrlResRbNum is set to RB24 or RB96 (indicating two common PDCCH symbols).
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	When the NRDUCellPdcch.OccupiedSymbolNum parameter is set to PART_SLOT_3SYM, the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	When both the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter and the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter are selected, multi-BWP1 PDCCH transmission algorithm must be disabled (by deselecting the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	In low-frequency NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when either of the following conditions is met: • NRDUCellPdcch.OccupiedSymbolNum is set to 2SYM or 3SYM (indicating two or three UE-specific PDCCH symbols) and NRDUCell.DlBandwidth is not set to CELL_BW_30M or CELL_BW_20M (indicating that the downlink bandwidth is not 30 MHz or 20 MHz in a cell). • NRDUCellCoreset.CommonCtrlResRbNum (indicating the number of RBs for common control resources in a cell) is set to RB24 or RB96 and NRDUCell.DlBandwidth is not set to CELL_BW_30M or CELL_BW_20M (indicating that the downlink bandwidth is not 30 MHz or 20 MHz in a cell).
Low-frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	PDCCH symbol number adaptation and rate-matching-pattern-configuration-based PDCCH rate matching are mutually exclusive, indicating that they cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	In NR TDD, rate-matching-pattern-configuration-based PDCCH rate matching cannot be enabled when any of the following conditions is met: • NRDUCellPdcch.OccupiedSymbolNum is set to 2SYM (indicating two UE-specific PDCCH symbols). • NRDUCellCoreset.CommonCtrlResRbNum is set to RB24 or RB96 (indicating two common PDCCH symbols). • Multi-BWP1 PDCCH transmission algorithm is enabled (by selecting the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter).
Low-frequency TDD	Configuration of multiple PDCCH search spaces	MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	"Configuration of multiple PDCCH search spaces" and "rate-matching-pattern-configuration-based PDCCH rate matching" are mutually exclusive.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH multi-symbol resource pool	MULTI_SYMBOL_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	If rate-matching-pattern-configuration-based PDCCH rate matching is enabled and the NRDUCellPdcch.OccupiedSymbolNum parameter is set to a value less than 18, PDCCH multi-symbol resource pool must be disabled.

Mutually exclusive relationships with power management functions under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Rate-matching-pattern-configuration-based PDCCH rate matching and NR inter-carrier dynamic power sharing are mutually exclusive, indicating that they cannot be both enabled.

Mutually exclusive relationships with spectrum management functions under RAN functions

None

Mutually exclusive relationships with functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	- In an NR TDD cell, if the REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter is selected, the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter can be selected only when any of the following conditions is met: NRDUCellRedCap.RedCapUeMaxAvail-Bandwidth is set to 20M and NRDUCellPdcch.OccupiedRbNum is set to a value greater than 4. NRDUCellRedCap.RedCapUeMaxAvail-Bandwidth is set to 20M, NRDUCellPdcch.OccupiedRbNum is set to a value less than 4, and NRDUCellPdcch.PdcchAggLevel1Strategy is set to CONFIG. NRDUCellRedCap.RedCapUeMaxAvail-Bandwidth is set to 40M, NRDUCellPdcch.OccupiedRbNum is set to a value less than 4, and NRDUCellPdcch.PdcchAggLevel1Strategy is set to CONFIG. - In an NR TDD cell, the NRDUCellCoreset.Com

RAT	Function Name	Function Switch	Reference	Description
				<i>monCtrlResRbNum</i> parameter cannot be set to RB24 if all these conditions are met: (1) the REDCAP_SW option of the NRDUCellFeatureSw. <i>UseCapbBasedFunSw</i> parameter is selected, (2) the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch. <i>RateMatchSwitch</i> parameter is selected, and (3) the NRDUCellPdcch. <i>OccupiedRbNum</i> parameter is set to a value less than 4.
Low-frequency TDD	Coordination between serving and cooperating cells with different numbers of PDCCH symbols	PDCCH_SYMBOL_NUM_DIFF_COORD_SW option of the NRDUCellPdcch. <i>PdcchAlgoEnhSwitch</i> parameter	<i>Co-MM (Low-Frequency TDD)</i>	"Rate-matching-pattern-configuration-based PDCCH rate matching" must be disabled before "coordination between serving and cooperating cells with different numbers of PDCCH symbols" is enabled.

Mutually exclusive relationships with functions related to inter-RAT coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR power sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When rate-matching-pattern-configuration-based PDCCH rate matching is enabled, power estimation on the NR side is inaccurate, affecting the LTE and NR power sharing function.

Mutually exclusive relationships with functions related to coverage improvement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink with independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Super uplink with independent SUL deployment must be disabled if all the following conditions are met: (1) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, (2) PDCCH uplink-to-downlink CCE ratio adaptation is disabled, and (3) NRDUCellPdcch.OccupiedRbNum is set to 1 or 2.

Mutually exclusive relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-sector power sharing	NR_INTER_SECTOR_POWER_SHARE_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	If rate-matching-pattern-configuration-based PDCCH rate matching is enabled or the ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter is selected for a cell, the NR_INTER_SECTOR_POWER_SHARE_SW and NR_INTRA_SECTOR_POWER_SHARE_SW options of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter must be deselected.
Low-frequency TDD	NR intra-sector power sharing	NR_INTRA_SECTOR_POWER_SHARE_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	
Low-frequency TDD	PDCCH rate matching enhancement	PDCCH_RATEMATCH_ENH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	If the PDCCH_RATEMATCH_ENH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is selected for any cell served by the BBU, rate-matching-pattern-configuration-based PDCCH rate matching and PDCCH rate matching enhancement cannot be enabled for other cells served by the BBU.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	For a cell specified by NRDUCellBiCarrier.NrDuCellId , the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter must be deselected if all the following conditions are met: (1) NRDUCellBiCarrier.PdcchRmPolicy is set to MINOR_CELL_PREFERRED for the cell, (2) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected for the cell, and (3) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected for the cell specified by NRDUCellBiCarrier.CoNrDuCellId .
Low-frequency TDD	Channel calibration in GP symbols in URLLC scenarios	URLLC_GP_SYM_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	If the URLLC_GP_SYM_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter is selected for a cell, the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter must be deselected for the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low Latency High Reliability Level 1 Sw	LOW_LAT_HIGH_RELIAB_LEVEL1_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	Rate-matching-pattern-configuration-based PDCCH rate matching does not take effect when the LOW_LAT_HIGH_RELIAB_LEVEL1_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter is selected.

– Rate-matching-pattern-configuration-free PDCCH rate matching

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In high-frequency NR TDD, the following two functions are mutually exclusive: rate-matching-pattern-configuration-free PDCCH rate matching and rate-matching-pattern-configuration-based PDCCH rate matching. This indicates that they cannot be both enabled.
Low-frequency TDD	DC component rate matching	DC_COMPONENT_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In low-frequency NR TDD, rate-matching-pattern-configuration-free PDCCH rate matching cannot be enabled if DC component rate matching is enabled (by selecting the DC_COMPONENT_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter).
Low-frequency TDD	Narrowband and TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	In low-frequency NR TDD, rate-matching-pattern-configuration-free PDCCH rate matching cannot be enabled if the NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink DCI extension-based scheduling	DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	"Pattern-free PDCCH rate matching activation rate increase" and "uplink DCI extension-based scheduling" are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQ_OPT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	"Pattern-free PDCCH rate matching activation rate increase" and "light-load time sequence optimization" are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW or FLEX_SUPER_ULPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	"Pattern-free PDCCH rate matching activation rate increase" and "super uplink" are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	"Pattern-free PDCCH rate matching activation rate increase" and "UL and DL Decoupling" are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Channel calibration in GP symbols in URLLC scenarios	URLLC_GP_SYM_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	If the URLLC_GP_SYM_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter is selected for a cell, the PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter must be deselected for the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	If NRDUCellPdcch.OccupiedSymbolNum is set to PART_SLOT_3SYM for a cell, the PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter must be deselected for the cell.
Low-frequency TDD	NR inter-sector power sharing	NR_INTER_SECTOR_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	If "rate-matching-pattern-configuration-free PDCCH rate matching" is enabled, "inter-TRP power sharing for high-speed cells" and "intra-TRP power sharing for high-speed cells" must be disabled.
Low-frequency TDD	NR intra-sector power sharing	NR_INTRA_SECTOR_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	

- SSB rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH rate matching UE compatibility rectification	PDCCH_RATE_MATCH_COMPATIBLE_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When PDCCH rate matching UE compatibility rectification is enabled, SSB rate matching cannot be enabled (by selecting the SSB_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter).
Low-frequency TDD	PDCCH RM adaptation for speed turbo cells	ST_PDCCH_RATE_MATCH_ADAPT_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None	For an NR FDD or SDL cell, if the ST_PDCCH_RATE_MATCH_ADAPT_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter is selected, SSB rate matching must be disabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None

- DC component rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB rate matching	SSB_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	SSB rate matching and DC component rate matching are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPECT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	LTE TDD and NR spectrum sharing and DC component rate matching are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Power saving BWP and DC component rate matching are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	PDCCH rate matching UE compatibility rectification	PDCCH_RATEMATCH_COMPATIBLE_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When PDCCH rate matching UE compatibility rectification is enabled, DC component rate matching cannot be enabled (by selecting the DC_COMPONENT_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter).
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	In NR TDD cells, RedCap and DC component rate matching cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DC I_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	DC component rate matching and single DCI are mutually exclusive, indicating that they cannot be both enabled.
Low-frequency TDD	Narrowband and TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	If DC component rate matching is enabled, the NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter must be deselected.

8.9.3.2.3 Function Impacts

- CSI-RS rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power reduction for CCUs	NRDUCellChnPower.UeSatPwrRankThld	<i>Power Control</i>	When power reduction for CCUs takes effect in low frequency bands, enabling CSI-RS rate matching may cause an unexpectedly high BLER in the slot with CSI-RS transmission. For this reason, it is recommended that the NZP_CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter be deselected in such scenarios. After the NZP_CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is deselected, REs that are not used for NZP CSI-RS transmission in the symbols to which NZP CSI-RS resources are mapped are not used for data scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When CSI-RS rate matching is in effect, the benefits delivered by NR inter-carrier power sharing may decrease.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When CSI-RS rate matching is in effect, the benefits delivered by NR multi-band power adaptive scheduling decrease.
High-frequency TDD	None	None	None	None

- TRS rate matching
 - TRS rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode set to DISTRIBUTED_MODE	<i>MIMO (TDD)</i>	It is recommended that MetaAAU beam sensing be enabled together with TRS rate matching.
Low-frequency TDD	PLLC primary cell	PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	If the PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected for a cell, rate-matching-pattern-configuration-based PDCCH rate matching must be disabled and TRS rate matching must be enabled for the cell.

- TRS rate matching optimization
 - None
- PDCCH rate matching

- Rate-matching-pattern-configuration-based PDCCH rate matching:

Impact relationships with spectrum management functions under RAN functions

None

Impact relationships with interference mitigation under RAN functions

None

Impact relationships with basic functions under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	PDCCH channel management	None	<i>Channel Management</i>	When rate-matching-pattern-configuration-based PDCCH rate matching is enabled, it is recommended that the NRDUCellPdcch.UIMaxCcePct parameter (indicating the maximum percentage of CCEs available for uplink scheduling) be set to 50. • If this parameter is set to a value less than 50, the CCEs available for uplink DCI may be insufficient. Consequently, uplink DCI transmission may fail, leading to resynchronization failures and repeated triggering of RRC reconfiguration over the air interface, resulting in service drops. • If this parameter is set to a value greater than 50, the CCEs available for downlink DCI may be insufficient, decreasing downlink throughput.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	When both flexible bandwidth (controlled by the FLEXIBLE_BANDWIDTH_S option of the NRDUCellFlexBw.FlexBwAlgoSwitch parameter) and rate-matching-pattern-configuration-based PDCCH rate matching are enabled for a cell, the following logic applies: $0 < \text{NRDUCellPdcch.OccupiedRbNum} \leq \text{Rounddown}((\text{NRDUCellFlexBw.CustDLStartRbIndex} + \text{NRDUCellFlexBw.CustDLAvailRbNum} - 1)/12) - \text{Roundup}(\text{NRDUCellFlexBw.CustDLStartRbIndex}/12)$

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	- For an FDD or SDL cell, if the LTE_NR_FDD_SPCT_SHR_SW and LTE_NR_SDL_SPCT_SHR_SW options of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter are deselected and rate-matching-pattern-configuration-based PDCCH rate matching is enabled, the following logic applies: - When NRDUCellPdcch.OccupiedSymbolNum is set to 2SYM, NRDUCellCoreset.CommonCtrlResRBNum must be set to RB24 or RB96. - When NRDUCellPdcch.OccupiedSymbolNum is set to 1SYM, NRDUCellCoreset.CommonCtrlResRBNum must be set to RB48.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	If (1) among the options of the NRDUCellIntrflIdent.IntrfOptimizationMethod parameter, only UL_NARROWBAND_INTRF_AVOID is selected, and (2) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, one of the following conditions must be met: • NRDUCellPdcch.OccupiedRbNum must be set to 0. • For a cell configured with two interference ranges, NRDUCellPdcch.OccupiedRbNum must be set to a value greater than or equal to 6. • For a cell configured with one interference range, NRDUCellPdcch.OccupiedRbNum must be set to a value greater than or equal to 5.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	If the CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter is selected and rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, the following rules apply: - For a cell configured with an interference range, if NRDUCellPdcch.OccupiedRbNum is not set to 0, the cell configuration must meet either of the following conditions: - If NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0: NRDUCellPdcch.OccupiedRbNum $\times 12 \leq \text{NRB} - N$ rounded up to a multiple of 6, where $N = \text{NRDUCellIntrfBand.EndRb}$ for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 + 1. - If NRDUCellIntrfBand.EndRb is set to a value equal to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0:

RAT	Function Name	Function Switch	Reference	Description
				NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRDUCellIntrfBand.StartRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0. For a cell configured with two interference ranges, if (1) NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0, and (2) NRDUCellIntrfBand.EndRb is set to a value equals to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1, NRDUCellPdcch.OccupiedRbNum cannot be set to 0 and must meet the following requirement: NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRDUCellIntrfBand.StartRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1 – <i>N</i> rounded up to a multiple of 6, where <i>N</i> = NRDUCellIntrfBand.EndRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 + 1. NRB indicates the number of RBs corresponding to the cell bandwidth

RAT	Function Name	Function Switch	Reference	Description
				specified in Table 5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	For a cell configured with two interference ranges, if (1) the CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter is selected, (2) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, (3) NRDUCellPdcch.OccupiedRbNum is not set to 0, and (4) the value of NRDUCellIntrfBand.StartRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1 is equal to the value of NRDUCellIntrfBand.EndRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 plus 1, the cell configuration must meet either of the following conditions: If NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0: $\text{NRDUCellPdcch.OccupiedRbNum} \times 12 \leq \text{NRB} - N$ rounded up to a multiple of 6, where $N = \text{NRDUCellIntrfBand.EndRb}$ for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1 + 1.

RAT	Function Name	Function Switch	Reference	Description
				If NRDUCellIntrfBand.EndRb is set to a value equal to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1: NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRDUCellIntrfBand.StartRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	If (1) NRDUCellIntrfIdent.DlCellIntrfAvoidPol is set to DL_INTRF_AVOID and (2) rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, there can be only one instance with NRDUCellIntrfBand.InterferenceDirection set to DOWNLINK or UPLINK_DOWNLINK and the configuration of NRDUCellPdcch.OccupiedRbNum must meet either of the following conditions: - If NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceDirection set to DOWNLINK or UPLINK_DOWNLINK: NRDUCellPdcch.OccupiedRbNum $\times 12 \leq \text{NRB} - N$ rounded up to a multiple of 6, where $N = \text{NRDUCellIntrfBand.EndRb}$ for the instance + 1. - If NRDUCellIntrfBand.EndRb is set to a value equal to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceDirection set to DOWNLINK or UPLINK_DOWNLINK: NRDUCellPdcch.OccupiedRbNum $\times 12 \leq \text{NRDUCellIntrfBand.StartRb}$ for the instance.

RAT	Function Name	Function Switch	Reference	Description
				NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	If the LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected and rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, the following conditions must be met: (1) NRDUCellPdcch.OccupiedRbNum ≠ 0, and (2) NRDUCellPdcch.OccupiedRbNum × 12 < 1 + gNBDLteNrSpctShrCg.NrReservedRbEndIndex corresponding to NRDUCellSpctCloud.NrSpctShrCellGrpld .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH - occupied frequency-domain resource range mapped onto OFDM symbols	NRDUCellPdcch.OccupiedRbNum	<i>Downlink Scheduling</i>	If (1) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter corresponding to NRDUCellBiCarrier.NrDuCellId , (2) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter corresponding to NRDUCellBiCarrier.CoNrDuCellId is selected, and (3) rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, NRDUCellPdcch.OccupiedRbNum corresponding to NRDUCellBiCarrier.CoNrDuCellId must be set to a value less than 14 for the cell.

Impact relationships with functions related to inter-RAT coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Channel adaptation optimization	CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	<i>LTE and NR Zero Bufferzone</i>	If the CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter is selected and rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, the following rules apply: - For a cell configured with an interference range, if NRDUCellPdcch.OccupiedRbNum is not set to 0, the cell configuration must meet either of the following conditions: - If NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0: NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRB – <i>N</i> rounded up to a multiple of 6, where <i>N</i> = NRDUCellIntrfBand.EndRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 + 1. - If NRDUCellIntrfBand.EndRb is set to a value equal to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0:

RAT	Function Name	Function Switch	Reference	Description
				NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRDUCellIntrfBand.StartRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0. For a cell configured with two interference ranges, if (1) NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0, and (2) NRDUCellIntrfBand.EndRb is set to a value equals to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1, NRDUCellPdcch.OccupiedRbNum cannot be set to 0 and must meet the following requirement: NRDUCellPdcch.OccupiedRbNum × 12 ≤ NRDUCellIntrfBand.StartRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1 – <i>N</i> rounded up to a multiple of 6, where <i>N</i> = NRDUCellIntrfBand.EndRb for the instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 + 1. NRB indicates the number of RBs corresponding to the cell bandwidth

RAT	Function Name	Function Switch	Reference	Description
				specified in Table 5.3.2-1 in 3GPP TS 38.104.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Channel adaptation optimization	CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter	<i>LTE and NR Zero Bufferzone</i>	For a cell configured with two interference ranges, if (1) the CHANNEL_ADAPT_OPT_SW option of the NRDUCellIntrfIdent.ChnMeasCtrlSw parameter is selected, (2) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, (3) NRDUCellPdcch.OccupiedRbNum is not set to 0, and (4) the value of NRDUCellIntrfBand.StartRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1 is equal to the value of NRDUCellIntrfBand.EndRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0 plus 1, the cell configuration must meet either of the following conditions: If NRDUCellIntrfBand.StartRb is set to 0 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0: $\text{NRDUCellPdcch.OccupiedRbNum} \times 12 \leq \text{NRB} - N \text{ rounded up to a multiple of 6, where } N = \text{NRDUCellIntrfBand.EndRb} \text{ for an instance with } \text{NRDUCellIntrfBand.InterferenceBandIndex} \text{ set to } 1 + 1.$

RAT	Function Name	Function Switch	Reference	Description
				If NRDUCellIntrfBand.EndRb is set to a value equal to NRB minus 1 for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 1: NRDUCellPdcch.OccupiedRbNum x 12 ≤ NRDUCellIntrfBand.StartRb for an instance with NRDUCellIntrfBand.InterferenceBandIndex set to 0. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.

Impact relationships with functions related to MIMO under RAN functions
None

Impact relationships with functions related to coverage improvement under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	SUL cannot be enabled when rate-matching-pattern-configuration-based PDCCH rate matching is enabled and NRDUCellPdcch.OccupiedRbNum is set to 1 or 2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is selected, the benefits delivered by rate-matching-pattern-configuration-based PDCCH rate matching decrease.
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UIDecouplingSwitch	<i>UL and DL Decoupling</i>	UL and DL Decoupling with SUL and NR FDD co-deployment cannot take effect when rate-matching-pattern-configuration-based PDCCH rate matching is selected and the NRDUCellPdcch.OccupiedRbNum parameter is set to 1 or 2 .
High-frequency TDD	Downlink PDSCH coverage enhancement	PDSCH_COVERAGE_ENHANCE_SW option of the NRDUCellChnPowerControlSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	If downlink PDSCH coverage enhancement is in effect and the transmit PSD of the base station is adjusted in the current TTI, then rate-matching-pattern-configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate matching cannot take effect in the current TTI.

Impact relationships with functions related to RAN services

None

Impact relationships with functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When both rate-matching-pattern-configuration-based PDCCH rate matching and power saving BWP are enabled, newly admitted UEs are not allowed to enter BWP2 as long as rate-matching-pattern-configuration-based PDCCH rate matching is in effect. The air interface resources of the UEs that have already switched to BWP2 are deleted through RRC reconfiguration messages, and such UEs are switched back to BWP1 from BWP2.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When the switch for rate-matching-pattern-configuration-based PDCCH rate matching is turned on, the RF channel intelligent shutdown function cannot take effect.
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	UEs with rate-matching-pattern-configuration-based PDCCH rate matching in effect do not switch to a sparse search space set group (SSSG).
High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Rate-matching-pattern-configuration-based PDCCH rate matching takes effect only for large-packet UEs. However, data convergence in symbols does not take effect for large-packet UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSwitch parameter	<i>UE Power Saving</i>	When both rate-matching-pattern-configuration-based PDCCH rate matching and time-domain power saving BWP are enabled, newly admitted UEs are not allowed to enter BWP2 as long as rate-matching-pattern-configuration-based PDCCH rate matching is in effect. The air interface resources of the UEs that have already switched to BWP2 are deleted through RRC reconfiguration messages, and such UEs are switched back to BWP1 from BWP2.

Impact relationships with functions related to carrier management under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	When CA SRS carrier switching is enabled, the increase in downlink throughput brought by rate-matching-pattern-configuration-based PDCCH rate matching slightly decreases.

Impact relationships with functions related to UCN under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordination between joint transmission and SSB rate matching	JT_SSB_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM is less likely to take effect if (1) "SSB rate matching", "rate-matching-pattern-configuration-based PDCCH rate matching", and "collaboration between joint transmission and PDCCH rate matching" are enabled for both the serving and cooperating cells and (2) SSB patterns or PDCCH rate matching patterns differ between the serving and cooperating cells. To increase this probability, it is recommended that the JT_SSB_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter be selected to enable coordination between joint transmission and SSB rate matching.
Low-frequency TDD	Coordination between joint transmission and SSB rate matching	JT_SSB_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter	<i>CoMP</i>	"Intra-base-station DL CoMP" is less likely to take effect if (1) "SSB rate matching", "rate-matching-pattern-configuration-based PDCCH rate matching", and "collaboration between joint transmission and PDCCH rate matching" are enabled for both the serving and cooperating cells and (2) SSB patterns or PDCCH rate matching patterns differ between the serving and cooperating cells. To increase this probability, it is recommended that the JT_SSB_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter be selected to enable coordination between joint transmission and SSB rate matching.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	- When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is selected, rate-matching-pattern-configuration-based PDCCH rate matching and intra-base-station DL CoMP (controlled by the INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter) can both take effect provided that they are both enabled. - When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is deselected, the following logic applies: If both rate-matching-pattern-configuration-based PDCCH rate matching and intra-base-station DL CoMP (controlled by the INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter) are enabled, only one will take effect on UEs in overlapping areas, operating adaptively based on conditions.
Low-frequency TDD	Collaboration between joint transmission and PDCCH rate matching	JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter	<i>CoMP</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	- When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is selected, rate-matching-pattern-configuration-based PDCCH rate matching and Co-MM can both take effect provided that they are both enabled. - When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is deselected, the following logic applies: If both rate-matching-pattern-configuration-based PDCCH rate matching and Co-MM are enabled, only one will take effect on UEs in overlapping areas, operating adaptively based on conditions.

Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Channel shutdown in low power consumption mode does not take effect when both low power consumption mode and rate-matching-pattern-configuration-based PDCCH rate matching are enabled and a custom backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink narrow band interference avoidance	UL_NARROWBAND_INTERF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	If (1) among the options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter, only UL_NARROWBAND_INTERF_AVOID is selected, and (2) rate-matching-pattern-configuration-based PDCCH rate matching is enabled, one of the following conditions must be met: • NRDUCellPdcch.OccupiedRbNum must be set to 0. • For a cell configured with two interference ranges, NRDUCellPdcch.OccupiedRbNum must be set to a value greater than or equal to 6. • For a cell configured with one interference range, NRDUCellPdcch.OccupiedRbNum must be set to a value greater than or equal to 5.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	If (1) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter corresponding to NRDUCellBiCarrier.NrDuCellId , (2) the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter corresponding to NRDUCellBiCarrier.CoNrDuCellId is selected, and (3) rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, NRDUCellPdcch.OccupiedRbNum corresponding to NRDUCellBiCarrier.CoNrDuCellId must be set to a value less than 14 for the cell.
Low-frequency TDD	Pattern-free PDCCH rate matching for speed turbo UEs	ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None	If rate-matching-pattern-configuration-free PDCCH rate matching is enabled and the ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter is deselected for a cell, rate-matching-pattern-configuration-based PDCCH rate matching must be enabled for the cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	If the LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected and rate-matching-pattern-configuration-based PDCCH rate matching is enabled for a cell, the following conditions must be met: (1) NRDUCellPdcch.OccupiedRbNum $\neq 0$, and (2) NRDUCellPdcch.OccupiedRbNum $\times 12 < 1 + \text{gNBDULteNrSpctShrCg.NrReservedRbEndIndex}$ corresponding to NRDUCellSpctCloud.NrSpectShrCellGrpId .
Low-frequency TDD	PLLC primary cell	PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	If the PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected for a cell, rate-matching-pattern-configuration-based PDCCH rate matching must be disabled and TRS rate matching must be enabled for the cell.

- Rate-matching-pattern-configuration-free PDCCH rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	In low-frequency TDD, when the DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected, rate-matching-pattern-configuration-free PDCCH rate matching does not take effect.
Low-frequency TDD	Multi-DCI Retransmission Adapt Switch	MULTI_DCI_RETRANS_ADAPT_SW option of the NrDuCellAlgoSwitch.MultiDciAlgoSwitch parameter	None	When the MULTI_DCI_RETRANS_ADAPT_SW option of the NrDuCellAlgoSwitch.MultiDciAlgoSwitch parameter is selected, rate-matching-pattern-configuration-free PDCCH rate matching does not take effect.
Low-frequency TDD	CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	When CA SRS carrier switching is enabled, the probability of rate-matching-pattern-configuration-free PDCCH rate matching taking effect decreases and the increase in downlink throughput slightly decreases.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	When the JT_PDCCH_NO_PATT_RM_COORD_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter is selected, rate-matching-pattern-configuration-free PDCCH rate matching and intra-base-station joint transmission can both take effect.
Low-frequency TDD	Scheduling in common PDCCH symbols for RedCap UEs	REDCAP_COMMON_PDCCH_SYMBOL_SCHEDULE_SWITCH option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	Rate-matching-pattern-configuration-free PDCCH rate matching does not take effect on UEs with scheduling in common PDCCH symbols for RedCap UEs in effect.
High-frequency TDD	None	None	None	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	UEs with rate-matching-pattern-configuration-free PDCCH rate matching in effect do not switch to a sparse SSSG.
High-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Rate-matching-pattern-configuration-free PDCCH rate matching takes effect only for large-packet UEs. However, data convergence in symbols does not take effect for large-packet UEs.
Low-frequency TDD	Pattern-free PDCCH rate matching for speed turbo UEs	ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None	If rate-matching-pattern-configuration-free PDCCH rate matching is enabled and the ST_UE_PDCCH_NO_PATTERN_RM_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter is deselected for a low-frequency cell, rate-matching-pattern-configuration-based PDCCH rate matching must be enabled for the cell.
High-frequency TDD	Downlink PDSCH coverage enhancement	PDSCH_COV_ENHANCE_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	If downlink PDSCH coverage enhancement is in effect and the transmit PSD of the base station is adjusted in the current TTI, then rate-matching-pattern-configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate matching cannot take effect in the current TTI.

- SSB rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3D Coverage Pattern (TDD)	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Beam Management</i>	In low-frequency NR TDD, when NRDUCellTrpBeam.MaxSsbPwrOffset is set to a value greater than 0, the SSB power increases, and the number of available PDSCH RBs in the SSB slots decreases. In this case, the SSB rate matching function causes benefits to decrease or even renders negative results.

- DC component rate matching

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None

8.9.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.9.3.4 Cells

When the **NRDUCellSulRelation.SulType** parameter is set to **FIX_SUL** and the **NRDUCell.FrequencyBand** parameter corresponding to the **NRDUCellSulRelation.SulNrDuCellId** parameter is set to **N199**, **N95**, or **N98**, the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter cannot be selected.

8.9.3.5 Others

- CSI-RS rate matching
None
- TRS rate matching
 - TRS rate matching
None
 - TRS rate matching optimization
When TRS rate matching optimization is enabled in a cell, the TRS periods must be the same between the cell and its neighboring cells. If the TRS periods are different, interference may occur, decreasing the average downlink UE throughput and average cell throughput.
- Rate-matching-pattern-configuration-based PDCCH rate matching

- Rate-matching-pattern-configuration-based PDCCH rate matching requires UEs to report their rate matching capabilities. The details are as follows:

When the rate matching capability identification optimization function is disabled (by deselecting the **RM_ABILITY_IDENTIFY_OPT_SW** option of the **NRDUCellPdsch.DLPdschAlgoSwitch** parameter), UEs need to report the **rateMatchingResrcSetSemi-Static** capability or both the **rateMatchingResrcSetSemi-Static** and **rateMatchingCtrlResrcSetDynamic** capabilities. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

When the rate matching capability identification optimization function is enabled, UEs need to report **rateMatchingResrcSetSemi-Static**, **rateMatchingResrcSetDynamic**, and **rateMatchingCtrlResrcSetDynamic** capability, or any combination of these capabilities. This means that the rate matching function can take effect for more UEs that report the rate matching capabilities in a cell. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

- It is recommended that the rate-matching-pattern-configuration-based PDCCH rate matching function be used only in single-UE peak rate scenarios and peak rate scenarios involving two to four UEs. In the following scenarios, this function must or should be disabled:
 - When the maximum DCI power offset in the common search space (specified by the **NRDUCellChnPwr.MaxCommonDciPwrOffset** parameter) is greater than 0, you are advised to disable the rate-matching-pattern-configuration-based PDCCH rate matching function because it may cause PDCCH power insufficiency if enabled.
 - You are advised to disable rate-matching-pattern-configuration-based PDCCH rate matching in specific scenarios, as this function fails to deliver peak rates for four MU-MIMO UEs in such scenarios. The aforementioned scenarios include:
 - LTE and NR are deployed on the same chip and 64T64R is used.
 - The UBBPfw provides CPRI connections.
 - If downlink MU-MIMO does not take effect for a heavily loaded low-frequency NR TDD cell that serves multiple UEs, it is good practice to disable rate-matching-pattern-configuration-based PDCCH rate matching.

- Rate-matching-pattern-configuration-free PDCCH rate matching

In low frequency bands, after this function is enabled, the downlink BLER increases for some UEs that support **multipleCORESET** or only **rateMatchingResrcSetSemi-Static**. Therefore, it is recommended that a relevant whitelist be configured on the base station so that rate-matching-pattern-configuration-free PDCCH rate matching does not take effect on such UEs.

- SSB rate matching

The SSB rate matching function requires that UEs report the rate matching capability in **rateMatchingResrcSetSemi-Static**. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

- DC component rate matching
DC component rate matching requires that UEs indicate their rate matching capability in *rateMatchingResrcSetSemi-Static* or *rateMatchingResrcSetDynamic*. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

8.9.4 Operation and Maintenance

8.9.4.1 Data Configuration

8.9.4.1.1 Data Preparation

[Table 8-10](#) and [Table 8-11](#) describe the parameters used for function activation and optimization, respectively.

Table 8-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Rate Match Switch	NRDUCellPdsch.Rate MatchSwitch	CSIRS_RATEMATCH_SW	Select this option.
Rate Match Switch	NRDUCellPdsch.Rate MatchSwitch	TRS_RATEMATCH_SW	Select this option in contiguous networking and deselect this option for isolated cells.
Rate Match Switch	NRDUCellPdsch.Rate MatchSwitch	TRS_RATEMATCH_OPT_SW	In low-frequency NR TDD peak rate or drive test scenarios, you are advised to select this option.
Rate Match Switch	NRDUCellPdsch.Rate MatchSwitch	PDCCH_RATEMATCH_SW	Select this option in peak-rate scenarios involving a single UE or 2–4 UEs is reached. It is recommended that this option be deselected on commercial networks. For details, see 8.9.3.5 Others .

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Switch	NRDUCellPdcch.PdccbAlgoSwitch	PDCCH_NO_PATERN_RATE_MATCH_SW	It is recommended that this option be deselected on commercial networks and be selected only in single-UE downlink peak rate scenarios.
Speed Turbo Algorithm Switch	NRDUCellServExp.SpeedTurboAlgoSwitch	PDCCH_NO_PAT_RM_RATIO_INC_SW	It is recommended that this option be deselected on commercial networks and be selected only in single-UE downlink peak rate scenarios.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	SSB_RATEMATCH_SW	NR TDD: Use the default value.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	DC_COMPONENT_RATEMATCH_SW	It is recommended that Huawei engineers configure this parameter based on network requirements.

Table 8-11 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Number of Occupied RBs	NRDUCellPdcch.OccupiedRbNum	None	This parameter is configurable when the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is selected. Set this parameter as required. When the NRDUCellPdcch.OccupiedRbNum parameter is set to a value less than 8 (corresponding to an actual configuration of 96 RBs), there is a risk that UEs cannot access the network due to insufficient PDCCH resources. The value of this parameter must be adapted to scenarios. This is a high-risk parameter. Modification of this parameter will interrupt services. Before the modification, refer to "Caution" for the corresponding parameter in the parameter reference.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DIPdschAlgoSwitch	RM_ABILITY_IDENTIFY_OPT_SW	Select this option.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	NZP_CSIRS_RATEMATCH_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	PDCCH_RATEMATCH_COMPATIBLE_SW	When the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter is selected in low frequency bands, the PDCCH_RATEMATCH_COMPATIBLE_SW option of the NRDUCellPdsch.RateMatchSwitch parameter can be selected to solve the compatibility issues of some UEs.

8.9.4.1.2 Using MML Commands

Before using MML commands, refer to [8.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling CSI-RS rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=CSIRS_RATEMATCH_SW-1;
//Enabling TRS rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=TRS_RATEMATCH_SW-1;
//Enabling TRS rate matching optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=TRS_RATEMATCH_OPT_SW-1;
//((Low-frequency cells) Enabling the rate-matching-pattern-configuration-based PDCCH rate matching
function with NRDUCellPdcch.OccupiedRbNum set to a value greater than 0 and the configured frequency-
domain range smaller than the number of RBs corresponding to the downlink system bandwidth specified
by NRDUCell.DlBandwidth
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=PDCCH_RATEMATCH_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedRbNum=2;
//Enabling rate-matching-pattern-configuration-free PDCCH rate matching in low frequency bands (This
function is not recommended on commercial networks and is recommended only in single-UE downlink
peak rate scenarios.)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_NO_PATTERN_RATE_MATCH_SW-1;
//((High-frequency cells) Enabling the PDCCH rate matching function (note that rate-matching-pattern-
configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate
matching are mutually exclusive)
//Enabling the rate-matching-pattern-configuration-based PDCCH rate matching function with
NRDUCellPdcch.OccupiedRbNum set to a value greater than 0 and the configured frequency-domain range
smaller than the number of RBs corresponding to the downlink system bandwidth specified by
NRDUCell.DlBandwidth
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=PDCCH_RATEMATCH_SW-1;
```

```

MOD NRDUCELLPDCCH: NrDuCellId=0, OccupiedRbNum=2;
//Enabling rate-matching-pattern-configuration-free PDCCH rate matching (This function is not
recommended on commercial networks and is recommended only in single-UE downlink peak rate
scenarios.)
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_NO_PATTERN_RATE_MATCH_SW-1;
//Enabling SSB rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=SSB_RATEMATCH_SW-1;
//Enabling DC component rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=DC_COMPONENT_RATEMATCH_SW-1;
//Enabling pattern-free PDCCH rate matching activation rate increase
MOD NRDUCELLSERVEXP: NrDuCellId=0, SpdTurboAlgoSwitch=PDCCH_NO_PAT_RM_RATIO_INC_SW-1;

```

Optimization Command Examples

```

//Enabling rate matching capability identification optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DIPdschAlgoSwitch=RM_ABILITY_IDENTIFY_OPT_SW-1;
//Turning on the switch for NZP CSI-RS rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=NZP_CSIRS_RATEMATCH_SW-1;
//Enabling PDCCH rate matching UE compatibility rectification
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=PDCCH_RATEMATCH_COMPATIBLE_SW-1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

//Disabling CSI-RS rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=CSIRS_RATEMATCH_SW-0;
//Disabling TRS rate matching optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=TRS_RATEMATCH_OPT_SW-0;
//Disabling TRS rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=TRS_RATEMATCH_SW-0;
//Disabling rate-matching-pattern-configuration-based PDCCH rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=PDCCH_RATEMATCH_SW-0;
//Disabling rate-matching-pattern-configuration-free PDCCH rate matching
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_NO_PATTERN_RATE_MATCH_SW-0;
//Disabling SSB rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=SSB_RATEMATCH_SW-0;
//Disabling DC component rate matching
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=DC_COMPONENT_RATEMATCH_SW-0;
//Disabling pattern-free PDCCH rate matching activation rate increase
MOD NRDUCELLSERVEXP: NrDuCellId=0, SpdTurboAlgoSwitch=PDCCH_NO_PAT_RM_RATIO_INC_SW-0;

```

8.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.9.4.2 Activation Verification

Observing Counters

- CSI-RS rate matching
After CSI-RS rate matching is enabled in single-UE peak rate scenarios, if the value of **N.ThpVol.DL.Cell** increases, this function has taken effect.
- TRS rate matching

- TRS rate matching

After TRS rate matching is enabled in contiguous networking with interference from neighboring cells, if the value of **N.ThpVol.DL.Cell** increases for CEUs, this function has taken effect.
- TRS rate matching optimization

None
- PDCCH rate matching

After PDCCH rate matching is enabled in single-UE peak rate scenarios, if the value of **N.ThpVol.DL.Cell** increases, this function has taken effect.
- SSB rate matching

After SSB rate matching is enabled in single-UE peak rate scenarios, if the value of **N.ThpVol.DL.Cell** increases, this function has taken effect.
- DC component rate matching

After DC component rate matching is enabled in single-UE scenarios, if the value of **N.ThpVol.DL.Cell** increases, this function has taken effect.

Observing Signaling Trace on the MAE-Access

Trace signaling over the Uu interface to check whether a function has taken effect.

1. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
2. In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.
3. Create a Uu interface tracing task.
4. Check whether the RRC_RECFCG message contains related IEs. If yes, the corresponding function has taken effect. If not, the corresponding function does not take effect.
 - CSI-RS rate matching, TRS rate matching

The same IEs in the RRC signaling are used to configure the ZP CSI-RS and ZP TRS. Specifically, check for the presence of the following IEs to determine whether CSI-RS rate matching or TRS rate matching takes effect:

 - *ZP-CSI-RS-Resource* IE, which specifies the time-domain position. In this IE, the **CSI-ResourcePeriodicityAndOffset** field indicates the time-domain slot position, and the **firstOFDMSymbolInTimeDomain** field indicates the time-domain symbol index.
 - *CSI-FrequencyOccupation* IE, which specifies the full-bandwidth puncturing position in the frequency domain.

Which rate matching function—CSI-RS rate matching or TRS rate matching—takes effect can be determined based on the **firstOFDMSymbolInTimeDomain** field because TRS and CSI-RS map to different time-domain symbol positions.
 - Rate-matching-pattern-configuration-based PDCCH rate matching

Check for the IEs related to *RateMatchPattern* (*ControlResourceSetId* specifying the PDCCH time-frequency resources).

- Rate-matching-pattern-configuration-free PDCCH rate matching
None
- SSB rate matching
Check for the IEs related to *RateMatchPattern* (*resourceBlocks* specifying the SSB frequency-domain information, with bit fields of only 20 RBs set to 1; *periodicityAndPattern* and *symbolsInResourceBlock* specifying the SSB time-domain information).
- DC component rate matching
Check for the IEs related to *RateMatchPattern* (*resourceBlocks* specifying the DC_COMPONENT frequency-domain information, with bit fields of one or two RBs set to 1; *periodicityAndPattern* and *symbolsInResourceBlock* specifying the DC_COMPONENT time-domain information).

8.9.4.3 Network Monitoring

Table 8-12 lists the counters for monitoring the performance of PDSCH resource mapping.

Table 8-12 Counters for monitoring the performance of PDSCH resource mapping

Description	Counter Name
Total volume of downlink data sent at the MAC layer in a cell	N.ThpVol.DL.Cell
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)

8.10 PDSCH PTRS Allocation

8.10.1 Principles

If there is noise (such as random white noise or flicker noise), components (such as RF components) cause random phase changes to output signals, creating phase noise. Phase noise may deteriorate signal to interference plus noise ratio (SINR) or error vector magnitude (EVM) at the receiver end, and produces many block errors. This impedes the ability of higher order modulation schemes to improve system capacity. Clock sources have higher frequency multiplication times in high frequency bands than in low frequency bands. This substantially increases phase noise, increasing the impact.

The PDSCH phase-tracking reference signal (PTRS) allocation scheme is introduced to reduce the impact of phase noise in such scenarios by using PTRS. This scheme is primarily used in high frequency bands.

As defined in section 5.1.6.3 "PT-RS reception procedure" in 3GPP TS 38.214, a PDSCH PTRS is determined by the time-frequency density and time-frequency position as follows:

- Time-frequency density
 - Time-domain density ($L_{\text{PT-RS}}$): every one, two, or four symbols, depending on both the MCS index threshold (1 to 4) for determining the downlink PTRS time-domain density and the DCI-indicated actual MCS index for scheduling, as described in [Table 8-13](#).

Table 8-13 PTRS time-domain density

Actual MCS Index for Scheduling	Time-Domain Density ($L_{\text{PT-RS}}$)
$l_{\text{MCS}} < \text{ptrs-MCS}_1$	PTRS is not present.
$\text{ptrs-MCS}_1 \leq l_{\text{MCS}} < \text{ptrs-MCS}_2$	4
$\text{ptrs-MCS}_2 \leq l_{\text{MCS}} < \text{ptrs-MCS}_3$	2
$\text{ptrs-MCS}_3 \leq l_{\text{MCS}} < \text{ptrs-MCS}_4$	1

In the formulas:

- l_{MCS} : actual MCS index for scheduling. The value is indicated by DCI.
- ptrs-MCS_1 : MCS index threshold 1 for determining the downlink PTRS time-domain density, which can be configured using the **NRDUCellPdsch.DIPtrsTimeDensityMcs1** parameter.
- ptrs-MCS_2 : MCS index threshold 2 for determining the downlink PTRS time-domain density, which can be configured using the **NRDUCellPdsch.DIPtrsTimeDensityMcs2** parameter.
- ptrs-MCS_3 : MCS index threshold 3 for determining the downlink PTRS time-domain density, which can be configured using the **NRDUCellPdsch.DIPtrsTimeDensityMcs3** parameter.
- ptrs-MCS_4 : MCS index threshold 4 for determining the downlink PTRS time-domain density. Depending on whether DL 256QAM or DL 64QAM is used for PDSCH PTRS modulation, ptrs-MCS_4 is either 28 or 29, respectively.
- Frequency-domain density ($K_{\text{PT-RS}}$): every two or four RBs, depending on both the RB number threshold (0 to 1) and the DCI-indicated actual number of scheduled RBs, as described in [Table 8-14](#).

Table 8-14 PTRS frequency-domain density

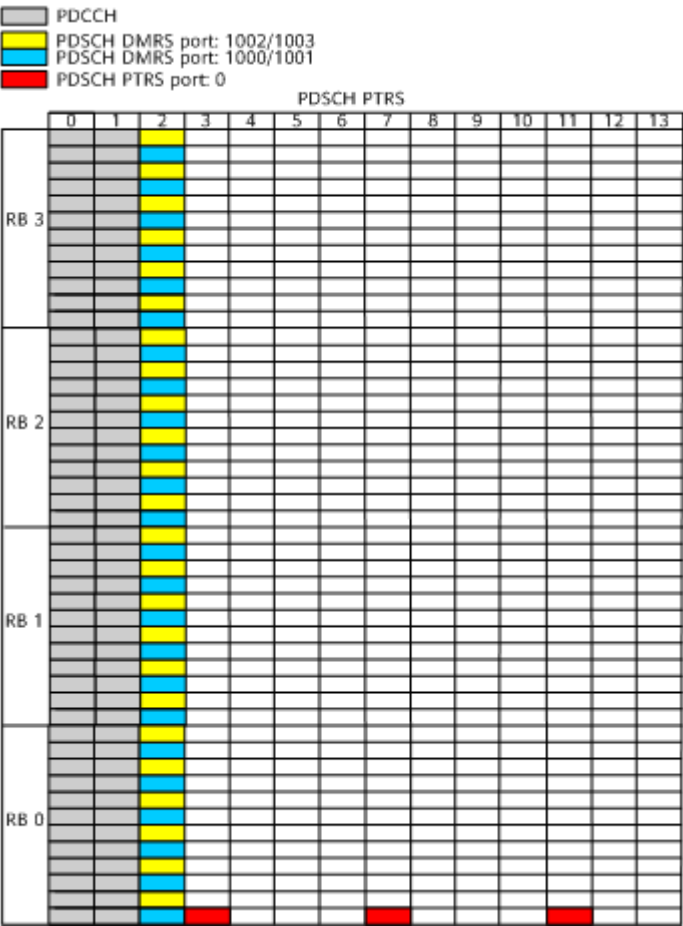
Actual Number of Scheduled RBs	Frequency-Domain Density ($K_{\text{PT-RS}}$)
$N_{\text{RB}} < N_{\text{RB0}}$	PTRS is not present.
$N_{\text{RB0}} \leq N_{\text{RB}} < N_{\text{RB1}}$	2
$N_{\text{RB1}} \leq N_{\text{RB}}$	4

In the formulas:

- N_{RB} : number of actually scheduled RBs. The number is indicated by the DCI.
- N_{RB0} : RB number threshold 0 for determining the downlink PTRS frequency-domain density, which can be configured using the **NRDUCellPdsch.DIPtrsFreqDensityNrb0** parameter.
- N_{RB1} : RB number threshold 1 for determining the downlink PTRS frequency-domain density, which can be configured using the **NRDUCellPdsch.DIPtrsFreqDensityNrb1** parameter.
- Time-frequency positions
 - The time-domain position starts from the first symbol for the PDSCH and skips the PDSCH DMRS symbols.
 - The frequency-domain position is related to C-RNTIs and PUSCH PTRS ports.
 - This version supports only one port.

Figure 8-11 shows the resource mapping of PDSCH PTRS with a 4-symbol time-domain density ($L_{\text{PT-RS}} = 4$), 4-RB frequency-domain density ($K_{\text{PT-RS}} = 4$), and a **00** start frequency-domain RE position.

Figure 8-11 Resource mapping of PDSCH PTRS



8.10.2 Network Analysis

8.10.2.1 Benefits

None

8.10.2.2 Impacts

The impacts between this function and other functions are described in [8.10.3.2.3 Function Impacts](#).

8.10.3 Requirements

8.10.3.1 Licenses

There are no license requirements for basic functions.

8.10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.10.3.2.1 Prerequisite Functions

None

8.10.3.2.2 Mutually Exclusive Functions

None

8.10.3.2.3 Function Impacts

None

8.10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

8.10.3.4 Others

None

8.10.4 Operation and Maintenance

8.10.4.1 Data Configuration

8.10.4.1.1 Data Preparation

Table 8-15 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 8-15 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL PTRS Time Density MCS Threshold 1	NRDUCellPdsch. DlPtrsTimeDensityMcs1	None	Use the default value.
DL PTRS Time Density MCS Threshold 2	NRDUCellPdsch. DlPtrsTimeDensityMcs2	None	Use the default value.
DL PTRS Time Density MCS Threshold 3	NRDUCellPdsch. DlPtrsTimeDensityMcs3	None	Use the default value.
DL PTRS Frequency Density RB Threshold 0	NRDUCellPdsch. DlPtrsFreqDensityNrb0	None	Use the default value.
DL PTRS Frequency Density RB Threshold 1	NRDUCellPdsch. DlPtrsFreqDensityNrb1	None	Use the default value.

8.10.4.1.2 Using MML Commands

Before using MML commands, refer to **8.10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the MCS threshold for the downlink PT-RS time-domain density
MOD NRDUCCELLPDSCH: NrDuCellId=0, DlPtrsTimeDensityMcs1=5;
MOD NRDUCCELLPDSCH: NrDuCellId=0, DlPtrsTimeDensityMcs2=5;
MOD NRDUCCELLPDSCH: NrDuCellId=0, DlPtrsTimeDensityMcs3=5;
//Configuring the RB number threshold for the downlink PT-RS frequency-domain density
MOD NRDUCCELLPDSCH: NrDuCellId=0, DlPtrsFreqDensityNrb0=3;
MOD NRDUCCELLPDSCH: NrDuCellId=0, DlPtrsFreqDensityNrb1=3;
```

Deactivation Command Examples

Deactivation command examples are not involved.

8.10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.10.4.2 Activation Verification

None

8.10.4.3 Network Monitoring

None

8.11 DCCH Type 1 Scheduling

8.11.1 Principles

In low-frequency scenarios, DCCH type 1 scheduling (controlled by the **DCCH_TYPE1_SCH_SW** option of the **NRDUCellDISchRes.DISchResAlgoSw** parameter) is introduced. When the number of UEs accessing a cell is greater than the UE number threshold for DCCH type 1 scheduling to take effect (specified by the **NRDUCellDISchRes.DcchType1SchUserNumThld** parameter), downlink resources can be allocated to small-packet MAC CEs using type 1 resource allocation. This function applies to low-frequency TDD cells. It is recommended that this function be enabled in heavy-load scenarios.

8.11.2 Network Analysis

8.11.2.1 Benefits

In low-frequency heavy-load scenarios, this function may increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**.

8.11.2.2 Impacts

The impacts between this function and other functions are described in [8.11.3.2.3 Function Impacts](#).

After this function is enabled, the downlink IBLER may increase, and the downlink PRB usage may decrease.

If this function is enabled in light-load scenarios, the values of the following indicators may decrease:

- Average downlink cell throughput
- **User Downlink Average Throughput (DU)**

8.11.3 Requirements

8.11.3.1 Licenses

There are no license requirements for basic functions.

8.11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.11.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DCCH type 1 scheduling	DCCH_TYPE 1_SCH_SW option of the NRDUCellDLSchRes.DLSchResAlgoSw parameter	<i>Downlink Scheduling</i>	The NRDUCellDLSchRes.DcchType1SchUserNumThld parameter specifies the UE number threshold for DCCH type 1 scheduling to take effect. This parameter takes effect only when DCCH type 1 scheduling is enabled.

8.11.3.2.2 Mutually Exclusive Functions

None

8.11.3.2.3 Function Impacts

None

8.11.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.11.3.4 Others

None

8.11.4 Operation and Maintenance

8.11.4.1 Data Configuration

8.11.4.1.1 Data Preparation

Table 8-16 describes the parameters used for function activation. This section does not describe parameters related to cell establishment or the activation of DSS or HDSS.

Table 8-16 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Scheduling Resource Algorithm Switch	NRDUCellDLSchRes.DLSchResAlgoSw	DCCH_TYPE1_SCH_SW	- It is recommended that this option be selected in downlink heavy-load scenarios. - It is recommended that this option be deselected in other scenarios.
DCCH Type1 Schedule User Number Threshold	NRDUCellDLSchRes.DcchType1SchUserNumThld	None	Use the default value.

8.11.4.1.2 Using MML Commands

Before using MML commands, refer to **8.11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling DCCH type 1 scheduling
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DLSchResAlgoSw=DCCH_TYPE1_SCH_SW-1;
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DcchType1SchUserNumThld=200;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling DCCH type 1 scheduling  
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DISchResAlgoSw=DCCH_TYPE1_SCH_SW-0;
```

8.11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.11.4.2 Activation Verification

If the **Cell Downlink Average Throughput (DU)** increases after this function is enabled, this function has taken effect.

8.11.4.3 Network Monitoring

After this function is enabled, check the performance indicators listed in [8.11.2 Network Analysis](#) to monitor the benefits and impacts of this function.

8.12 Optimized Scheduling of Tail RBGs

8.12.1 Principles

In low-frequency scenarios, optimized scheduling of tail RBGs (controlled by the **TAIL_RBG_SCH_OPT_SW** option of the **NRDUCellDISchRes.DISchResAlgoSw** parameter) is introduced to allocate the last RBG of the PDSCH to an appropriate UE, thereby reducing data packet truncation caused by insufficient resources. This function applies to low-frequency TDD cells.

8.12.2 Network Analysis

8.12.2.1 Benefits

After this function is enabled in low-frequency scenarios, the proportion of samples with low downlink average UE throughput in the cell may decrease. (This proportion = $(N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1) / \text{SUM}(N.Thp.DL.Samp.Indexk)$, where k is in the range of 0 to 14)

8.12.2.2 Impacts

The impacts between this function and other functions are described in [8.12.3.2.3 Function Impacts](#).

After this function is enabled, the delay of the first packet may increase, and the downlink PRB usage may decrease.

8.12.3 Requirements

8.12.3.1 Licenses

There are no license requirements for basic functions.

8.12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.12.3.2.1 Prerequisite Functions

None

8.12.3.2.2 Mutually Exclusive Functions

None

8.12.3.2.3 Function Impacts

None

8.12.3.3 Hardware

For details, see [4.3.3 Hardware](#).

8.12.3.4 Others

None

8.12.4 Operation and Maintenance

8.12.4.1 Data Configuration

8.12.4.1.1 Data Preparation

[Table 8-17](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment or activation of DSS or HDSS.

Table 8-17 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Scheduling Resource Algorithm Switch	NRDUCellDLSchResAlgoSw	TAIL_RBG_SCH_OPT_SW	- It is recommended that this option be selected if the proportion of small-packet UEs is high. - In other scenarios, the default value is recommended.

8.12.4.1.2 Using MML Commands

Before using MML commands, refer to [8.12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling optimized scheduling of tail RBGs
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DLSchResAlgoSw=TAIL_RBG_SCH_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling optimized scheduling of tail RBGs
MOD NRDUCELLDLSCHRES: NrDuCellId=0, DLSchResAlgoSw=TAIL_RBG_SCH_OPT_SW-0;
```

8.12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.12.4.2 Activation Verification

If the proportion of samples with low average downlink UE throughput in a cell decreases after this function is enabled, this function has taken effect.

8.12.4.3 Network Monitoring

After this function is enabled, check the performance indicators listed in [8.12.2 Network Analysis](#) to monitor the benefits and impacts of this function.

9 DMRS Resource Allocation

9.1 Principles

DMRS resource allocation is aimed at PDSCH demodulation by UEs. Section 7.4.1 "Reference signals" in 3GPP TS 38.211 V15.4.0 defines PDSCH DMRS, in terms of DMRS symbols in the time domain and DMRS configuration types in the frequency domain.

- DMRS symbol
 - Front-loaded DMRS
 - Number of front-loaded DMRS symbols: One or two symbols are supported in the current version, in accordance with the specifications. The **NRDUCellPdsch.DlDmrsMaxLength** parameter specifies the maximum length of symbols carrying downlink front-loaded DMRS.
 - When **NRDUCellPdsch.DlDmrsMaxLength** is set to **1SYMBOL**, downlink front-loaded DMRS spans one symbol to minimize DMRS overhead.
 - When **NRDUCellPdsch.DlDmrsMaxLength** is set to **2SYMBOL**, downlink front-loaded DMRS adaptively spans one or two symbols to preferentially ensure user experience.
 - When **NRDUCellPdsch.DlDmrsMaxLength** is set to **2SYMBOL_FOR_COV_ENHANCEMENT**, downlink front-loaded DMRS adaptively spans one or two symbols to preferentially ensure the coverage performance of cell edge UEs.
 - Number of front-loaded DMRS positions: One front-loaded DMRS position is supported.

 NOTE

The number of DMRS positions is not equivalent to the number of DMRS symbols. The former indicates the number of available DMRS positions, whereas the latter indicates the number of symbols occupied by one DMRS position.

- In single-symbol DMRS scenarios, one DMRS position spans one symbol.
- In dual-symbol DMRS scenarios, one DMRS position spans two symbols.

– Additional DMRS

- Number of additional DMRS symbols: Additional DMRS spans the same number of symbols as front-loaded DMRS.
 - When front-loaded DMRS spans one symbol, additional DMRS at each position spans one symbol.
 - When front-loaded DMRS spans two symbols, additional DMRS at each position spans two symbols.
- Number of additional DMRS positions: **pos0**, **pos1**, **pos2**, and **pos3** are defined in specifications. **pos0**, **pos1**, and **pos2** as well as adaptive number of additional DMRS positions are supported in the current version. The **NRDUCellPdsch.DLAdditionalDmrsPos** parameter specifies the additional DMRS position in the downlink. The time-domain positions of additional DMRS symbols in a slot are determined based on the PDSCH symbol configuration, and do not require special indication.
 - When **NRDUCellPdsch.DLAdditionalDmrsPos** is set to **NOT_CONFIG**, downlink additional DMRS configuration is **pos0**, indicating that no downlink additional DMRS symbol is configured.
 - When **NRDUCellPdsch.DLAdditionalDmrsPos** is set to **POS1**, a maximum of one additional DMRS position is allowed in the downlink.
 - When **NRDUCellPdsch.DLAdditionalDmrsPos** is set to **POS2**, a maximum of two additional DMRS positions are allowed in the downlink.
 - When the **NRDUCellPdsch.DLAdditionalDmrsPos** parameter is set to **POS_ADAPTIVE**, the gNodeB adaptively configures the number of additional DMRS positions based on the UE moving speed in low-frequency scenarios or based on the UE channel conditions in high-frequency scenarios, reducing unnecessary reference signal overheads while ensuring demodulation performance. The configuration takes effect only after the RRC reconfiguration procedure is complete.

In low-frequency NR TDD scenarios, the following optimization functions related to the number of additional DMRS positions are introduced.

- For low-frequency NR TDD cells where downlink MU-MIMO (controlled by the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter) is enabled, load-based downlink additional DMRS position adaptation (controlled by the **DL_LOAD_BASED_POS_ADAPT_SW** option of the **NRDUCellDmrs.DLPosAdaptOptAlgoSw** parameter) is

introduced. After the load-based downlink additional DMRS position adaptation function is enabled: If there are many UEs to be scheduled in a cell, the number of downlink additional DMRS positions is not adaptively adjusted but pos1 configuration is always used to ensure the pairing performance of MU-MIMO UEs; if there are few UEs to be scheduled in a cell, pos0 or pos1 is adaptively selected as the configuration for the number of downlink additional DMRS positions. Load-based downlink additional DMRS position adaptation is supported only in low frequency bands in NR TDD.

- The function of restricting the application scenarios of adaptive number of downlink additional DMRS positions is introduced for low-frequency NR TDD cells to resolve the scenario-specific performance differences between UEs using pos0 configuration. This function is controlled by the **DL_POS_ADAPT_SCENARIO_SW** option of the **NRDUCellDmrs.DIPosAdaptOptAlgoSw** parameter and applies to low-speed cells in low frequency bands in NR TDD.
 - DMRS configuration type
- The DMRS configuration type is specified by the **NRDUCellPdsch.DIDmrsConfigType** parameter, which can be set to **TYPE1**, **TYPE2**, or **TYPE_ADAPTIVE**.
- **Figure 9-1** shows the DMRS when **NRDUCellPdsch.DIDmrsConfigType** is set to **TYPE1** (using front-loaded DMRS of type A PDSCH as an example).

Figure 9-1 PDSCH DMRS configuration type 1

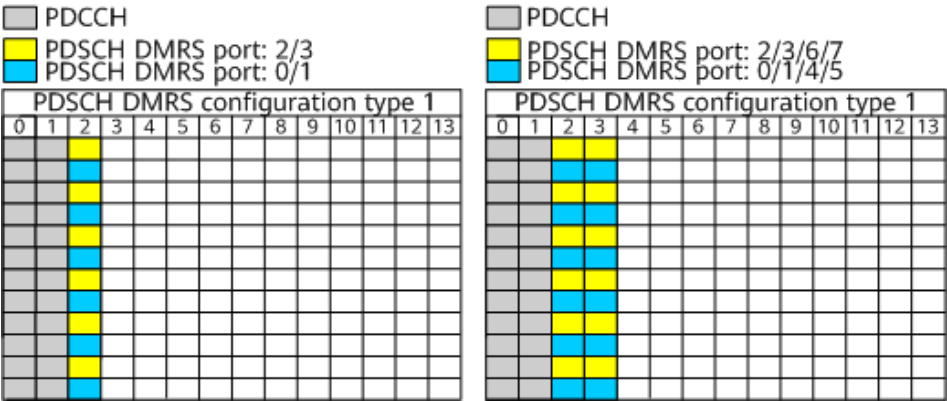


Table 7.4.1.1.2-1: Parameters for PDSCH DM-RS configuration type 1 in 3GPP TS 38.211 V15.6.0 defines two CDM groups for DMRS configuration type 1:

- The first group (CDM group 0) includes DMRS REs marked in blue in the preceding figure. The corresponding DMRS ports are as follows:
 - 0/1 (front-loaded DMRS spans one symbol)
 - 0/1/4/5 (front-loaded DMRS spans two symbols)
- The second group (CDM group 1) includes DMRS REs marked in yellow in the preceding figure. The corresponding DMRS ports are as follows:

- 2/3 (front-loaded DMRS spans one symbol)
- 2/3/6/7 (front-loaded DMRS spans two symbols)
- **Figure 9-2** shows the DMRS when **NRDUCellPdsch.DlDmrsConfigType** is set to **TYPE2** (using front-loaded DMRS of type A PDSCH as an example).

Figure 9-2 PDSCH DMRS configuration type 2

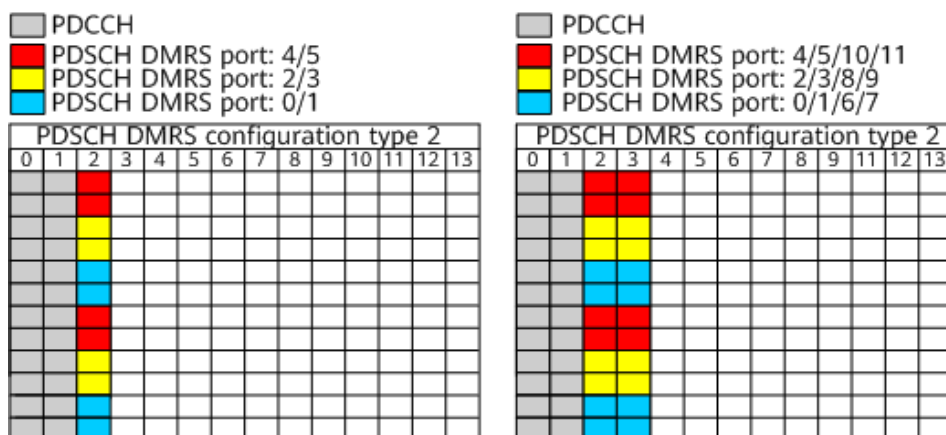


Table 7.4.1.1.2-2: Parameters for PDSCH DM-RS configuration type 2 in 3GPP TS 38.211 V15.6.0 defines three CDM groups for DMRS configuration type 2:

- The first group (CDM group 0) includes DMRS REs marked in blue in the preceding figure. The corresponding DMRS ports are as follows:
 - 0/1 (front-loaded DMRS spans one symbol)
 - 0/1/6/7 (front-loaded DMRS spans two symbols)
- The second group (CDM group 1) includes DMRS REs marked in yellow in the preceding figure. The corresponding DMRS ports are as follows:
 - 2/3 (front-loaded DMRS spans one symbol)
 - 2/3/8/9 (front-loaded DMRS spans two symbols)
- The third group (CDM group 2) includes DMRS REs marked in red in the preceding figure. The corresponding DMRS ports are as follows:
 - 4/5 (front-loaded DMRS spans one symbol)
 - 4/5/10/11 (front-loaded DMRS spans two symbols)
- When **NRDUCellPdsch.DlDmrsConfigType** is set to **TYPE_ADAPTIVE**:
 - In low-frequency scenarios, the DMRS configuration type is adaptively selected between type 1 and type 2 based on the channel quality of UEs.
 - In high-frequency scenarios, the DMRS configuration type is adaptively selected between type 1 and type 2 based on the resource overhead and downlink data transmission performance of UEs.

 NOTE

- The effective values of the **NRDUCellPdsch.DlDmrsMaxLength**, **NRDUCellPdsch.DlAdditionalDmrsPos**, and **NRDUCellPdsch.DlDmrsConfigType** parameters are determined by the UE capability. For example, if the **NRDUCellPdsch.DlDmrsConfigType** parameter is set to **TYPE2**, the effective value is **TYPE1** for UEs that support only type 1 or **TYPE2** for UEs that support type 2.
- The **NRDUCellPdsch.DlDmrsCdmGroupNumber** parameter on the base station specifies the number of CDM groups that can be used during PDSCH DMRS port allocation. The effective number of CDM groups is equal to the smaller of two values, the value of the **NRDUCellPdsch.DlDmrsCdmGroupNumber** parameter or the number of CDM groups supported by the DMRS configuration type (specified by the **NRDUCellPdsch.DlDmrsConfigType** parameter).
- DMRS sequence
The gNodeB complies with 3GPP Release 16 to use downlink low-PAPR DMRS sequence for UEs that support such sequence. This function requires UEs to report the **lowPAPR-DMRS-PDSCH-r16** field. This function is controlled by the **DL_LOW_PAPR_DMRS_SW** option of the **NRDUCellPdsch.DlPdschAlgoSwitch** parameter.
 - When this option is selected, the low PAPR DMRS sequence is used for UEs with the downlink low-PAPR DMRS capability.
 - When this option is deselected, this function does not take effect.

9.2 Network Analysis

9.2.1 Benefits

DMRS resource allocation for UEs is aimed at PDSCH demodulation. DMRS resource overheads can be properly configured based on service scenarios to maximize system capacity.

Enabling load-based downlink additional DMRS position adaptation increases the MU pairing probability when the cell load is relatively heavy, thereby improving the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**).

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

9.3 Requirements

9.3.1 Licenses

There are no license requirements for basic functions.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	In single-beam scenario of mmWave 3D coverage pattern (with NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_103(Single Beam Scenario)), adaptive number of downlink additional DMRS positions cannot be enabled.
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetWorkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Adaptive number of downlink additional DMRS positions cannot be enabled for hyper cells.
High-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIS_SWITCH option of the NRDUCellAIgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	When the low latency and high reliability function is enabled, adaptive number of downlink additional DMRS positions cannot be enabled.
Low-frequency TDD	None	None	None	None

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Downlink additional DMRS in a high-speed cell can occupy only one symbol, that is, NRDUCellPdsch.DlDmrsMaxLength must be set to 1SYMBOL ; one or two downlink additional DMRS positions are supported by a high-speed cell, that is, NRDUCellPdsch.DlAdditionalDmrsPos can be set to POS1 or POS2 .
Low-frequency TDD	Downlink DMRS type control	DL_DMRS_TYPE_CTRL_SW option of the NRDUCellVoiceAlgo.VoiceUserAlgoSw parameter	<i>VoNR</i>	With downlink DMRS type control, PDSCH DMRS configuration type 1 is always used for voice service UEs when NRDUCellPdsch.DlDmrsConfigType is set to TYPE_ADAPTIVE .

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

The downlink low-PAPR DMRS function has the following hardware requirements:

- Base station models
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- Boards and RF modules
All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
This function has the following requirements on RF modules and baseband processing units:
If RRUs are used as RF modules, NR TDD-capable low-frequency UBBPh or later series baseband processing units can be used.
If AAUs are used as RF modules:
 - AAU5668m/AAU5368/AAU5656/AAU5356/AAU5368md/AAU5668/AAU5832a/AAU5665md: All NR TDD-capable baseband processing units can be used.
 - Other AAUs: NR TDD-capable low-frequency UBBPh or later series baseband processing units can be used.

For details about the hardware requirements of other functions, see [4.3.3 Hardware](#).

9.3.4 Others

The downlink low-PAPR DMRS function requires UEs to report the **lowPAPR-DMRS-PDSCH-r16** field.

Other requirements for the other functions: None

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-1 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 9-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink DMRS Configuration Type	NRDUCellPdsch.DlDmrsConfigType	None	Use the default value.
Downlink DMRS Maximum Length	NRDUCellPdsch.DlDmrsMaxLength	None	Use the default value.
Downlink Additional DMRS Position	NRDUCellPdsch.DlAdditionalDmrsPos	None	Use the default value.
DL DMRS CDM Group Number	NRDUCellPdsch.DlDmrsCdmGroupNumber	None	Use the default value.
DL Pos Adaptation Optimization Algo Switch	NRDUCellDmrs.DlPosAdaptOptAlgoSw	DL_LOAD_BASED_POS_ADAPT_SW	Select this option.
DL Pos Adaptation Optimization Algo Switch	NRDUCellDmrs.DlPosAdaptOptAlgoSw	DL_POS_ADAPT_SCENARIO_SW	Use the default value.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	DL_LOW_PAPR_DMRS_SW	Select this option.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the downlink DMRS configuration type to type 2
MOD NRDUCELLPDSCH: NrDuCellId=0, DlDmrsConfigType=TYPE2;
//Setting the maximum number of downlink DMRS symbols to two
MOD NRDUCELLPDSCH: NrDuCellId=0, DlDmrsMaxLength=2SYMBOL;
//Configuring the downlink additional DMRS position (POS1 used as an example)
MOD NRDUCELLPDSCH: NrDuCellId=0, DlAdditionalDmrsPos=POS1;
//Configuring the CDM groups of downlink DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlDmrsCdmGroupNumber=CDM_GROUP_NUM3;
//Turning on the switch for load-based downlink additional DMRS position adaptation
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_LOAD_BASED_POS_ADAPT_SW-1;
//Turning on the switch for restricting the application scenarios of adaptive number of downlink additional DMRS positions (available only when DlAdditionalDmrsPos is set to POS_ADAPTIVE)
MOD NRDUCELLPDSCH: NrDuCellId=0, DlAdditionalDmrsPos=POS_ADAPTIVE;
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_POS_ADAPT_SCENARIO_SW-1;
//Enabling downlink low-PAPR DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=DL_LOW_PAPR_DMRS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

High-frequency NR TDD: None

Low-frequency NR TDD:

```
//Turning off the switch for load-based downlink additional DMRS position adaptation
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_LOAD_BASED_POS_ADAPT_SW-0;
//Turning off the switch for restricting the application scenarios of adaptive number of downlink additional DMRS positions
MOD NRDUCELLDMRS: NrDuCellId=0, DlPosAdaptOptAlgoSw=DL_POS_ADAPT_SCENARIO_SW-0;
//Disabling downlink low-PAPR DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=DL_LOW_PAPR_DMRS_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

This function is a basic function and takes effect by default. Activation verification is not involved.

9.4.3 Network Monitoring

The indicators listed in [9.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

10 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

11 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

12 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

13 Reference Documents

- 3GPP TS 38.214: "NR; Physical layer procedure for data"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *QoS Management*
- *Channel Management*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *MIMO (TDD)*
- *Interference Avoidance*
- *High Speed Mobility*
- *UE Power Saving*
- *Super Uplink*
- *Hyper Cell*
- *Cell Combination*
- *CoMP*
- *Multi-RAT Coordinated Energy Saving*
- *URLLC*
- *Network Slicing*
- *Power Control*
- *Energy Conservation and Emission Reduction*
- *Beam Management*
- *LTE and NR Power Allocation*
- *EPC-based NSA Basics*
- *Co-MM (Low-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *mmWave Beam Management (High-Frequency TDD)*

- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *CA Experience Improvement*
- *3D Networking Experience Improvement*
- *RedCap*
- *Uplink Scheduling*
- *NR Inter-Carrier Dynamic Power Allocation*
- *5G Networking and Signaling*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Green Site*
- *VoNR*
- *Multi-Frequency Smart Aggregation*
- *LTE and NR Zero Bufferzone*
- *UL and DL Decoupling*
- *mmWave Super Coverage (High-Frequency TDD)*